

BIOMIMICRY: EXPLORING PEPTOIDS AND RECOMBINANT PROTEIN
POLYMERS FOR BIOMEDICAL APPLICATIONS

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Abstract

Nature has been evolving unparalleled molecular designs with extraordinary activity, simplicity, efficiency, and durability. The process of biomimicry takes inspiration from Nature. In this dissertation, I focused on the development of functional analogues of natural peptides and protein polymers for several applications that continue to challenge the biomedical field.

Five applications were discussed: The first three sections focus on bioactive peptides that are generally shorter than 50 amino acids. The goal was to preserve the bioactivity of the natural peptides while reducing their natural sensitivity to proteases and fast *in vivo* clearance. The last two sections focus on developing protein polymer-based hydrogels. The goal was to generate biomimicking scaffolds for tissue regenerations.

(i) Poly-*N*-substituted glycines, or peptoids, provide a biostable scaffold that can display a great diversity of side chains in highly tunable sequences via facile solid-phase synthesis. Current chemotherapeutics in oncology are often limited by side effect profiles and selection for drug resistance. Herein, I present a library of anticancer peptoids that mimic the cationic, amphipathic structural features of host defense peptides and explore their structure-activity relationship, and killing mechanisms. Several peptoids were found with broad cytotoxicity against cancer cells as well as the ability to overcome multidrug resistance. An initial *in vivo* study with a primary, orthotopic human breast cancer xenotransplantation model demonstrated anticancer efficacy of one of the studied peptoids.

(ii) Cell-penetrating peptides have found numerous applications in biology and medicine as molecular transporters. We developed a library of cationic, amphipathic peptoids as a novel class of transporters and investigated the relationships between their structures, cellular uptake efficiency, and the associated cytotoxicity. Both guanidinium heads and bulky, aromatic hydrophobic residues were found to render the cationic, amphipathic constructs more permeable. Moreover, different internalization mechanisms were observed for peptoid transporters with distinct structures. One peptoid was identified as a promising transporter with excellent cellular uptake efficiency and low cytotoxicity.

(iii) Bombesin (BBN) peptide can bind with high affinity and specificity to the GRP receptors (GRPR) which are upregulated in invasive prostate cancer. BBN(7-14) provides a promising basis for developing radiometallated diagnostic or therapeutic radiopharmaceuticals to target GRPR positive prostate cancer. I report a design of a 4-arm PEG-based platform with multivalent BBN(7-14) for targeted delivery to the GRPR positive prostate cancer. The PEG-BBN conjugates displayed comparable tumor uptake as the free BBN while having both a lower liver uptake and higher tumor-to-blood ratio in a biodistribution study.

(iv) Regenerative medicine is in need of bioactive extracellular matrix-like scaffolds that can interact with cells and allow tissue regeneration in a well-controlled manner. Via “controlled cloning”, matrix metalloproteases (MMP) degradation sites were built into multiple sites along a protein polymer that can be enzymatically crosslinked into a previously established non-bioactive hydrogel systems. The incorporation of MMP degradation sites greatly improved cell infiltration into the hydrogel.

(v) Lastly, a novel *in situ* forming hybrid, biomimetic hydrogel comprising bioactive recombinant protein polymers, hyaluronic acid (HA) and polyethylene glycol (PEG) was developed. In a preliminary study, this hydrogel was found to be biocompatible and bioactive, which opens up future studies with this hydrogel system.

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Selected Abbreviations

Fmoc fluorenyl-methoxy-carbonyl

ACN acetonitrile

AMP antimicrobial peptide

VEGFR2 vascular endothelial growth factor receptor 2

CPP Cell penetrating peptide

PTD protein transduction domain

Tat trans-activator of transcription

Antp Antennapedia protein

VEC murine vascular endothelial cadherin

MAP model amphipathic peptide

RNA ribonucleic acid

DNA deoxyribonucleic acid

FACS Fluorescence-activated cell sorting

PCI photochemical internalization

ROS reactive oxygen species

ECM extracellular matrix

GAG glycosaminoglycans

TG transglutaminase

PEG poly(ethylene glycol)

MMP matrix metalloproteases

IPTG isopropyl- β -D-thiogalactopyranoside

CIP calf intestinal phosphatase

SDS-PAGE sodium dodecyl sulfate polyacrylamide gel electrophoresis

MALDI-TOF matrix-assisted laser desorption/ionization-time-of-flight

3D three-dimensional

2D two-dimensional

HA hyaluronic acid

PEG-DA PEG-diacrylate

MSC mesenchymal stem cell

bFGF basic fibroblast growth factor

DTNB 5,5'-Dithio-*bis*-(2-nitrobenzoic acid)

Chapter 1

Introduction

1.1 An overview of biomimicry

The term “biomimicry” comes from the Greek words “Bios”, meaning life, and “mimesis”, meaning to imitate. Biomimicry is to learn from nature’s elegant creatures, materials or processes which have been refined over the course of evolution, and come up with biologically-inspired designs, adaptations or derivations with similar or even improved properties. Biomimicry involves the identification and understanding of functions, structures and principles of various objects evolved in nature, and the further design and fabrication of various materials or devices. This field is highly interdisciplinary (1-3).

Biomimicry has a long history and has influenced a variety of fields in people’s lives. About 3000 years ago, the Chinese already tried to make artificial silk. In the twentieth century, aircraft were developed with designs learned from nature (1). State-of-the-art technology is enabling biomimicry to occur not only on the macroscale but also on the nanoscale and microscale, which has led to the development of a broad range of bio-inspired materials and devices (2). Superhydrophobicity, self-cleaning, drag reduction in fluid flows, energy conversion and conservation, high adhesion, reversible adhesion, high mechanical strength, biological self-assembly, thermal insulation and self-healing are some examples of the properties that current studies are looking for via biomimicry (1). Advances in technology and knowledge of nature are also continually expanding what is possible in the field of biomimicry.

In particular, proteins and peptides built from a series of amino acids, which are essential molecules that participate in virtually every process in living organisms, are important targets in biomimicry (3-6). Hence, the major work presented here focuses on

synthetic peptidomimetics to mimic small peptides and genetically engineered recombinant protein polymers to utilize and combine various bioactive domains into one protein molecule.

1.2 Peptidomimetic scaffolds

Peptides are short, sequence- and length-specific oligomers composed of amino acids. They are distinguished from proteins on the basis of molecular size, typically comprising fewer than 50 amino acids (4). There are many natural bioactive peptides in living cells, which assume functions like cellular receptor ligands involved in signaling transduction, endogenous antibiotics, *etc* (7). Moreover, even in larger proteins, despite the overall structural complexity, their functions can be mediated just by a small number of key protein domains. These domains often just involve one or more of the well-defined structural elements, the structures and functions of which can be captured by relatively short peptide sequences (3). Therefore, short peptide sequences from proteins have also been studied extensively to understand the protein structures or to modulate protein-protein interactions.

Nature has refined these peptide sequences and activities through evolution. Recognizing the wisdom that Nature has offered, the field of synthetic peptides has attracted great interest in order to study protein structures and functions or to develop bioactive peptides for fundamental research or for therapeutic applications (8). For example, many therapeutic peptides have been developed to contribute to the treatment of diabetic, osteoporotic, oncologic, antiviral, antibacterial, and other indications (8). Unfortunately, natural peptides with great bioactivity are inferior drug candidates because

of their sensitivity to proteases, low oral bioavailability, potential immunogenicity and poor metabolic stability *in vivo* (9).

One of the efforts to ameliorate the disadvantages of peptides and generate effective therapeutics based on peptides is to employ non-natural peptide mimics, ‘Peptidomimetics’ (6). Synthetic peptidomimetics can typically be produced in an efficient and cost-effective manner. They are designed to transcend their natural counterparts with improved chemical diversity, possibly reduced immunogenicity and more favorable therapeutic profiles with improved stability and permeability. In addition, in a peptidomimetic system, molecular parameters can generally be precisely controlled to simplify the structure-activity study, and the resulting conclusions are also transferable back to the natural system from where more broadly applicable knowledge may result.

Peptidomimetics can mimic the peptide primary structure through use of amide bond isosteres or modifications of the peptide backbone including chain extensions or heteroatom incorporations (4). In addition to this primary structure mimicry, peptidomimetic oligomers can also be designed to exhibit well-defined secondary structural elements such as helices, turns and sheet-like structures. These peptidomimetic oligomers are known as “foldamers”, which are typically oligomeric or polymeric molecules (up to ~50mers) that mimic a variety of simple secondary structures via non-natural backbones (4, 10). In biological systems, structure and function are intimately intertwined, as protein functions are generally controlled via accurate protein interactions that are mediated via proper protein structures (3). Thus, properly designed structural mimics of a peptide or a key protein domain will often be able to perform functions in a similar fashion. As shown in Figure 1.1 for the peptide primary structure mimics, some

examples are azapeptides, oligocarbamates and oligoureas. Well-characterized foldamers include β -peptides, α/β -peptides, γ -peptides and poly-*N*-substituted glycines (peptoids) (2, 4, 6, 10). In addition to the above bioinspired aliphatic foldamers which resemble natural peptide backbones (Figure 1.1), there are also aryl foldamers which contain aromatic rings in the backbone (6, 10). Among these foldamers, the following sections will focus on β -peptides, α/β -peptides and peptoids which have demonstrated potential for use as pharmaceutical agents.

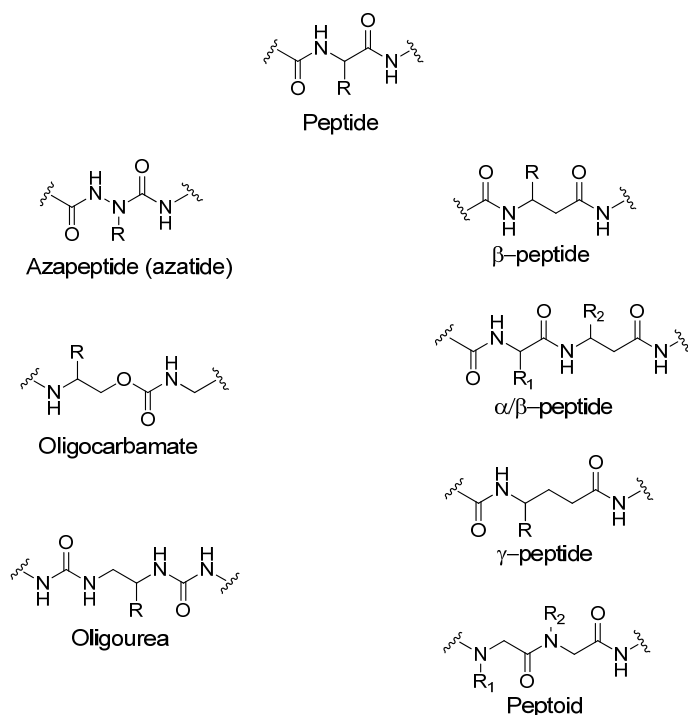


Figure 1.1 Structures of representative sequence-specific peptidomimetic oligomers.

1.2.1 Synthesis of β -peptides, α/β -peptides and peptoids

Progression in synthetic peptide chemistry is the fundamental driver of peptide and peptidomimetic research (11). The development of solid-phase synthesis which enables

the easy removal of byproducts and the washing between subsequent couplings, together with improvements in protecting group designs and coupling agents, have greatly simplified peptide synthesis and increased the product quality and yield (12). Advances in purification and analysis techniques further make the synthesis of sequence-specific peptides up to 50 monomers easily achievable with the use of standard chemistry and careful handling. Methods to synthesize peptidomimetics are typically adapted from solid-phase synthetic methods used for peptides.

β -amino acids (Figure 1.1) contain an extra methylene group between the amide groups in the backbone in comparison to α -amino acids, and are often referred to as β_3 -substituted residues. Typically, β -peptides can be constructed from β^3 -substituted residues, or from cyclohexane rings to give rise to cyclized residues in which the β^2 and β^3 positions are part of a five-membered ring (6, 13). The β^3 -substituted residues are under intense investigation because of the availability of monomers through homologation of α -amino acids. Therefore, short sequences of β -peptides can be made using the standard fluorenyl-methoxy-carbonyl (Fmoc) protecting group-based solid-phase synthesis (14, 15). However, the yield to synthesize β -peptides longer than eight residues using this chemistry is low due to incomplete deprotection and the need for extended coupling times (10, 12). Thus, sometimes two β -peptides are further coupled with each other to yield a longer β -peptide. A microwave-assisted synthesizing method has been reported to enable the efficient synthesis of β -peptide combinatorial libraries (16).

α/β -peptides which contain both α - and β -amino acids in a heterogeneous backbone have been reported with several advantages over their homogenous backbone

counterparts, including accessibility to many new molecular shapes and improved side chain diversification (16). They can be composed of α -peptide segments and β -peptide segments, or of α - and β -amino acid monomers interspersed in a regular pattern, like α - β - α - β -, α - α - β - α - β -, α - β - β - α - β -, to name just a few (6). Like what we have stated about the β -peptide chemistry, having to synthesize every building block is a practical barrier to the development of new foldamers. For α/β -peptides, with all different types of α -amino acids commercially available, their synthesis is greatly simplified with improved efficiency (6). The α -amino acids present can provide various side chain functional groups and secondary structures, and a few types of rigidified unnatural building blocks can be incorporated to promote a specific folded conformation and improve the molecular stability against proteases.

Peptoid synthesis was revolutionized by Zuckermann *et al.*, who reported a solid-phase submonomer synthetic approach as shown in Figure 1.2 (17). This novel synthetic protocol allows a variety of side chains to be incorporated into peptoids at modest cost and effort. Any functional group with a primary amine can be incorporated with high coupling efficiency (Figure 1.2). Moreover, most reagents used in the synthesis are readily available and inexpensive. Sequence-specific peptoids up to 50 residues in length can be synthesized in high yields (comparable to the efficiency of standard Fmoc peptide chemistry) (2).

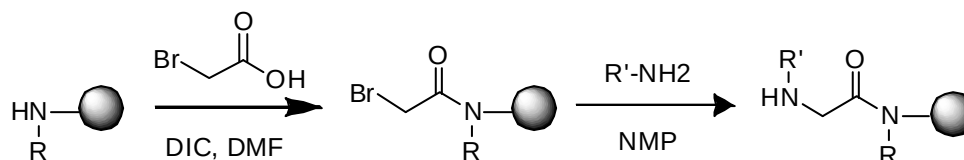


Figure 1.2 The solid-phase submonomer synthesis scheme of peptoids

Peptoids are highly suitable for use in commercializable medical applications (18, 19). Their solid-phase synthesis can be performed in a standard automated peptide synthesizer, and it is also easy to create large combinatorial libraries to screen thousands of compounds for bioactivity (18). Furthermore, one can easily synthesize peptide-peptoid hybrids by alternating between this submonomer method and the Fmoc-based monomer chemistry within one single synthesis sequence on the automated synthesizer. Like α/β -peptides, peptide-peptoid hybrids may achieve an improved balance of good bioactivity, structural properties and an appropriate rate of biodegradation.

Enabling a facile and efficient synthesis of these peptidomimetics in specific sequences with diverse side chains provides the groundwork to probe their applications. The establishment of these enabling technologies started decades ago, and much effort is still ongoing in order to make the synthesis even easier and cheaper to broaden the potential real-life applications of these foldamers.

1.2.2 Folding of β -peptides, α/β -peptides and peptoids

One of the initial purposes of studying foldamers is to mimic a variety of secondary structures of peptides, including helices, turns and sheet-like structures, while gaining a high resistance to protease degradation. β -peptides have more conformational freedom than natural peptides because of the additional methylene unit present in the backbone. In natural peptides, due to hydrogen bonding, the main conformations are the 3_{10} - and α -helices, with 10 atoms and 13 atoms in the hydrogen bonded rings (Figure 1.3). For β -peptides, the placement and stereochemistry of side chains along the backbone will greatly influence the foldamer structure. Among the β -peptide helical conformations, the 14-helix is the most common (Figure 1.3). Peptides composed of β^3 -substituted amino

acids have a high propensity to form 14-helices (6). γ -branched β^3 -substituted amino acids (such as $\beta^3\text{hVal}$, $\beta^3\text{hIle}$), and incorporation of conformationally constrained cyclic residue with six-membered rings can promote the formation of 14-helices (10, 20-22). As shown in Figure 1.3, there are 14 atoms in the hydrogen-bonded ring $\text{N-H}(i)\cdots\text{O}=\text{C}(i+2)$ and approximately three-residue per turn. The β -peptide 12-helix can be formed with constrained cyclic residues of five-membered rings (10, 22). Pyrrolidine-containing cyclic β -amino acids can be incorporated into the 12-helices to improve the water solubility (23) (Figure 1.3). Other helices as well as turn and sheet-like conformations are also possible in β -peptides (6, 10).

In α/β -peptides with heterogeneous backbones, the presence of β^3 -substituted amino acids can encourage turn formation in peptides (6). A single β -amino acid is typically well-tolerated in an α -helix. Alternating α - and β -amino acids in a regular pattern has more significant impact on the structure. For example, the 3_{10} -helix expands from the 10-atom hydrogen bonded ring to 11-atom ring in $(\alpha\text{-}\beta)_n$ peptides, whereas in $(\alpha\text{-}\alpha\text{-}\beta)_n$ peptides, the α -helix can expand from 13 to 14 atoms in the hydrogen bonded ring (6, 24). Different combinations of α - and β -amino acids in a heterogeneous backbone can result in many new structures as summarized in several review papers (25, 26), and these heterogeneous peptides offer good controls over the molecular display and a great structural diversity for functional studies.

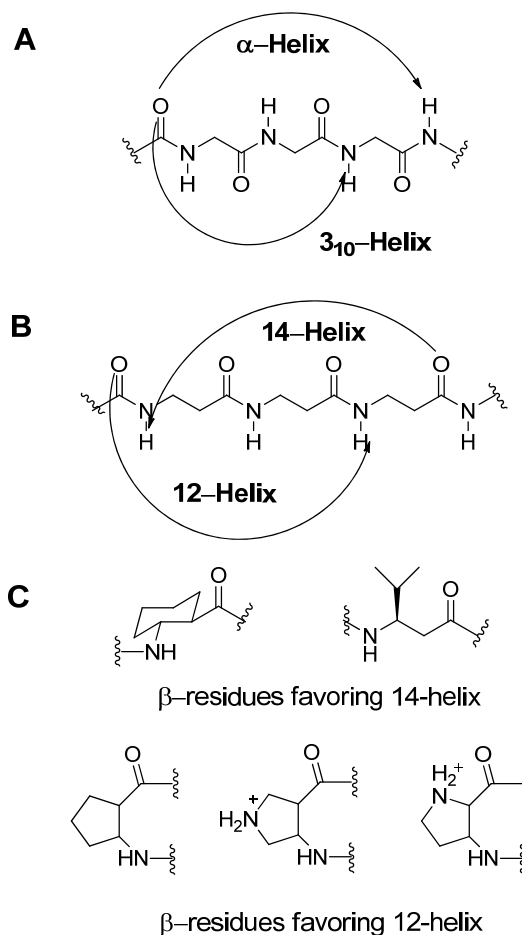


Figure 1.3 Illustration of hydrogen bonding in natural peptide and β -peptide helices. (A) the 3_{10} - and α -helix that can be formed in a natural peptide due to intrachain hydrogen bonding; (B) the 14-helix and 12-helix that can be formed in β -peptides; (C) examples of some β -residues that can facilitate the folding of helices (10).

Peptoids, in contrast to peptides and α/β -peptides, have side chains attached to the backbone amide nitrogen rather than to the carbons in the backbone, and thus the peptoid backbone lacks chirality and the intrachain hydrogen bonding (27). It is known that peptoids can be easily driven to form polyproline type-I-like helical structures via a periodic incorporation of bulky, α -chiral side chains (28, 29). These polyproline type-I-like peptoid helices have a periodicity of approximately three residues per turn and a helical pitch of 6.0–6.7 Å (28, 30). Helical conformations for peptoids can be readily

formed in a variety of solvents, including organic solvents, and the handedness of the helix can be predictably controlled by using the appropriate enantiomers (30, 31). The threefold periodicity of the peptoid helix allows an easy display of X, Y, and Z residues in separate faces using the $(X-Y-Z)_n$ trimer repeat peptoids (30).

Not many secondary structures other than the polyproline type-I-like helix have been reported in peptoids. Unlike β -peptides and α/β -peptides, the peptoid backbone is almost devoid of hydrogen bonding donors, which may be an important factor precluding the formation of hairpin/sheet-type structures (31). A threaded loop structure has been reported with a nonamer of *Nspe* (a chiral peptoid monomer analogue of phenylalanine) with exceptional thermal stability and conformational homogeneity in acetonitrile (ACN). This structure is chain-length specific and relies on bulky, chiral side chains and chain-terminating functional groups for stability (31). Cyclic peptoids have been reported with an easy synthesis via highly efficient head-to-tail macrocyclization reactions (32). A diverse array of peptoid sequences from pentamers to 20mers can be converted to macrocyclic products within 5 min at room temperature.

Ongoing studies are still seeking to discover new secondary structures of these foldamers and achieve better understanding and control of their folding.

1.2.3 Some applications of β -peptides, α/β -peptides and peptoids

In addition to the design of conformationally constrained frameworks to address questions of folding, these foldamers have been used to probe several biological applications. In some applications, molecular amphipathicity, the presentation of polar and apolar groups in separate sectors is a critical property (6). For example, antimicrobial peptides (AMPs) are one such application which generally requires

amphipathic structures, and thus is one of the most active areas of peptidomimetics (33). AMPs are a class of short (typically containing fewer than 40 amino acids) and naturally occurring peptides that help defend organisms against a broad spectrum of bacterial invaders (34). Their killing mechanisms are generally believed to occur at the membrane, not involving specific receptor-mediated binding but causing general membrane damage. There are also some AMPs that have been reported with intracellular targets, and more and more studies indicate that modes of action of AMPs may be more complicated than the prevailing understanding of them being lytic (34, 35). AMPs are attractive as lead molecules to be developed into antimicrobial therapeutic agents, due to their fast action, broad-spectrum activity and no reporting of resistance developed in bacteria (34). However, short peptides suffer from proteolytic susceptibility and poor *in vivo* bioavailability. The protease-stable foldmers are thus able to incorporate the advantages AMPs provide while circumventing their shortcomings.

DeGrado and co-workers first reported the design of cationic, amphipathic β -peptides as antimicrobial compounds, which had potent activity against bacteria but also caused extensive hemolysis (14). By employing a less hydrophobic β -amino acid, β -HAla, instead of β -HVal, the new compounds displayed much reduced hemolytic effects and also retained good antimicrobial potency (15). Gellman's group first reported a β -17 peptide, a 12-helix β -peptide with 17 residues containing cyclic β -residues with five-membered rings, which have relatively rigid conformations but still possess potent antimicrobial activity while exhibiting minimal hemolysis (36). They also synthesized a series of β -17 analogues with molecular amphipathicity and varied the ratio of cationic to hydrophobic residues. Their data suggested that molecular amphipathicity was critical

for the antimicrobial activity and a ratio of 40% cationic to hydrophobic residues led to the optimal selectivity of β -17 (37). In addition to helping identify the structural influences on the antimicrobial activity and selectivity, studies of antimicrobial β -peptides and α/β -peptides are also informative with respect to the mechanism of action of these cationic, amphipathic compounds, which also suggests that their killing occurs at the membrane (38). A study of antibacterial α/β -peptides demonstrated a role for bacterial lipids in the antibacterial activity of the compound (39).

Our laboratory has focused on designing peptoids as potent and selective antimicrobial agents (30, 40, 41). Peptoids provide an ideal peptidomimetic scaffold to design cationic, amphipathic compounds by forming the polyproline type-I-like helix with roughly three residues per turn. In addition, the ease of peptoid synthesis and great diversification of approachable side chains enable the creation of large compound libraries for functional screening. Our group has reported several antimicrobial peptoids as potent and selective antibiotics with broad-spectrum activity. A number of observations have been made regarding the effects of structural variations on the antimicrobial and hemolytic activities of these peptoids (30, 41, 42):

- Loss of amphipathicity resulted in a decrease in antimicrobial and hemolytic activity
- Longer variants were more toxic to bacteria and more hemolytic
- Shorter variants were less hemolytic, but also less toxic to bacteria
- Use of bulkier hydrophobic sidechains increased toxicity and hemolytic activity
- Structures of opposite chirality (e.g. using the *Nrpe* sidechains ((*r*)-*N*-1-phenylethylglycines) instead of *Nspes* to create oppositely handed helices) showed similar activities

- Replacing cationic residues with anionic residues destroyed antibiotic and hemolytic activity
- Use of achiral sidechains significantly reduced hemolytic activity with only minimal or no reduction of antibiotic activity, thereby increasing selectivity.

Whereas AMPs' functions are related to the overall molecular architecture such as the amphipathicity and hydrophobicity, many other applications require more precise residue arrangements. These foldamers have been used to mimic a variety of peptides for various applications. Cell penetrating peptides are another class of compounds of significant interest in peptidomimetic research, which will be discussed in detail in Chapter 1.4. Other applications include developing peptidomimetic inhibitors of protein-protein interactions, inhibitors of fat and cholesterol uptake and RNA binding (2, 6). The Kodadek group reported a two-color, cell-based screen to identify specific receptor-binding compounds in a combinatorial library of peptoids displayed on beads, and they have isolated vascular endothelial growth factor receptor 2 (VEGFR2)-binding peptoids. A dimeric derivative of one of these lead compounds has been shown to be an antagonist of VEGFR2 activity both *in vitro* and *in vivo* (18).

It has been decades since the first report of cyclic peptides containing β -amino acids, the studies of the secondary structure of β -peptides, and Zuckerman's ground-breaking work on peptoids. The research of foldamers has been focused to enable their efficient synthesis, to increase the available structural diversity, to manipulate their folding, and to explore their applications in a couple of areas. With all this fundamental work, biomimicry is now less constrained by technological capability but primarily by the ability to identify new targets. Interdisciplinary collaborations will greatly benefit the

progress of biomimicry and enable this promising area to make more practical contributions to people's life.

1.3 Cell penetrating peptides—mechanisms and their peptidomimetics

Cell penetrating peptides (CPPs), also known as molecular transporters, are of great research interest due to their ability to translocate across cellular membranes and to deliver associated cargo into the cells (43-45). The plasma membrane forms a barrier between the cytosol and the extracellular space. Depending on cell type, plasma membranes are composed of various types of lipids such as unsaturated and saturated phospholipids, sphingolipids, and cholesterol with many types of embedded proteins (46). For most polar molecules, these membranes represent impermeable barriers. It is remarkable that some proteins and the peptides derived from them can possess the ability to cross this barrier and further to mediate the import of membrane-impermeable, biologically active molecules into the cells *ex vivo* and in whole organisms (47, 48).

This research area was initiated by the finding that the HIV-1 trans-activator of transcription (Tat), a protein of 101 amino acids which is essential for HIV-1 replication, can be rapidly taken up by the cells (49), and only a small fraction of the protein 47-57, was actually responsible for the internalization. This Tat (47-57) domain was further proven to be able to transport many cargo into cells for various applications (50-52). Later, *Drosophila* Antennapedia protein, Antp (43-58) was found with similar cell-penetrating functions (53). These protein domains that can mediate the translocation of proteins are also referred to as protein transduction domains (PTDs). After the development of synthetic peptides, the term CPPs and molecular transporters were used

to refer to both the naturally occurring PTDs and their functional homologues from organic synthesis.

In general, studies in the evolving field of CPPs have been focusing on the following three research areas: 1) Defining the structural properties of CPPs that enable their cellular translocation and effective endosomal escape, which has led to several new classes of CPPs and their synthetic analogues to achieve optimal activity. 2. Elucidating the internalization mechanisms of CPPs since much controversy exists in the literature. 3. Exploiting their cell penetrating abilities to deliver a wide range of impermeable molecules into cells to modify cellular functions for therapeutic applications, or to image/illuminate intracellular processes.

1.3.1 Structural characteristics of CPPs

CPPs are typically short, cationic peptides (with ≤ 30 amino acids) (44). Various peptides have been reported with cell-penetrating abilities but they display very distinct structures. As shown in Table 1.1, these peptides can be categorized into several classes based on their structure (54). The first class of CPPs has a high content of cationic residues like arginines, such as the Tat-(47-57), and oligoarginines developed based on Tat. The second group is cationic and amphipathic CPPs, which have both basic amino acids and hydrophobic residues, like Antp-(43-58), pVEC-(615-632) (murine vascular endothelial cadherin) (55), the model amphipathic peptide (MAP) (56), and Transportans (57). The basic residues in this group can be arginines, lysines or histidines. The third group is another class of CPPs, which are extremely hydrophobic, like the membrane translocating sequence peptide which was derived from the hydrophobic region of the secretion signal sequence of the Kapor fibroblast growth factor (58).

As discussed above, there are various CPPs with their own unique characteristics. Among them, arginine-rich peptides are the most frequently utilized carriers that have been used to deliver a variety of cargos, such as oligonucleotides, small hydrophobic drugs, magnetic beads and even liposomes (54, 59). Many proteins and peptides that have arginine-rich domains have been found with cell-penetrating capabilities, such as many RNA/DNA binding peptides (60).

Class	Peptides	Sequences
Cationic	Tat (47-57)	YGRKKRRQRRR
	Oligoarginine	R _n (n~ 7-11)
Cationic, amphipathic	Antp (43-58)	RQIKIWFQNRRMKWKK
	pVEC-(615-632)	TLILRRIRKQAHHS
	MAP	KLALKLALKALKAALKLA
	Transportan	GWTLSAGYLLGKINLKALAALA KKIL
Hydrophobic	Membrane translocating sequence peptide	AAVALLPAVLLALLP

Table 1.1 Examples of cell-penetrating peptides, categorized based on their structural properties.

In addition, many CPPs are found to be cationic, amphipathic peptides (Table 1.1). Interestingly, several antimicrobial peptides, the key structural characteristics of which are also being cationic and amphipathic, have also been reported with cellular translocation. Antimicrobial peptides (AMPs) are usually associated with membrane toxicity depending on their concentration, but several of them have been found to translocate into the cells quite efficiently without causing permanent membrane damages (61, 62). Thus, it is interesting to explore how different the CPPs and AMPs actually are from each other, and there may be opportunities to develop novel CPPs based on the AMPs.

1.3.2 Internalization mechanisms of CPPs

1.3.2.1 Introduction of uptake mechanisms in cells

The plasma membrane of mammalian cells consists of a lipid bilayer with embedded proteins. These lipids, most of which are phospholipids, have their hydrophilic heads facing the cytosolic and extracellular environments with their hydrophobic tails shielded with the bilayer. The plasma membrane is differentially permeable. For example, the lipid membrane generally has very low permeability for ions and most polar molecules, but some hydrophobic molecules can cross the membrane just by simple diffusion. The plasma membrane finely regulates what enters and exits the cells, and employs a number of transport mechanisms (46):

1) Passive transport by diffusion. Diffusion processes can be driven by the compounds' concentration gradients and the potential differences across the membrane. The molecules can diffuse directly through the membrane or via certain channels in the membrane. The majority of small hydrophobic drugs can enter the cells by direct diffusion through the plasma membrane. There are a set of rules to guide drug designs to achieve good cellular diffusion of small drugs. For efficient cellular uptake via diffusion, the molecular weight of the compound should not be larger than 500 Da, and a proper molecular hydrophobicity is required. The balance of molecular hydrophilicity and hydrophobicity is reflected by the octanol-water partition coefficient of the compound, which can be determined experimentally to give rise to the logP value as a measure of molecular hydrophobicity (63). However, many drugs or probes like oligonucleotides and proteins are too large or too hydrophilic to cross the membrane by passive diffusion.

2) Active transport via transmembrane transporters. For the uptake of nutrients, ions and signaling molecules against concentration gradients or the potentials, there are

various forms of active transport, such as the ATP-dependent pumps or endocytosis. Membrane transporters are specialized proteins or protein complexes that form membrane spanning channels which enable the solute molecules to cross the phospholipid bilayer. Transporters are usually quite specific, recognizing and transporting only a limited number of substances, often just a single substance. A common example is the ubiquitous Na^+/K^+ ATPase pump (46). Membrane transporters are important determinants for the pharmacokinetic, safety and efficacy behavior of drugs. Drug molecules can be chemically modified to be recognized by transporters to facilitate the drug absorption and disposition in various organs (64, 65).

3) Active transport via endocytosis. Endocytosis permits the internalization of polar substrates in a highly regulated and substrate specific process. The substrate adsorbs to small patches of the plasma membrane, further invaginates in an energy-dependent process and then pinches off into the cytoplasm. Endocytosis pathways can be subdivided into four major categories, including clathrin-mediated endocytosis, caveolae-dependent endocytosis, macropinocytosis, and phagocytosis (46). Several other endocytotic pathways were well reviewed by Doherty *et al* (66).

Clathrin-mediated endocytosis is mediated by small clathrin-coated vesicles (CCVs), which form domains on the plasma membrane termed clathrin-coated pits. Coated pits can concentrate large extracellular molecules that bind to different receptors responsible for the receptor-mediated endocytosis (66).

Caveolae are the most commonly reported non-clathrin-coated membrane buds, which consist of the cholesterol-binding protein caveolin (Vip21) with a bilayer enriched in cholesterol and glycolipids. Caveolae are small flask-shape pits in the membrane,

being especially abundant in smooth muscle, type I pneumocytes, fibroblasts, adipocytes, and endothelial cells (67).

Macropinocytosis, which usually occurs in highly ruffled regions of the plasma membrane, is the invagination of the cell membrane to form a pocket, which then pinches off into the cell with a vesicle filled with a large volume of extracellular fluid and molecules. The vesicle then travels into the cytosol and fuses with other vesicles such as endosomes and lysosomes (68).

Phagocytosis is the process by which cells bind and internalize larger substances ($\sim \geq 0.75\mu\text{m}$ in diameter), such as small-sized dust particles, cell debris, and micro-organisms. This process involves the uptake of larger membrane areas than other endocytotic pathways (66).

After the internalization, the standard endocytic pathway also involves either recycling the vesicles back to the surface (via early endosomes and recycling endosomes) or sorting them for degradation (via late endosomes and lysosomes) (69). Early endosomes are the first station which receives most types of vesicles coming from the cell surface. They have a mild acid pH. They are principally the sorting organelles where many ligands dissociate from their receptors and from which many of the receptors are sorted back to the cell surface, or to degradation. Late endosomes are acidic (pH~5.5), and mediate a final set of sorting events prior to the delivery to lysosomes, which are the last compartments of the endocytic pathway. Lysosomes are acidic (pH~4.8), containing a high content of lysosomal membrane proteins and many types of lysosomal hydrolases. They are generally regarded as the principal hydrolytic compartment of the cell, to break

down waste products, fats, carbohydrates, proteins, and other macromolecules into simple compounds (70).

1.3.2.2 Internalization mechanisms proposed for CPPs

It is a mystery as to how these purely cationic or cationic amphipathic peptides can efficiently go through the plasma membrane even with various associated cargo molecules. There are many controversies in the reported CPPs' cellular uptake behavior and their internalization mechanisms. Initially, misleading results were obtained due to not being able to distinguish the internalized CPPs from those associated with the plasma membranes in the fluorescence-activated cell sorting (FACS) experiments and also by artifacts resulting from cell fixation in confocal microscopy imaging (71, 72). With these two issues addressed, the internalizations of CPPs were still found to be complicated, highly dependent on their sequences, the experimental conditions, and their associated cargos (59). The alteration of CPPs by the attachment of cargo may change the internalization pathways depending on the cargo's size and hydrophobicity, and may affect the efficiency of the delivery system for either the cytoplasmic or nuclear targeting. Therefore, it is important to understand the internalizations and intracellular processing of various CPPs in order to develop an effective delivering system.

Many evidence suggest that the internalization of some CPPs are mediated via non-endocytotic pathways, also known as transduction, with their uptake to be energy-independent and receptor-independent (72, 73). For example, oligoarginines and Tat-(47–57) have unique translocation with the following properties found: (i) a rapid internalization, (ii) entry into cells at 4°C with almost the same efficiency as at 37°C, indicating the energy-independent process (iii) a low sensitivity to typical clathrin-

dependent endocytosis inhibitors, and (iv) a low cytotoxicity (54). However, with most efforts now directed to illustrate the details in endocytosis, the transduction process is not well-understood, limited to two hypothetical models with relatively little experimental evidence. These models include direct membrane penetration and inverted micelle. Both models follow a 3-step process: (a) membrane interaction, (b) membrane permeation and (c) release of CPPs into the cytosol (59, 74). The main difference between transduction and endocytosis lies in step 2 and 3, whereas transduced CPPs are directly localized in cytoplasm while CPPs are typically confined in vesicles in endocytosis and may or may not be released into the cytosol.

On the other hand, endocytosis pathways discussed above have also been implicated to be involved in the internalization of many CPPs with various cargos (75-77). Endocytosis is a collective term for a regulated cellular process of macromolecule internalization that is energy dependent and is characterized by vesicle formation. The involvement of different endocytosis pathways is specified by either looking for protein markers that co-localize with the CPP or by using chemical compounds known to inhibit a specific type of endocytic process (78). The fact that certain inhibitors may affect more than one endocytotic pathway can further complicate interpretation of the results (78). The common feature for all types of endocytosis is that it is an energy driven process. At 4°C, cellular processes including ATP hydrolysis are arrested, so internalization by endocytosis can be blocked. Moreover, at 4°C, the membrane fluidity is also affected, and therefore temperature dependence assays are typically carried out to evaluate the internalization of CPPs (79).

Dowdy *et al* have implicated macropinocytosis as a pathway of internalization for the Tat peptide (80) and Tat-Cre fusion protein (81). Macropinocytosis has been proposed as an attractive pathway to explain the uptake because it involves internalization of extracellular fluids and since CPPs are concentrated on the surface of the cell via electrical interactions, they will indiscriminately be internalized together with the extracellular fluid. However, most evidence suggests that the uptake mechanisms of CPPs are far more complicated than simply employing a single pathway. Khalil *et al* showed that liposomes modified with a high density of octaarginine were preferably internalized by macropinocytosis, but liposomes modified with a low density of octaarginine resulted in uptake via clathrin-mediated endocytosis (82). In contrast, Zaro *et al* indicated that the uptake of ^{125}I -oligoarginine was not mediated via macropinocytosis, which may be due to the smaller size of ^{125}I in contrast to previously reported larger cargos such as Cre-protein (83). Moreover, in a comprehensive study, Duchardt *et al* proved that the Antp peptide, nona-arginine (R_9) and Tat peptide simultaneously use three endocytic pathways: macropinocytosis, clathrin-mediated endocytosis and caveolae/lipid-raft-mediated endocytosis. Antp differs from Tat and R_9 by the extent by which the different import mechanisms contribute to uptake. Tat and R_9 employ different internalization mechanisms depending on their concentrations. At higher concentrations, their uptake occurs by a mechanism that originates from spatially restricted sites of the plasma membrane and leads to a rapid cytoplasmic distribution of the peptides. Endocytic vesicles could not be detected, suggesting an endocytosis-independent mode of uptake for Tat and R_9 at high concentrations. To summarize, various endocytosis pathways are not mutually exclusive; they may contribute unequally

to the overall internalization with one pathway more dominant than others, depending on the CPP's sequence, associated cargos and their working concentrations.

1.3.2.3 Intracellular processing of CPPs

The internalization mechanisms of CPPs govern their intracellular processing, kinetics and final fate of the CPPs and their cargo (59). If the CPP enters the cells via transduction, it could directly interact with its targets in the cytoplasm or be imported to the nucleus, be degraded by cytoplasmic proteases, or it may be redirected out of the cell either intact or after degradation. On the other hand, if it is internalized via endocytosis, depending on its endocytosis pathways, it could be targeted for lysosomal degradation, may be able to escape lysosomal degradation and enter the cytoplasm and possibly the nucleus, or be taken to the Golgi apparatus or the endoplasmic reticulum, or transcytosed out of the cell. Again, the pathways followed highly depend on the CPP's sequence, type of cargo attached, and the nature of linkage between the cargo and CPP.

For the associated cargos to be effective, CPPs need to escape the endosome and liberate the cargos. The N-terminal domain of influenza virus hemagglutinin-2 (HA2), a pH-dependent fusogenic peptide that induces lysis of membranes at a low pH, has been utilized to enhance the endosomal escape of Tat- and oligoarginine-associated cargos (81). In addition, photochemical internalization (PCI), a strategy which generated reactive oxygen species (ROS) by photosensitizers upon irradiation to disrupt endosome, has also been used to enhance the endosomal escape of CPPs (84).

1.3.3 Synthetic CPP mimics

Rapid degradation of CPP is undesirable as premature degradation of the carrier may prevent its cargo from reaching the targets, though the carrier should also be able to

release the attached cargo after internalization. Stable synthetic mimics of CPPs provide a feasible strategy to achieve better *in vitro* and *in vivo* cargo delivery. D-amino acids, β -peptides, and peptoids have been utilized to design efficient molecular transporters with improved molecular stability. β -homoarginine (85), β -homolysine (86), and Tat-derived β -peptides (87) exhibited remarkable cellular translocation. As shown in Figure 1.3, β -peptides can form 14-helix with the side chains positioned in an angle of $\sim 120^\circ$ to each other. This facilitates the side chains of similar properties to be aligned in the same face to form amphipathic helices.

Peptoids have also been investigated to develop efficient transporters. Wender *et al* reported that R₉ was more efficient than Tat-(49-57) at cellular uptake, and the D-arginine oligomer (r₉) exhibited an even greater uptake rate enhancement. Remarkably, polyguanidine peptoid derivatives also displayed efficient cellular uptake, especially the peptoid containing long spacers between its guanidine head group and backbone, such as (N-hxg)₉ with the six-methylene spacers which exhibited significantly enhanced cellular uptake compared to Tat-(49-57) and even to r₉ (Figure 1.4A) (88). Rothbard *et al* also indicated that the spacing between the guanidine head groups was also important for transport efficiency with the 6-aminocaproic acid as the optimal spacer (Figure 1.4B) (89). Moreover, replacing the amide backbone in CPPs with carbamate also resulted in the increased backbone flexibility and thus enhanced cellular uptake (Figure 1.4C) (90). Schröder *et al* designed peptoid transporters with poly-amine head groups and found that these peptoids mainly located in the cytoplasm after uptake in contrast to the polyguanidine peptoid which preferentially accumulated in nucleus (Figure 1.4D) (91, 92). Foged *et al* developed a α -peptide/ β -peptoid library, evaluated their cellular uptakes

and membrane-destabilising properties, and found guanidinylated chimeras with higher uptake efficiency than Tat-(47-57) (one example shown in Figure 1.4E) (93).

Synthetic transporter research aims to develop more efficient and stable transporters with better controls of their structures and higher chemistry diversity of available functional groups. Studies using these cell-penetrating peptidomimetics may also provide insights into the structural influences on CPPs' efficiency and translocation mechanisms.

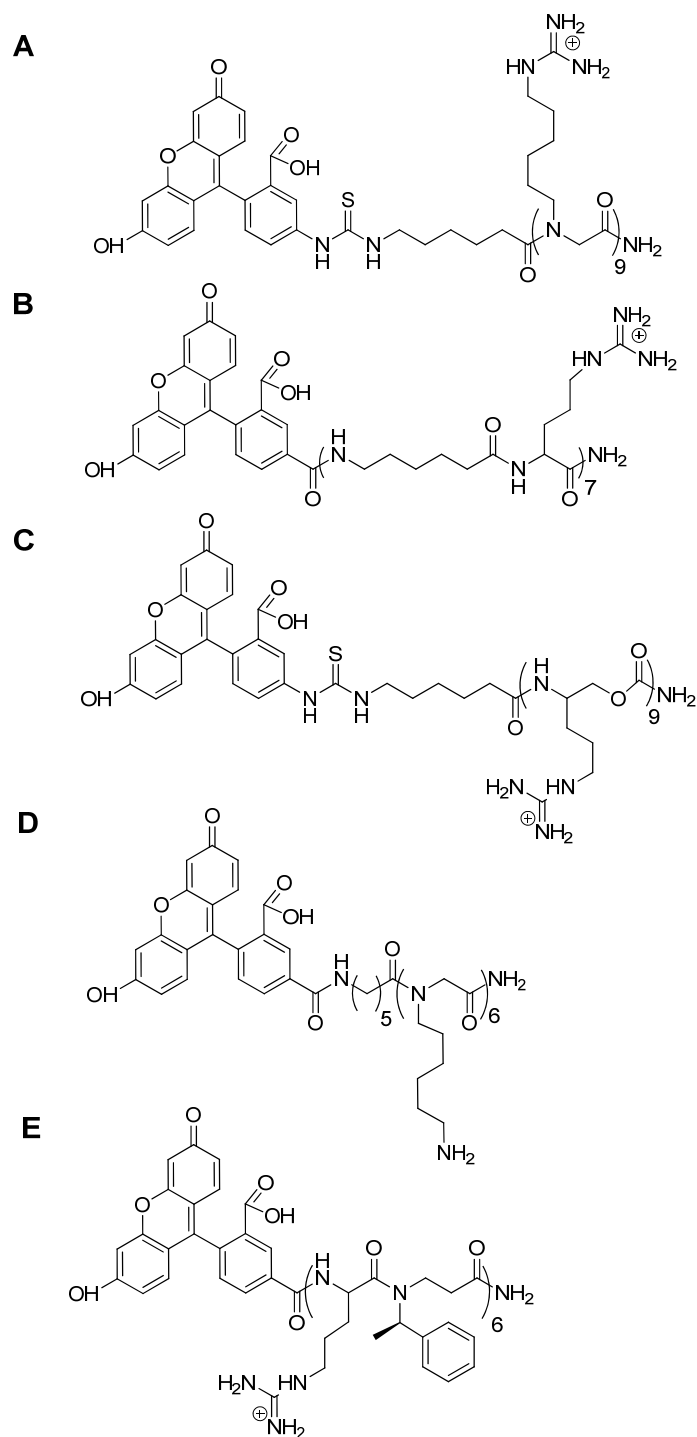


Figure 1.4 Examples of synthetic CPP mimics. (A) Fluorescently labeled (*N*-hxg)₉ with the six-methylene spacers (88); (B) Oligoarginines with 6-aminocaproic acid as the backbone spacer (89); (C) Oligocarbamates with polyguandinine head groups (90); (D) a peptoid transporter with polyamine head groups (91, 92) ; (E) a α -peptide/ β -peptoid chimera transporter (93).

1.4 Genetically engineered protein polymers

1.4.1 Recombinant protein polymers and their biomedical applications

Proteins play essential roles in virtually every biological process, and are one of the most important building blocks and bioactive macromolecules in living organisms, including proteins that have structural or mechanical functions, such as actin and myosin in muscle and the proteins in the cytoskeleton, extracellular matrix proteins that provide support for cells and tissues, enzymes that catalyze biochemical reactions and are vital to metabolism, proteins that are important in cell signaling, immune responses, cell adhesion, and *etc.*. Due to their diverse functions, proteins can be used for a variety of research and biomedical applications, such as growth factors, antibodies, protein-based biomaterials and medicines (94). State-of-the-art technology can purify desired proteins from their native sources, such as collagen, the most abundant protein in mammals which can be purified from animal tissues to achieve a cost-effective production (95), or genetically engineer proteins with desired functions, express them in a wide range of organisms and obtain highly pure recombinant protein polymers.

Short peptides are generally synthesized on solid-phase, while large recombinant protein polymers can be produced via genetic engineering. The properties of these polymers is easily customized by designing appropriate DNA sequences that code for the desired, final protein sequence (96) which is produced in biological systems, such as the bacterium *Escherichia coli*. Compared to proteins purified from natural resources, recombinant protein polymers produced via genetic engineering have several advantages: 1) molecular cloning allows easy design and production with the ability to combine various bioactive protein domains into just one single protein polymer, with good control

over the properties of the final product; 2) recombinant proteins eliminate the risk of containing transmissible disease agents (e.g. virus or prion) that can associate with proteins purified from animal tissues (97); and 3) many proteins are hard to get in large quantities from their natural resources, and thus recombinant proteins can be made in bulk by scaling up their production in the expression systems.

A variety of recombinant proteins have been designed as protein-based medicines, which have provided breakthrough therapies for many diseases (94). Some protein medicines are designed to replace or supplement a specific bioactive protein to solve its deficiency or abnormal activity in some diseases. For example, recombinant insulin is dosed to treat patients of diabetes (98), and several types of interferons have been made and administered as antiviral, antiseptic and anticarcinogenic drugs, and to treat some autoimmune diseases (99-101). Moreover, protein therapeutics also include novel engineered proteins with desired functions that can be introduced to the body or certain locations to achieve the desired effect. One such example is the monoclonal antibodies targeting certain receptors or membrane markers of particular cells, which are undergoing extensive studies to provide innovative treatments for many diseases, especially for cancer (102). The anti-human epidermal growth factor receptor (HER2) antibody trastuzumab (Herceptin) has been proven to be able to increase the clinical benefit of first-line chemotherapy in metastatic breast cancer that overexpresses HER2 (103). Moreover, the development of monoclonal antibodies that possess superior tumor targeting provides a novel way to deliver cytotoxic agents specifically to antigen-expressing tumors. A trastuzumab-conjugated drug showed greater activity compared

with nonconjugated trastuzumab while maintaining selectivity for HER2-overexpressing tumor cell (104).

Protein-based biomaterials have also emerged as a fast-growing application for recombinant protein polymers, tackling challenges in regenerative medicine and drug delivery, which is the major focus of the second part in this dissertation and will be discussed in detail in the following section.

1.4.2 The *de novo* design of recombinant protein polymers

Various bioactive protein domains comprising certain amino acid sequences that can reliably fold into specific three-dimensional structures with distinct functions have been identified from natural proteins, and many of them can be designed to generate novel biomaterials such as bioactive scaffolds for tissue engineering, drug delivery systems, biosensors, *etc.* (105-108).

The *de novo* designed protein polymers are typically modular proteins, consisting of repeated protein modules which can be derived from natural protein domains or be non-natural and designed for a particular structure and/or function. Various functional protein domains with proper secondary structures can be incorporated into protein polymers, such as the cell adhesion motifs (e.g. RGD), growth factor-tethering motifs, and protease degradable sequences (109, 110). The functionalities and their displaying pattern in a protein polymer can be exactly controlled by cloning a particular DNA sequence. In our group, DNA with any defined sequence can be cloned and easily elongated to the desired chain length via a cloning method developed previously, called “controlled cloning” (111). Briefly, a short peptide sequence (<150bp) that encodes desired protein motifs is derived from annealing of chemically synthesized oligonucleotides, which is then

amplified by the polymerase chain reaction (PCR) and cloned as a monomer of the building block/protein module. Based on these DNA monomers, the “controlled cloning” method can generate long repeats of a single monomer DNA sequences or ligate two or more different monomer sequences to form proteins with multiple functionalities. DNA plasmids with the correct encoding sequences are then expressed in the chosen expression system and proteins can be purified using state-of-the-art technology.

Many recombinant protein polymers have been engineered to combine various bioactive domains and display desired functions. Some examples of groups that are active in generating protein-based biomaterials are summarized in Table 1.2. Elastin-like proteins (ELPs) have been extensively studied and are one of the best characterized proteins used for tissue engineering applications. The ELPs feature pentamer consensus sequences, VPGXG, where X can be any amino acid. They can undergo a reversible inverse phase transition, to form insoluble β -spiral aggregates below their transition temperature and become soluble in aqueous solutions again with more ordered molecular structure above the transition temperature. Thus, purification of ELPs can be greatly simplified by changing the temperature during the purification process. Taken together, the inherent elasticity of ELPs, along with their reversible phase transition, has made ELPs attractive in the development of responsive biomaterials, self-assembled hydrogels, reversibly swelling hydrogels, and hydrogels with tunable mechanical properties (112, 113).

Research Group	Example Sequences
Tirrell (Caltech)	ELPs containing REDV sequence (113) Coiled-coil self-assemblies of (AG) ₃ PEG and SGDLENEVAQLEREVRSLEDEAAELEQKVSRLKNEIEDL KAE domains (114)
Chilkoti (Duke)	Elastin-like polypeptides (ELPs = VPGXG where X is any amino acid except proline) (115, 116)
Conticello (Emory)	Elastin-like polypeptides (117, 118)
Cappello (Protein Polymer Technologies, Inc)	Silk-elastin-like proteins (SELP) (GVGV ₄ PEGV ₃ GAGAGS (119, 120)
Kopecek (Utah)	Coiled-coil self-assembling hydrogels (AG) ₃ PEG and VSSLESK domains (121, 122)
Kaplan (Tufts)	Spider silk mimics SGRGGLGGQGAGAAAAAGGAGQGGYGGGLGSQGT(123)
Kiick (Delaware)	alanine-rich protein polymers AAAEQAAAAQAAQAAQAAQ (124, 125)
Hubbell (EPFL)	Fibrinogen (LRGDFSSANNRDNTY) and collagen (GPQGIAGQRGV) motifs including RGD, plasmin sites, MMP cleavage sequence (126-128)

Table 1.2 Examples of some groups that have been developing recombinant protein polymers as novel biomaterials.

Moreover, the leucine zipper domain which is a motif commonly found in transcription factors is also widely adopted in biomaterial developments due to its ability to self-assemble into coiled-coil bundles that are reversible with changes in pH, temperature, and ionic strength. They have been used to develop self-assembled protein-based hydrogels with reversible physical crosslinking and to combine with different building blocks to create multifunctional hydrogels (112, 114). In addition, silk-like proteins (SLPs) have also been developed with the aim to achieve the exceptional

material properties and mechanical strength as that found in natural silk (129). The primary sequences in SLPs are usually composed of repeats of Ala and Gly, and the most common secondary structure of SLPs is an anti-parallel β -sheet. Robust materials have been generated from SLPs, and their biomedical applications are well reviewed by Kaplan *et al* (129, 130). The Hubbell group and the Kiick group have also engineered various bioactive protein modules into protein polymers to develop hybrid hydrogels for tissue engineering applications (124-128).

1.5 An overview of the biomimicry niche in this study

The Barron lab's contributions to the biomimicry field lie in the development of peptoid-based mimics and enzyme-crosslinkable protein polymer hydrogels. Predecessors in the lab have developed peptoids mimics for antimicrobial applications (30, 40, 42) and lung surfactant proteins (131-133), and a "controlled cloning" method to develop monodisperse protein polymers for multivalent displaying and the modular enzymatically crosslinked protein polymer hydrogels (111, 134, 135). The aims of the work in this dissertation were established from these platforms in the Barron lab. Specifically, the objectives of this work were: (1) Develop potent, selective cationic, amphipathic peptoids for anticancer applications, understand how their cytotoxicity can be manipulated, and explore their killing mechanisms; (2) Design efficient cationic, amphipathic peptoid transporters and explore how different structures may influence their uptake efficiency; (3) Evaluate a PEG-based platform for multivalent delivery of Bombesin to GRPR/BB in prostate cancer; (4) Generate genetically engineered protein polymers with matrix metalloprotease (MMP) degradation sites to incorporate into a enzyme crosslinkable hydrogel system; and (5) develop a hybrid biomimetic hydrogel

with a biocompatible crosslinking method based on the genetically engineered protein polymers.

In **Chapter 2**, based on previous knowledge that several cationic, amphipathic host defense peptides have been identified with anticancer activity and the ability to overcome the multidrug resistance developed in cancer cells, we present a library of anticancer peptoids that mimics the cationic, amphipathic structural feature of the host defense peptides and explore the relationships between the structure, anticancer activity and selectivity of these peptoids. Several peptoids have been identified with potent cytotoxicity against a broad range of cancer cell lines at low micromolar concentrations including cancer cells with multidrug resistance. The structure-activity relationships derived from this analysis reaffirms previous knowledge of peptides and further extends the designing heuristics to guide future development of anticancer peptidomimetics. For the killing mechanism studies, the cytotoxicity of peptoids involves plasma membrane perturbations. Furthermore, peptoid 1, the most potent peptoid synthesized, significantly inhibited tumor growth in a human breast cancer xenotransplantation model without any noticeable toxicity in mice. This work opens up further *in vivo* evaluations of these peptoids in cancer animal models.

Chapter 3 introduced another application of cationic, amphipathic compounds by developing a different library of cationic, amphipathic peptoids as novel molecular transporters. The search for highly efficient, stable transporters and the understanding of their internalization mechanisms are two areas under extensive investigation. We investigated the relationships between the peptoid structure, uptake efficiency and their associated cytotoxicity. For these cationic, amphipathic transporters, both guanidinium

heads and bulky, aromatic hydrophobic residues were found to render them more permeable. Moreover, the internalization mechanisms of these cationic, amphipathic peptoid transporters were distinct from those of polyguanidine constructs (Arg₈ and Nbtg₈). Future work involves using the developed transporters and evaluating their delivery efficiency for certain applications.

In **Chapter 4**, to take advantage of the targeting/binding ability of small peptides and to enhance *in vivo* pharmacokinetics of small peptide-based targeting probes, we proposed to use 4-arm PEG as a multivalent platform of these small peptides. Gastrin-releasing peptide receptor (GRPR/BB₂) is up-regulated in numerous cancers especially in prostate cancer as an important marker for malignant transformation. Bombesin (BBN), a homologue of gastrin-releasing peptide can bind to GRPR/BB₂ with high affinity and thus has been used to develop molecular probes for targeted tumor imaging. We designed and synthesized a 4-arm PEG-BBN conjugate with facile coupling chemistry, allowing further molecular radiolabelling. The binding affinity and specificity of the PEG-BBN conjugates were verified. Their *in vivo* tumor imaging ability and biodistribution was further evaluated via microPET in PC3 tumor-bearing mouse models, which indicated their good, specific tumor uptake.

In **Chapter 5** and **Chapter 6**, the focus is shifted from studying relatively small peptides via synthetic methods to generating protein polymer-based biomaterials via genetically engineering for hydrogel development.

In **Chapter 5**, we incorporated biofunctionalities into a previously established enzyme-crosslinkable hydrogel system, in which a lysine substrate protein polymer can

be crosslinked with a glutamine substrate protein polymer into hydrogels by transglutaminase. By using a “controlled cloning” method, a MMP degradable module was cloned into the lysine substrate backbone, giving rise to three types of MMP-K block polymers, two of which have been successfully expressed and purified. These two MMP-K proteins have been confirmed to have high water solubility and good expression yield. The functionality of the MMP degradation sites in the polymer and their further incorporation into the hydrogel has been confirmed. Via this method, more bioactive modules can be incorporated into the system.

In **Chapter 6**, a hybrid, biomimetic hydrogel was developed, comprising the MMP-K protein polymers developed in Chapter 5, thiolated HA and PEG crosslinkers. The Michael-type addition reaction was employed as the crosslinking chemistry. As a preliminary study, thiolated MMP-K proteins and thiolated HA were crosslinked with 8-arm PEG-vinylsulfone into hydrogels, and this gelation was found to be biocompatible with low toxicity to encapsulated cells. Moreover, the presence of proteins in the hydrogel promoted cell adhesion and the three-component hydrogel (with HA, proteins, PEG) led to the best cell proliferation. This preliminary study indicated a promising application of this hybrid, biomimetic hydrogel as a scaffold for tissue engineering. Future work including more material characterizations and evaluating the hydrogel for a specific application is necessary.

The work discussed herein only scratches the surface of what promises to be a very fruitful research area. Possible future directions of each project are discussed in **Chapter 7**.

Chapter 2

**Learning from host-defense peptides - cationic,
amphipathic peptoids with potent anticancer
activity**

2.1 introduction

Chemotherapy remains a must-have treatment for many cancer patients, especially in those with advanced diseases (136). Classic chemotherapeutical agents typically target the rapid proliferation of tumor cells by interrupting the synthesis or the functions of DNA, RNA or proteins. The lack of specificity of these drugs usually leads to many severe side effects. Moreover, it is very common for patients to develop multidrug resistance (MDR) through the efflux of drugs from cancer cells and become unresponsive to multiple chemotherapeutics (137). In order to overcome drug resistance and provide more effective treatment, new selective tumor-targeting drugs employing novel therapeutic mechanisms must be developed.

Antimicrobial peptides, also known as host-defense peptides, represent a class of naturally occurring compounds that have recently been explored for their anticancer activity (138-142). Natural products have been playing an important role in developing chemotherapeutics with a substantial amount of anticancer agents in use being either natural or derived from natural products from various sources (143). Antimicrobial peptides are evolutionarily ancient weapons found throughout the animal and plant kingdoms (144). A hallmark of this class is that the molecule can adopt a structure in which clusters of cationic and hydrophobic residues are spatially organized in discrete sectors, and this cationic, amphipathic structural feature is critical for their activity and selectivity (34). Most host-defense peptides are believed to be membranolytic, with cationic residues selecting for anionic cellular membranes via electrostatic interactions and hydrophobic regions responsible for membrane permeation and disruption (145). Magainin 2 and its analogs were first found in 1993 to display selective cytotoxicity

towards carcinoma cells *in vitro* and were proven to be as effective as doxorubicin *in vivo* via intraperitoneal delivery in ovarian cancer mouse models (146). Over the last two decades, a growing number of studies have shown that some cationic, amphipathic peptides, including both natural host defense peptides and synthetic antimicrobial peptides, exhibit a broad spectrum of cytotoxic activity against cancer cells and are effective in reducing tumor burdens in several cancer animal models (138-142). The selectivity of these peptides towards cancer cells is not well understood and is hypothesized to result from some altered membrane properties of cancer cells compared to normal tissue cells, e.g., more negative charges on outer membrane leaflets, more microvilli, higher transmembrane potentials, or higher membrane fluidity (138-142).

This class of cationic, amphipathic peptides possesses many features ideal for anticancer applications, including 1) high water solubility, 2) broad, potent cytotoxicity against cancer cells, and 3) the ability to overcome multidrug resistance developed in cancer cells (35, 146, 147). However, the clinical use of peptide-based drugs has been limited due to their rapid degradation and clearance *in vivo*. Non-natural peptidomimetics can circumvent this proteolytic sensitivity while retaining the beneficial features of peptides (2). A group of stable diastereomeric lytic peptides (containing both D- and L-forms of lysines and leucines) was developed based on antimicrobial model amphipathic peptides and investigated for anticancer applications (148-150). Poly-N-substituted glycines, or peptoids, comprise another class of protease-resistant peptidomimetics, with side chains attached to the backbone nitrogen rather than to the α -carbon as in peptides (151, 152). Peptoids can be easily synthesized on solid phase in a sequence-specific manner on an automated peptide synthesizer. Virtually any desired

chemical functionality available as a primary amine can be incorporated into peptoids via a submonomer method, which provides great chemical “design diversity” in peptoid sequences (152).

Peptoids are ideal architectures to mimic the cationic, amphipathic structural feature. Though lacking backbone chirality and intrachain hydrogen bonding, peptoids can readily form helical structures with a periodic incorporation of bulky, α -chiral side chains, giving rise to polyproline type-I-like helices with approximately three residues per turn and a helical pitch of 6.0 – 6.7 Å (27, 28, 152). This threefold periodicity of peptoids allows the cationic, amphipathic structure to be easily recapitulated in three-faced helices, simply using peptoids comprising trimer repeats, (X-Y-Z)_n, which will display X, Y, and Z residues on separate faces. Accordingly, our group developed antimicrobial peptoids by mimicking magainin-2 (40). A library of antimicrobial peptoids that typically adopt cationic, amphipathic helical structures was synthesized and studied for the relationships between their structures and their antimicrobial activity and selectivity (30, 42).

Here, inspired by anticancer peptides, we developed a new library of cationic, amphipathic peptoids and screened their anticancer activity and selectivity *in vitro*. We demonstrated for the first time that cationic, amphipathic peptoids can exhibit potent, fast cytotoxicity at low micromolar concentrations to a broad range of human cancer cell lines, and some peptoids gained higher *in vitro* selectivity with a slight compromise in potency. Moreover, actions of these peptoids were not influenced by multidrug resistance, killing primarily via plasma membrane disruptions. Finally, the therapeutic efficacy of the most potent peptoid derivative was validated in a breast cancer xenotransplantation model established with human patient tumor cells.

This study was prepared as a manuscript entitled “Learning from host-defense peptides - cationic, amphipathic peptoids with potent anticancer activity” with the authors: Wei Huang, Jiwon Seo, Stephen B. Willingham, Ann M. Czyzewski, Mark L. Gonzalgo, Irving L. Weissman and Annelise E. Barron.

2.2 Materials and methods

2.2.1 Peptoid synthesis and purification

Peptoids were synthesized using an ABI 433A peptide synthesizer (Applied Biosystems, Inc.) on Rink amide MBHA resin (EMD Biosciences, Gibbstown, NJ) using the submonomer protocol (30, 152). Briefly, the amine on the nascent chain is bromoacetylated or chloroacetylated followed by S_N2 displacement of bromide or chloride by a primary amine to form the side chain. Resin-bound peptoids were then exposed to a mixture of trifluoroacetic acid (TFA): triisopropylsilane: water (95:2.5:2.5, volume ratio) for 10 minutes to cleave peptoids from the resin. Crude peptoids were purified by reversed-phase high performance liquid chromatography (RP-HPLC) (Waters Corporation) using a C18 column and a linear acetonitrile/water gradient. A final purity > 95% as measured by analytical RP-HPLC (Waters Corporation) was achieved, and the identity of each peptoid was confirmed using electrospray ionization mass spectrometry (ESI/MS). Pexiganan was synthesized by standard Fmoc chemistry on an ABI 433A peptide synthesizer (EMD Biosciences). Unless indicated otherwise, all reagents were purchased from Sigma Aldrich (St. Louis, MO). Among the submonomers used, *N*spe was derived from (S)-*N*-(1-phenylethyl)amine; *N*pm from benzylamine (1-phenylethylamine); *N*His from histamine (2-[4-imidazolyl]ethylamine); *N*Leu from isobutylamine; *N*Lys from *N*-*tert*-butoxycarbonyl-1,4-butanediamine (CNH Technologies,

MA). Guanidinylation of *N*Lys was carried out according to the reported procedure (88). When *N*His was used in the peptoid sequence, chloroacetic acid was used (153). 5(6)-Carboxyfluorescein was used to label the N-terminus of peptoid 1 (154) .

2.2.2 Cell cultures

MRC-5 cells were purchased from American Type Culture Collection (ATCC) and were grown in media suggested by ATCC supplemented with 10% FBS (Hyclone, US sources) and antibiotics (Sigma). MCF-7, MCF-7/TxT50 and OVCAR-3 were kindly provided by Professor Branimir Sikic's lab at Stanford University, and were cultured in McCoy's 5A media (GIBCO) with 10% FBS and antibiotics. Primary dermal fibroblasts were gifts from Lifeline Cell Technology and were cultured < 6 passages according to the protocol Lifeline provides. All the cells were cultured in cell incubators (Thermo Scientific) at 37°C with 5% CO₂.

2.2.3 The MTS assay

This assay is a colorimetric method for determining the number of viable cells in proliferation. Aliquots of 100µl media containing 1×10^4 cells were distributed into each well of a 96-well plate (BD Falcon). The following day, when cell density reaches about ~40% confluency, the cell media were removed and replaced with serial dilutions of peptoid stocks in culturing media. For peptoid dilutions, peptoid stocks were initially diluted in media at 100µM and then diluted by half in series using a multichannel pipette, and maintained in 100µl media for each concentration with triple repeats. After peptoid solutions were transferred onto cells in 96-well plates, cells were incubated at 37°C for certain time periods. Then 20µl of the CellTiter 96® Aqueous Non-Radioactive cell proliferation assay (Promega) reagent which contains a tetrazolium compound, [3-(4,5-

dimethylthiazol-2-yl)-5-(3-carboxymethoxyphenyl)-2-(4-sulfophenyl)-2H-tetrazolium, inner salt; MTS(a)], was added to each well and cells were further incubated for 2h to metabolize. The absorbance of formazan products were measured at 490nm in a microplate reader (Molecular Devices). Percentage of cell viability = $(A - A_{\text{testblank}})/(A_{\text{control}} - A_{\text{blank}}) \times 100$, where A is the absorbance of the test well and A_{control} the average absorbance of wells with cells not treated with peptoids. $A_{\text{testblank}}$ (media, MTS, and diluted peptoids) and A_{blank} (media and MTS) were background absorbances measured in the absence of cells.

2.2.4 The Guava® Viacount® assay

This assay measures cell viability by using two DNA-binding dyes, one membrane-permeant, the other membrane-impermeant. Briefly, cells were plated and peptoids were diluted as described in MTS assays. After peptoid treatments, media supernatant with floating cells were collected. Cells were washed once, trypsinized and neutralized with previously collected media supernatant with floating cells. Cells were centrifuged at 1800rpm for 3min, and cell pellets were resuspended in 180μl of fresh media. 20μl of Guava® Viacount® reagents (Millipore) was added to each cell suspension, gently mixed and incubated at room temperature (protected from light) for 5min before being submitted to the Guava Easycyte Plus® flow cytometry system (Millipore) to measure the cellular fluorescent signals stained with dyes in Viacount® reagents and to quantify cell viability.

2.2.5 The LDH assay

This assay is used to measure the membrane integrity as a function of the amount of cytoplasmic LDH released into the medium. Experiments were carried out according to

the protocol of the *In Vitro* Toxicology Assay Kit, Lactic Dehydrogenase (LDH) based (Sigma-Aldrich). Briefly, cells were plated as described before, and peptoids were diluted similarly but in culturing media without phenol red to reduce background signal. After peptoid treatments, media supernatant were collected and centrifuged to remove any cell debris, and analyzed for LDH activity in a 96-well plate using the kit, absorbance at 490nm and 690nm measured using a microplate reader. All the following absorbance difference = $A_{490\text{nm}} - A_{690\text{nm}}$. Percentage of LDH leakage = $(A - A_{\text{testblank}}) / (A_{\text{lysis}} - A_{\text{blank}}) \times 100$, where A is the average absorbance difference of the test wells and A_{lysis} the average absorbance difference of wells with cells treated with cell lysis solution provided by the kit for 45min. $A_{\text{testblank}}$ (media, diluted peptoids and LDH measuring reagents) and A_{blank} (media and LDH measuring reagents) were background absorbance differences measured in the absence of cells.

2.2.6 Confocal imaging

MCF-7 cells seeded in 35mm glass-bottom dishes (MatTek Corporation) were treated with 8 μ M carboxyfluorescein-peptoid1 (CF-peptoid 1) diluted in culturing media with 10% FBS for 1h. Supernatants were removed, and cells were washed with PBS three times before being imaged via a Leica SP2 AOBS confocal laser scanning microscope.

2.2.7 Peptoid evaluation in a human breast cancer xenograft mice model

This part was carried out by Stephen Willingham in Irving Weissman lab at Stanford. 80,000 cells from a dissociated second generation metastatic breast cancer tumor were suspended in Medium 199 containing 25% Matrix Matrigel (Becton Dickinson 354248) and injected into the mammary fat pad of 4-8 week old NOD.Cg-Prkdc^{scid} Il2rg^{tm1Wjl}/SzJ

(NSG) mice. After two weeks, 100 μ l PBS containing 110 μ M peptoid-1 or control peptide was injected into the mammary fat pad three times a week, roughly at a dose of 1mg/kg. Tumor volumes were determined using direct caliper measurements following twelve weeks of treatment.

2.3 Results

2.3.1 Peptoid Design and *in vitro* Screening

Peptoids were used as the peptidomimetic scaffold to capture the cationic, amphipathic nature of anticancer peptides as well as to improve molecular stability and increase chemical diversity. The design of anticancer peptoids were derived from previous antimicrobial peptoids and were optimized to improve the activity and selectivity of peptoids against anionic membranes (30, 40, 42). Peptoid 1, [H-(NLys-Nspe-Nspe)₄-NH₂], composed of NLys (the peptoid analog of Lys) and Nspe (the peptoid analog of Phe, Figure 2.1), is the most active antimicrobial dodecamer we had previously developed yet it had some minor hemolytic effects at high concentrations (Figure 2.1A). Several novel variants were designed based on peptoid 1 but with different charges, amphipathicity, length, and helicity to change their activity or selectivity. The sequences of these variants, the solvent composition at RP-HPLC elution as a relative measure of molecular hydrophobicity, molecular net charges and their charge-to-length ratio are summarized in Table 2.1. Variants marked with an asterisk were previously reported to possess antimicrobial activity (30, 42). Pexiganan, a clinically-relevant peptide analog of magainin 2 (155), and LL-37, the only known human cathelicidin which is a non-selective antimicrobial peptide (156), were also tested for comparison.

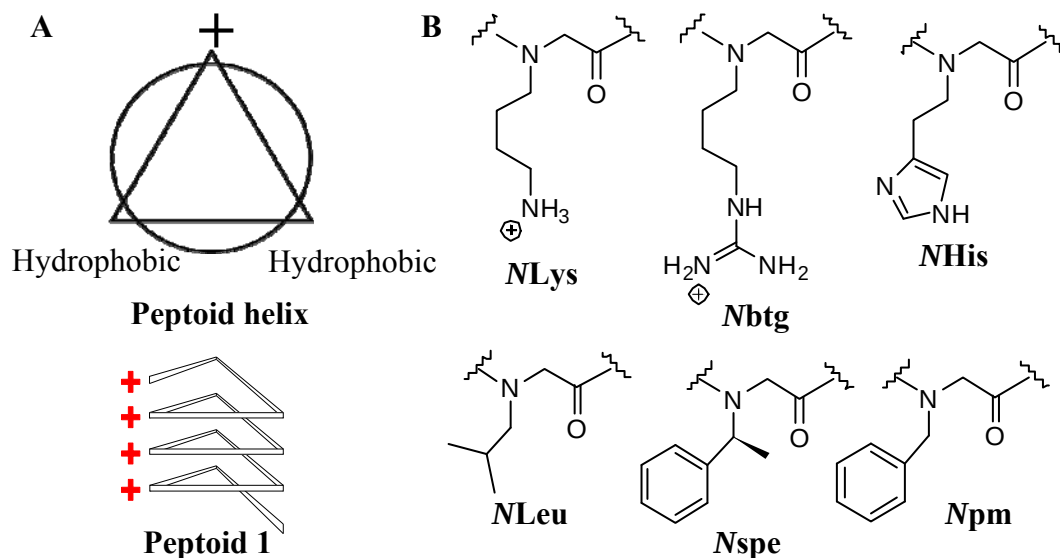


Figure 2.1 The cationic, amphipathic structure of peptoids and monomers. (A), top-view of the cationic, amphipathic structure of peptoids (upper), and side-view of the helical structure of peptoid 1 (bottom); (B), peptoid monomer side-chain structures with shorthand names.

The activity and selectivity of these peptoids were first evaluated against cancer cells *in vitro*. Their cytotoxicity was tested in three cancer cell lines, MCF-7 (human breast cancer), LNCaP (human prostate cancer), and OVCAR-3 (human ovarian cancer). To estimate the selectivity of these peptoids *in vitro*, hemolytic activities of some peptoids were cited from previously published work (Table 2 **column), and the MRC-5 cell line (human fibroblasts derived from fetal lung tissues) and primary dermal fibroblasts were also tested as normal cell controls. Briefly, cells were treated with peptoids diluted in culturing media for certain time periods, and then cell viability was measured via MTS assays. These peptoids as well as pexiganan and LL-37 were more potent in media with low serum concentrations (data not shown), but results reported here are all tested in media supplemented with 10% FBS.

	Compound	Sequence	MW (Da)	HPLC elution (% ACN)	Net Charge	CTLR
	Peptoid 1 *	H-(NLys-Nspe-Nspe) ₄ -NH ₂	1819	65.1	+4	0.33
Negative control	1_{11mer}-NLeu	H-(NLys-NLeu-NLeu) ₃ -NLys-NLeu-NH ₂	1321	51.0	+4	0.36
Cationic, amphipathic variants	1-NLys_{5,11} *	H-(NLys-Nspe-Nspe-NLys-NLys-Nspe) ₂ -NH ₂	1753	51.2	+6	0.50
	1-Nbtg	H-(Nbtg-Nspe-Nspe) ₄ -NH ₂	1987	65.0	+4	0.33
	1-NHis_{1,7}	H-(NHis-Nspe-Nspe-NLys-Nspe-Nspe) ₂ -NH ₂	1767	62.5	+2~+4**	0.17~0.33
Length variants	1_{13mer} *	H-(NLys-Nspe-Nspe) ₄ -NLys-NH ₂	1948	62.8	+5	0.38
	1_{11mer} *	H-(NLys-Nspe-Nspe) ₃ -NLys-Nspe-NH ₂	1658	63.5	+4	0.36
	1_{9mer} *	H-(NLys-Nspe-Nspe) ₃ -NH ₂	1368	60.0	+3	0.33
Achiral variants	1_{achiral} *	H-(NLys-Npm-Npm) ₄ -NH ₂	1701	59.8	+4	0.33
	1_{achiral}-Nspe₂ *	H-NLys-Nspe-Npm-(NLys-Npm-Npm) ₃ -NH ₂	1721	60.8	+4	0.33
	1_{achiral}-Nspe_{2,12}	H-NLys-Nspe-Npm-(NLys-Npm-Npm) ₂ -NLys-Npm-Nspe-NH ₂	1735	62.0	+4	0.33
	1-Npm_{2,3,8,9} *	H-(NLys-Npm-Npm-NLys-Nspe-Nspe) ₂ -NH ₂	1763	63.3	+4	0.33
Peptide comparators	Pexiganan	GIGKFLKKAKKFGKAFVKILKK-NH ₂	2477	50.2	+9	0.41
	LL 37	LLGDFFRKSKEKIGKEFKRIVQRIKDFLRNLVPRTES	4495	66.0	+6	0.16

Table 2.1 Sequences and molecular properties of peptoids and comparator peptides. See Figure 2.1 for the structures of the peptoid monomers indicated in each sequence. HPLC elution is reported as the percentage of compound elution in the water/acetonitrile (0.1% trifluoroacetic acid) solvent in analytic HPLC. A linear water/acetonitrile (0.1% trifluoroacetic acid) gradient of 5%-95% acetonitrile over 30min was run on a C18 column. Net charge indicates molecular charges at neutral pH. (“**”, NHis has 10% probability to be charged around neutral pH, so the net charge of 1-NHis_{1,7} is between +2 to +4). CTLR stands for charge-to-length ratio, which is defined as the ratio of the total number of cationic residues to the total number of residues in each sequence. Peptoids labeled with asterisk have been reported previously to possess antimicrobial activities (30, 40, 42).

These anticancer peptoids were found to exert fast killings in all tested cancer cell lines, and their activities were highly dependent on their structures and working

concentrations. A majority of cell death occurred within only 4h of peptoid treatments, and increased cytotoxicity was observed with longer treatments. The cytotoxicity curves of peptoid 1 in MCF-7 cells for different treatment times are shown in Figure 2.2A. Table 2.2 summarizes the compound activity for 72h treatment to better correlate with cytotoxicity of apoptosis-inducing chemotherapeutics which are typically evaluated for 72h treatment *in vitro*.

As shown in Table 2.2, cytotoxicity of these peptoids varied slightly in different cancer cell lines, with lethal concentration (LC_{50}) in the low micromolar range. In addition, several peptoids were found with selectivity towards cancer cells, killing cancer cells efficiently while being relatively safe to MRC-5, primary dermal fibroblasts, and red blood cells at certain concentration ranges. The cytotoxicity curves of $1_{achiral}$ in MCF-7 cells and primary dermal fibroblasts for 24h treatment are compared in Figure 2.2B, indicating a therapeutic concentration window. Better selectivity (indicated by high LC_{50} in MRC-5 and primary dermal fibroblasts and high HC_{10} against red blood cells) was typically achieved with compromised potency against cancer cells, which will be further discussed in structure-activity studies. The highest selectivity ratio (LC_{50} in primary dermal fibroblast divided by LC_{50} in cancer cells) we have observed for peptoids was ~ 3 for $1_{achiral}$ (Figure 2.2B); however, we did not observe any selectivity of Doxorubicin (data not shown) and LL37 (Table 2.2) in this *in vitro* screening system. We understand that this *in vitro* two-dimensional system cannot mimic *in vivo* tumor environments perfectly, and that MRC-5 and primary dermal fibroblasts may not be good controls to estimate the *in vivo* systemic toxicity of these peptoids. Therefore, we grouped our drug hits into two categories: (1) Highly potent peptoid. Peptoid 1 is the best candidate with

great anticancer potency, good water solubility, ease of synthesis and relatively low hemolytic activity, though it has similar cytotoxicity against cancer cells and fibroblasts cultured *in vitro*; (2) Selective peptoids with good potency. 1_{11mer} , $1_{achiral}$ and $1_{achiral-Nspe_{2,12}}$ are three good candidates which retain good potency and also display some selectivity in this *in vitro* screening system.

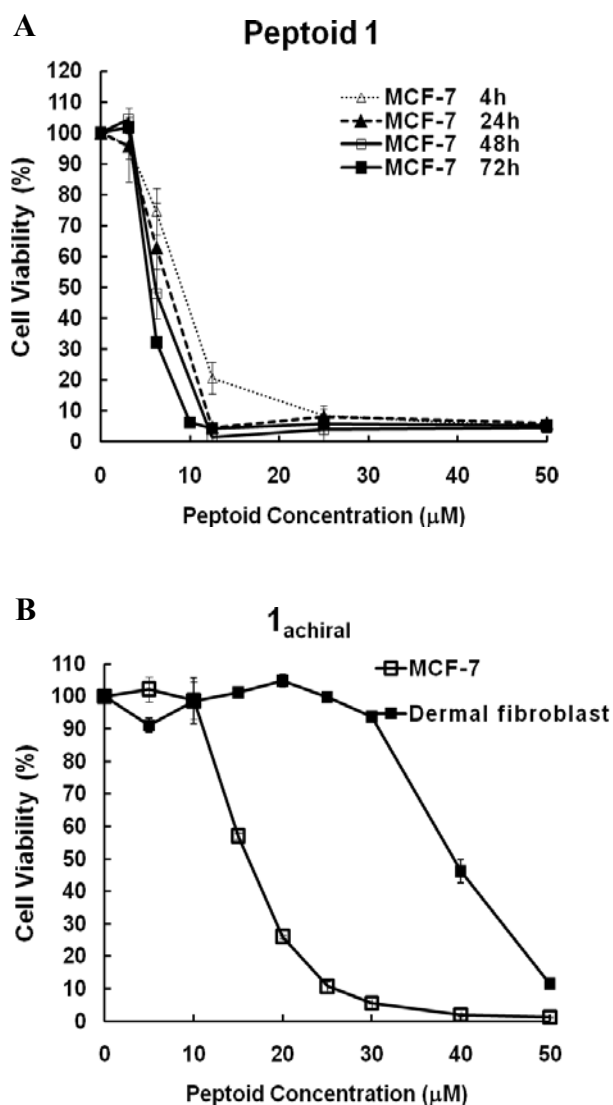


Figure 2.2 Cell viability curves of peptoids. A, cell viability of MCF-7 cells treated with peptoid 1 for different time periods. Cell viability was measured with MTS

assays; B, cell viability curve of 1_{achiral} in MCF-7 and primary dermal fibroblast cells. 1_{achiral} was more toxic to MCF-7 cells.

	Compounds	LC ₅₀ (μM)					HC ₁₀ (μM) ** Red blood cells
		MCF-7 (breast cancer)	LNCaP (prostate cancer)	OVCAR3 (ovarian cancer)	MRC-5 (fetal lung fibroblast)	Primary dermal fibroblast	
	Peptoid 1 *	5	5	6	8	8	21
Negative control	1_{11mer}-NLeu	>100	>100	>100	>100	>100	--
Cationic, amphipathic variants	1-NLys_{5,11} *	31	25	37	41	34	>100
	1-Nbtg	5	8	8	10	8	--
	1-NHis_{1,7}	19	23	30	30	30	--
Length variants	1_{13mer} *	8	8	9	10	14	21
	1_{11mer} *	11	10	10	24	19	103
	1_{9mer} *	30	27	34	40	32	150
Achiral variants	1_{achiral} *	13	11	16	30	39	183
	1_{achiral}-Nspe₂ *	13	11	16	22	36	160
	1_{achiral}-Nspe_{2, 12}	12	10	16	21	32	--
	1-Npm_{2,3,8,9} *	9	10	11	18	21	80
Peptide comparators	Pexiganan	8	6	13	21	19	70
	LL 37	21	27	24	24	26	--

Table 2.2 Peptoid Cytotoxicity in cancer cell lines and control cells. LC₅₀ means lethal concentrations causing 50% of the cell death. The maximum peptoid and peptide concentration tested was 100μM. Cell viability was measured via MTS assay in cells treated with compounds for 72h, and LC₅₀ of peptoids was derived from peptoids' viability curves. HC₁₀ **, concentration causing 10% hemolysis of human RBCs, all the numbers were cited from (42). "--", data not measured. Peptoids labeled with asterisk have been reported previously to possess antimicrobial activities (30, 40, 42) .

2.3.2 Structure-Activity Relationship Studies

2.3.2.1 Cationic, amphipathic structure

We intended to study how hydrophobic and cationic residues as well as the structural amphipathicity influenced peptoid potency and selectivity (Table 2.2). Knowing that

aromatic side chains are critical for the biological activities of cationic, amphipathic peptoids (40), an 11-mer with *N*Leu as hydrophobic residues was designed as a negative control and was confirmed to be inactive, with LC_{50} above 100 μ m in all the cells tested. Furthermore, hydrophobicity and structural amphipathicity were found to be important for peptoid activity. Reduced potency against mammalian cells was observed with 1-*N*Lys_{5,11} which had reduced hydrophobicity with only 6 *N*spe residues per molecule and reduced amphipathicity with cationic residues present in hydrophobic faces (42). Other cationic residues were also employed in peptoids. Guanidinium heads have been reported to enhance the membrane interactions of cell penetrating peptides via hydrogen bonding (157). We synthesized 1-*N*btg, or [*N*-(4-butyl)guanidine] glycine], by guanidinylating *N*Lys in peptoid 1. The activity of 1-*N*btg was similar to peptoid 1 with no improvement on selectivity, indicating a minor role of hydrogen bonding in the anticancer peptoid-membrane interactions. Using histidines (pKa ~6.1) as cationic residues has been reported to result in the pH-dependent activity of anticancer peptides with enhanced selectivity against tumor environments *in vivo* which are significantly more acidic than normal tissues (158). 1-*N*His_{1,7} was synthesized with *N*His replacing *N*Lys at position 1 and 7 in peptoid 1, which would reduce peptoid cationic charges in neutral pH. Decreased cytotoxicity against both cancer cell lines and fibroblasts was observed with 1-*N*His_{1,7}. We did not observe a pH-dependent activity of this *N*His-*N*Lys hybrid peptoid *in vitro* (data not shown), consistent with the previous result of histidine-containing peptide which only showed enhanced selectivity *in vivo* (158). 1-*N*His_{1,7} could exhibit better selectivity *in vivo* yet needs to be further validated.

2.3.2.2 Chain length variants

The chain length is believed to influence not only the activity but also modes of action of cationic, amphipathic peptides (159, 160). In a previous study, increasing the length beyond a 12mer, such as peptoid 1_{15mer}, [H-(NLys-Nspe-Nspe)₅-NH₂], did not benefit antibacterial potency but resulted in substantially higher hemolytic activity, indicating that long chain lengths may lead to undesirable systemic toxicity of anticancer peptoids (40). Thus we limited the chain length of peptoids to 13 residues at most, and studied 1_{9mer} which has the same charge-to-length ratio (CTLR) as peptoid 1 at 0.33, 1_{11mer} with CTLR at 0.36, and 1_{13mer} with CTLR at 0.38 (42). As shown in Table 2.2, the relatively low toxicity of 1_{9mer} against mammalian cells indicated that peptoids have to reach certain chain lengths to gain high potency. 1_{11mer} was found to be more selective than peptoid 1 with reduced toxicity against MRC-5 and primary dermal fibroblasts and significantly reduced hemolytic activity, though its potency against cancer cells was slightly reduced as well. 1_{13mer} was slightly less active than peptoid 1 yet with no big improvements on its selectivity.

2.3.2.3 Achiral monomer variants

The molecular chirality of peptoids is derived from the chirality of the side chains rather than that of the backbone (27). 1_{achiral} with less hydrophobic, achiral Npm residues lost helical signals in circular dichroism (CD) spectroscopy, which suggests a lack of stable secondary structure, and increased replacement of Nspe with Npm in peptoid 1 decreased peptoid helical intensities in CD. The achiral monomer variants had significantly reduced hemolytic activities compared to peptoid 1 (42). In our *in vitro* screening, 1_{achiral}, 1_{achiral}-Nspe₂, 1_{achiral}-Nspe_{2,12}, 1-Npm_{2,3,8,9} all displayed better

selectivity than peptoid 1 with reduced toxicity to MRC-5 cells and primary dermal fibroblasts. 1_{achiral} was the most selective peptoid in this group with good potency against cancer cells (Figure 2.2B).

2.3.3 Anticancer peptoids overcome multidrug resistance developed in cancer cells

Several anticancer peptides, such as magainin 2 (146) and buforin IIb (35, 147), have been reported to overcome multidrug resistance developed in cancer cells. Interestingly, activities of our anticancer peptoids were also found to be unaffected by the multidrug resistance in cancer cells. The resistant MCF-7/TxT50 cell line which was selected by increasing exposure to docetaxel is resistant due to the high expression of the ABCB1/MDR1, P-glycoprotein (161). As shown in Figure 3A, MCF-7/TxT50 cells were confirmed to be resistant to Docetaxel compared to wild-type MCF-7 cells. However, peptoid 1 and 1_{achiral} had similar activities in both MCF-7 and the resistant MCF-7/TxT50 cells (Figure 3B-C), indicating a different mode of action of peptoids from the hydrophobic small molecule chemotherapeutic agents.

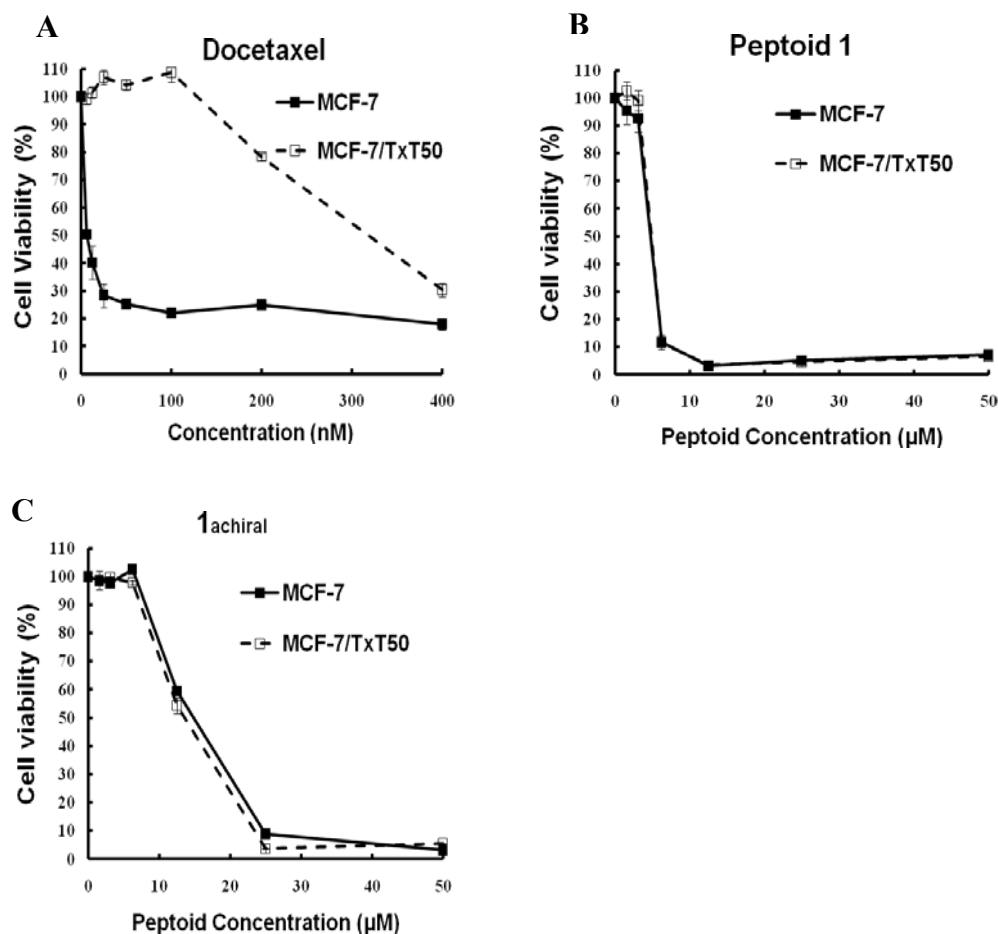


Figure 2.3 Activity of docetaxel and peptoids in MCF-7 and MCF-7/TxT50 cells. MCF-7/TxT50 cells were selected by docetaxel screening and are resistant due to overexpression of MDR1. Figures show the cell viability of MCF-7 and MCF-7/TxT50 cells treated with docetaxel (A), peptoid 1 (B) and 1_{achiral} (C) for 72h. Cell viability was measured with MTS assays.

2.3.4 Primary cytotoxicity of anticancer peptoids in cancer cells involves plasma membrane damage

These anticancer peptoids exert fast killing in cancer cells and are not affected by the multidrug resistance. To study how these peptoids interact with cells, we synthesized CF-peptoid 1 with fluorescent labeling on the *N*-terminus of peptoid 1. Via live cell confocal imaging, peptoid 1 was found to penetrate into cells efficiently with a dot-like

cytoplasmic distribution, even at low concentrations without causing any noticeable cytotoxicity (Figure 2.4A), indicating that these peptoids can interact with plasma membranes and translocate into cells. To evaluate if plasma membranes are damaged upon peptoid interactions and to determine how much membrane damage accounts for the cell death caused by peptoid 1, cells were treated with peptoids at various concentrations for 5min, 30min and 60min. The cells were then either cultured in fresh media for another 48h before testing via the MTS assay, which measures cell viability by tracking cellular metabolism (Figure 2.4B), or immediately tested via two cytotoxicity assays that evaluate plasma membrane intactness, the Guava Viacount assay (Figure 2.4C) and the lactate dehydrogenase (LDH) assay (Figure 2.4D). The Guava Viacount assay quantifies dead cells with a cell-impermeable DNA binding dye that can only stain cells with damaged membranes and cellular signals are measured by Guava flow cytometry. The LDH assay measures membrane damage by quantifying the leakage of the cytoplasmic enzyme LDH. Cell viability curves quantified by the Guava Viacount assay right after peptoid treatment were similar to those measured 48h later with MTS assays, suggesting that plasma membranes were damaged upon peptoid treatments and most of the cell deaths caused by peptoids were due to plasma membrane damage. Moreover, in the LDH assay, a fast release of LDH into culturing media was also observed upon peptoid treatment. With LDH leakage caused by cell lysis solution as the 100% control, 50 μ M of peptoid 1 (Figure 2.4D) and melittin (Figure 2.5) caused ~50% of enzyme leakage. A linear correlation with $r^2=0.955$ was observed between LDH leakage and cell viability quantified via the MTS assay (Figure 2.4E). Similar results were observed with melittin which is generally accepted to be lytic (Figure 2.5), further confirming the

primary toxicity of peptoids being on plasma membranes. Moreover, typical apoptotic DNA ladders were not observed in cells after 24h of peptoid 1 treatment (Figure 2.6). Taken together, these anticancer peptoids killed cancer cells through plasma membrane damage. However, there could be intracellular targets as well, since peptoids can translocate into cells and are widely distributed in the cytoplasm, and we have observed increased cell death with longer peptoid treatment in some concentration ranges (Figure 2.2A).

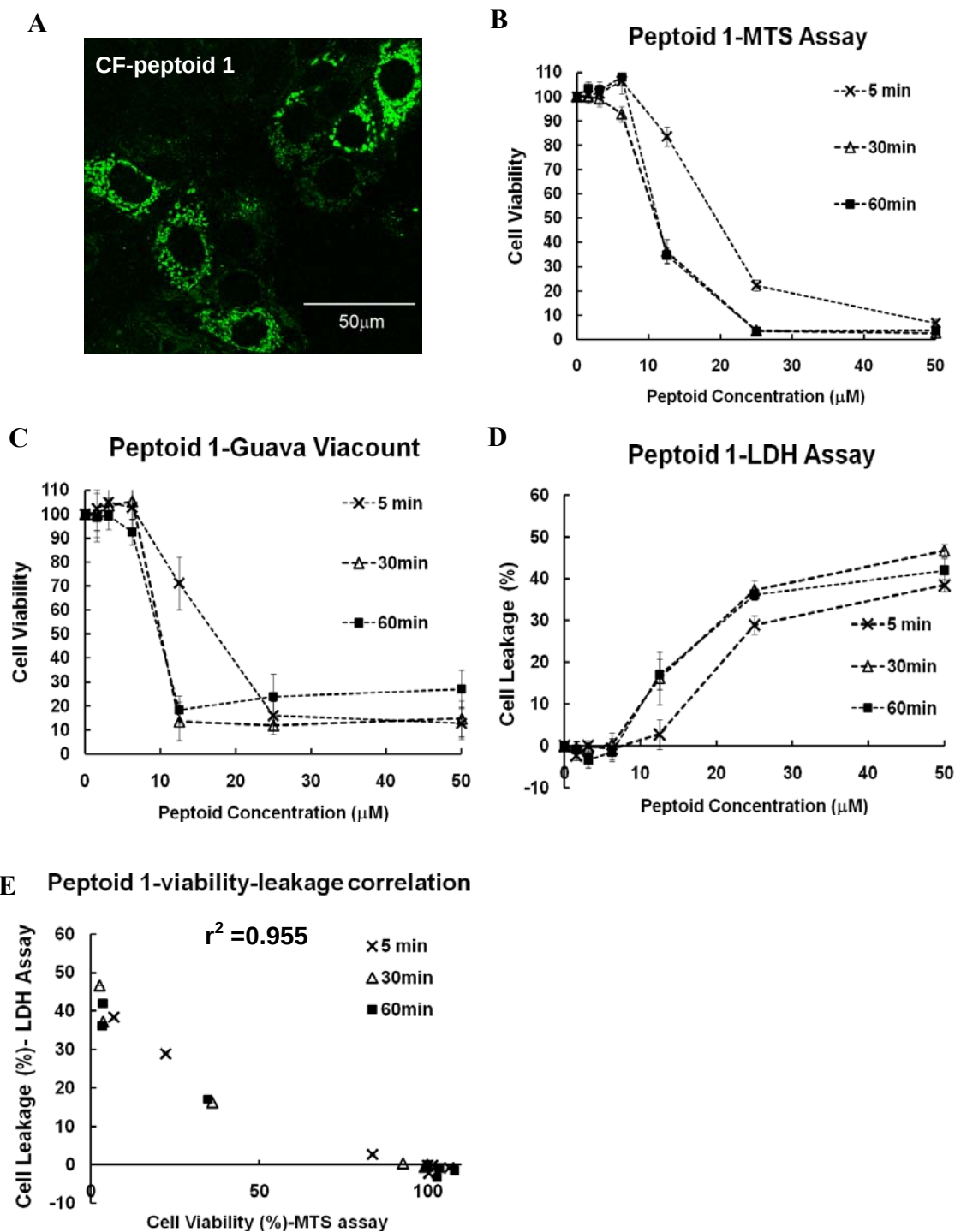


Figure 2.4 Killing mechanisms. A, live cell confocal images. MCF-7 cells were treated with 8μM of CF-peptoid 1 for 1h, and cells were imaged with ×63 oil lens. MCF-7 cells were treated with peptoids for indicated time and cell viability was measured with

MTS assays after another 48h incubation with fresh media (B), or measured via the Guava assay immediately after peptoid treatment (C), or quantified with LDH leakage immediately (D). E, correlation of LDH leakage and cell viability upon peptoid treatment measured with MTS assays, with $r^2 = 0.955$.

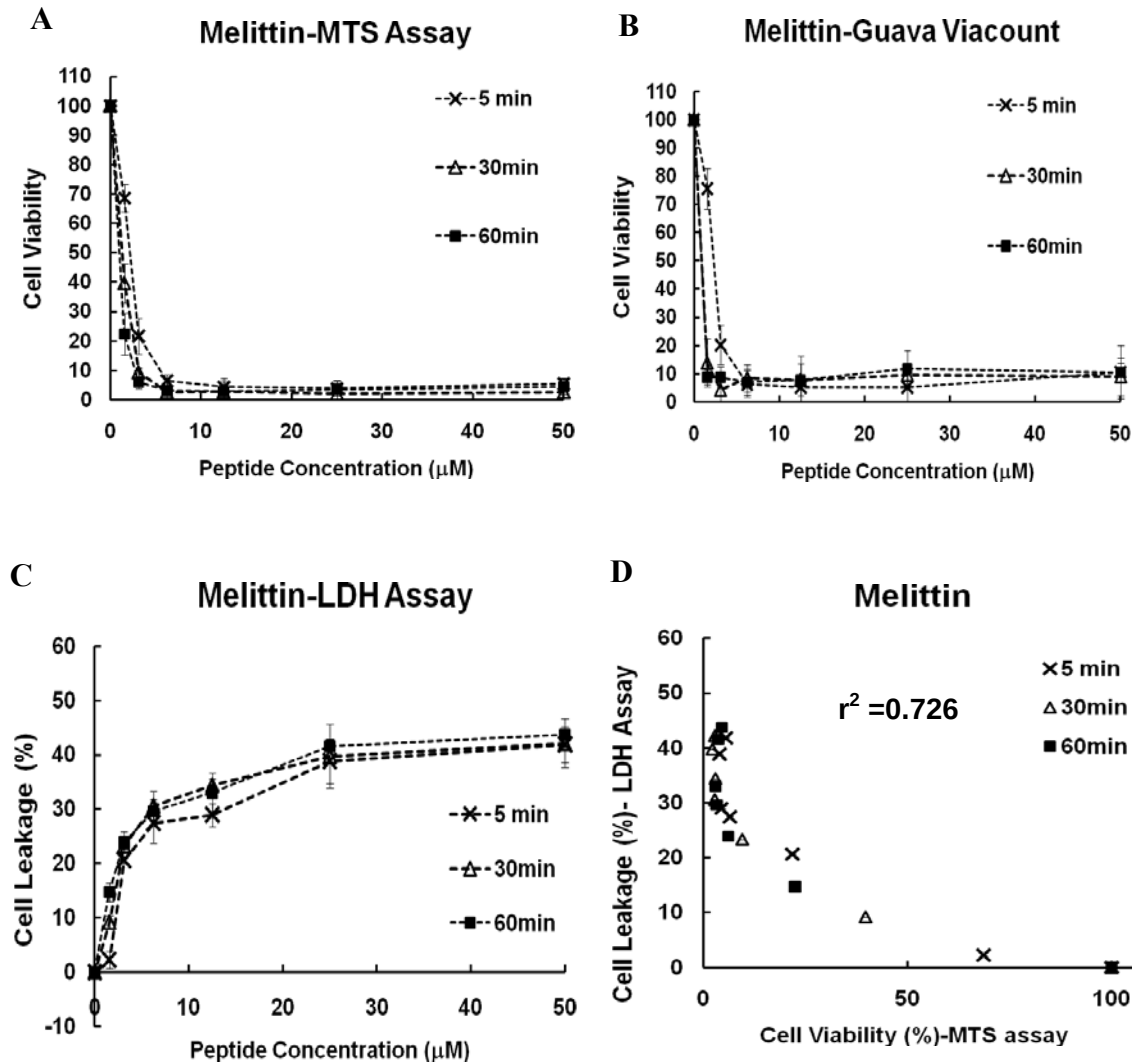


Figure 2.5 Evaluation of melittin as a comparison. MCF-7 cells were treated with melittin for indicated time and cell viability was measured with MTS assays after another 48h incubation with fresh media (A), or measured via the Guava assay immediately after treatment (B), or quantified with LDH leakage immediately (C). D, correlation of LDH leakage and cell viability upon melittin treatment measured with MTS assays, with $r^2 = 0.726$.

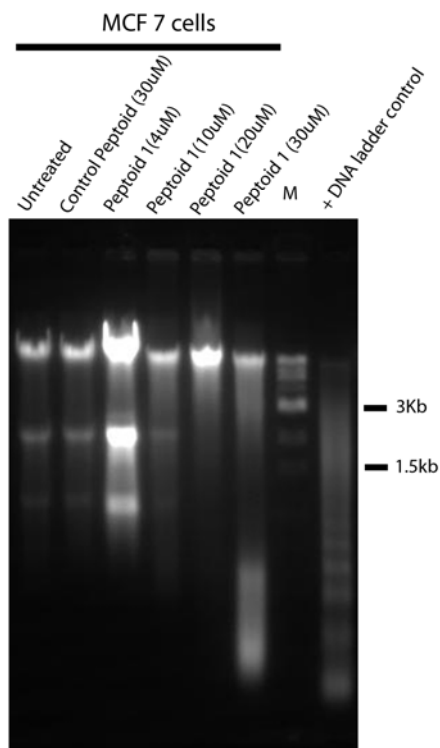


Figure 2.6 DNA ladder assay. MCF-7 cells were treated with control peptoid and peptoid 1 at the indicated concentrations for 24h. Cells and floated cells were collected and combined for each sample. DNA was extracted using the apoptotic DNA ladder Kit (Roche), stained with GelStar[®] (Lonza) and was run in 1% agarose gel. Experiments were done according to the kit, with the +DNA ladder control provided in the kit.

2.3.5 Peptoid 1 inhibits tumor growth in a clinically relevant orthotopic xenotransplantation model

As a preliminary study of the *in vivo* efficacy and safety of peptoids, the most potent peptoid, peptoid 1, was evaluated in an orthotopic xenograft mouse model. Human breast cancer cells were implanted in the mammary fat pad of immunocompromised mice. After two weeks, peptoid 1 or an inactive negative control peptoid $1_{11\text{mer}}\text{-NLeu}$ was injected into the mammary fat pad at a dose of $\sim 1\text{mg/kg}$ three times a week. Peptoid 1 significantly inhibited the tumor growth compared to the control peptoid (Figure 2.7). Notably, the therapeutic effect of Peptoid 1 occurred without any obvious acute toxicity.

These results indicate that peptoid derivatives may be effective therapeutic agents for the treatment of human tumors.

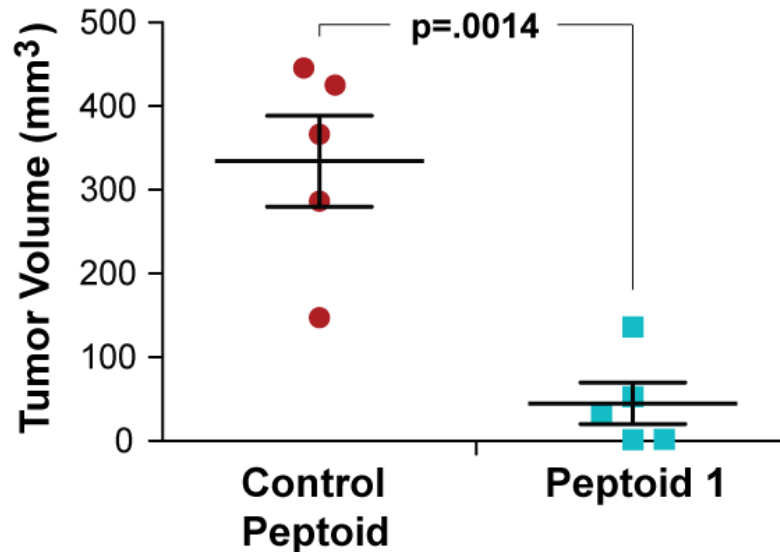


Figure 2.7 Reduction of tumor volumes in a human breast cancer xenograft mouse model treated with peptoid 1. Cancer cells (8×10^4 cells) isolated from the second generation xenografts of human breast cancer tissues were implanted s.c. into the lower left mammary fat pads of 5-6-week-old immunocompromised NSG (NOD.Cg-Prkdc^{scid} Il2rg^{tm1Wjl}/SzJ) female mice. Two weeks after cell implantation, 100µl of peptoid 1 and control peptoids were injected into xenografts at 110µM in PBS (~1mg/kg) three times a week, up to 8 weeks. Tumor sizes were measured by a caliper. $P=0.0014$, t test.

2.4 Discussion

Several cationic, amphipathic antimicrobial peptides or their derivatives have recently been reported with anticancer activity, displaying selective cytotoxicity against cancer cells *in vitro* and being effective in several *in vivo* xenograft models (138, 139, 142). Moreover, a recent report indicated a naturally occurring anticancer role of a host defense peptide, cathelicidin, in natural killer cells' antitumor functions in mice, most strongly supported by the experiment that cathelicidin knockout (*Camp*^{-/-}) mice permitted faster

tumor growth than wild type controls (162). All these findings suggest that these cationic, amphipathic peptides, which are used by nature in host defense, represent a very interesting and promising group of candidates to be developed for anticancer applications.

In this study, we exploited the protease resistance and the propensity for helix formation of certain peptoids to develop stable anticancer peptidomimetics, expanding available monomer chemistry and capturing the cationic, amphipathic structural feature. Several anticancer peptoids were developed successfully with potent cytotoxicity toward a broad range of human cancer cell lines and their killing was not affected by multidrug resistance. Moreover, the most potent hit, peptoid 1 which showed little selectivity in our *in vitro* screening system, greatly reduced tumor formations via intratumor injections in a human breast cancer xenograft mouse model without causing any noticeable weakness or loss of body weight in mice, suggesting the *in vivo* anticancer efficacy and good local tolerance of peptoids and also demonstrating their potential for development into clinically useful chemotherapeutics.

Our structure-activity studies revealed that it was critical for peptoids to have the cationic, amphipathic structure and to reach a certain chain length to obtain good potency against cancer cell lines. Peptoids needed to retain certain proportions of hydrophobic residues in their structures, namely a certain size of the hydrophobic arc in order to be potent. A typical ratio of cationic and hydrophobic residues per molecule was ~1:2 in peptoids with good potency. Increasing cationic residues (as in **1-NLys_{5,11}** with 1:1 cationic to hydrophobic ratio) or reducing cationic charges (as in **1-NHis_{1,7}**, with ~90% of NHis base in uncharged form around neutral pH) both weakened their activity. Moreover, using Nbtg which has a guanidinium head to potentially enhance membrane

interactions via hydrogen bonding instead of *N*Lys in peptoid 1 did not increase peptoid activity significantly, suggesting a minor role of hydrogen bonding in their membrane interactions. Instead, bulkier aromatic side chains in hydrophobic residues are found to be critical for peptoid potency. The cytotoxicity of **1**_{11mer}-*N*Leu was low and was used as the inactive control. Taken together, the results with our peptoids support current explanations for the actions of cationic, amphipathic peptides: cationic residues select for anionic cellular membranes via electrostatic interactions; hydrophobic regions lead to membrane disruptions.

To evaluate peptoid selectivity *in vitro*, MRC-5 and primary dermal fibroblasts were tested as normal control cells, and hemolytic activities of several peptoids were also cited from previous work (42). Though there are many reports about changed membrane properties of cells once they become cancerous (138, 142, 159, 163), it could be tricky for peptides or peptoids to distinguish the slight differences. In our study, we did find several peptoids with selectivity towards cancer cells, killing several cancer cell lines efficiently while being relatively safe to normal control cells and red blood cells in some concentration ranges, such as achiral monomer peptoid variants with less stable secondary structures. The hemolytic activities of peptoids are generally really low. These results prove the successful mimicry of anticancer peptides using non-natural peptoids.

The selectivity ratio of the peptoids in our *in vitro* screening system was not strikingly high. However, *in vitro* systems have limitations in mimicking *in vivo* environments. In addition, we did not observe any selectivity of doxorubicin in this *in vitro* screening system as the tested cancer cell lines and the normal control cells all divide and

proliferate *in vitro*. Promisingly, peptoid 1 which kills cancer cells, MRC-5 and primary dermal fibroblasts almost equally efficiently *in vitro* was found to be effective and well tolerated in our preliminary xenograft mouse model, without causing any observable toxicity in mice. Taken together, these anticancer peptoids are a very promising class of agents to be further evaluated *in vivo* for anticancer applications, and we understand that the *in vivo* toxicity associated with peptoids will depend on how peptoids are administrated.

There are several methods to further improve selectivity or reduce toxicity. First, we can engineer the peptoids at the molecular level by attaching “tumor homing” moieties to peptoids to enhance their accumulation in tumors and reduce their nonspecific interactions (164-166), or by developing peptoid-based prodrugs, the full activity of which need to be activated by tumor related enzymes (167, 168). Second, systemic toxicity of peptoids can be tuned by the careful choice of delivery methods. Mitomycin C, a DNA crosslinking agent with antitumor antibiotic activity is administrated as a single instillation within 6 hours of bladder tumor resection to minimize its toxicity and is proven to be effective in reducing recurrence (169). The nonselective melittin was incorporated into nanocarriers with favorable pharmacokinetics, named “nanobees”, and was selectively accumulated in multiple tumor targets, dramatically reducing tumor growth without causing any apparent signs of toxicity (170). Thus, a proper delivery method can be explored to further improve the *in vivo* behaviors of these peptoids.

Cationic, amphipathic peptides are generally believed to be membranolytic *in vitro*, accumulating on lipid membranes and subsequently disrupting the membrane structural integrity, either by forming pores in the membrane or acting like detergents and

dissolving the membrane altogether (159). We have observed membrane damage upon peptoid treatments, indicated by the Guava viacount assay and LDH assay, and there was a good correlation between immediate membrane damage and the cytotoxicity caused by peptoids. Interestingly, these peptoids can also penetrate into cells even at low concentrations. We even observed some co-localization of peptoids with mitochondria in the cytoplasm (data not shown). Therefore, we do not exclude potential intracellular targets of these peptoids, which is backed by the observation that longer peptoid treatments resulted in increased cell death at some concentration ranges. Altogether, we believe that these peptoids can accumulate on cell membranes and interact with membranes in a concentration dependent manner. At low concentrations, peptoids penetrate into cells in a way that does not disrupt plasma membranes or from which cells can recover, and they may disrupt intracellular organelles depending on their intracellular concentrations. At high concentrations, the way peptoids interact with plasma membranes causes membrane damage, which contributes to peptoids' observed, fast cytotoxicity.

Our preliminary *in vivo* experiment with peptoid 1 indicated the promising anticancer applications of these peptoids. Currently, the efficacy of the peptoid in bladder cancer and ovarian cancer mouse models is being evaluated. This group of cationic, amphipathic peptoids, because of their broad spectrum of anticancer activity, unique modes of action, resistance to MDR and stability towards protease degradation, has a great potential to complement existing chemotherapy.

2.5 Conclusion

Cationic, amphipathic host defense peptides represent a promising group of agents to be developed for anticancer applications. Poly-*N*-substituted glycines, or peptoids, are a class of biostable, peptidomimetic scaffold that can display a great diversity of side chains in highly tunable sequences via facile solid-phase synthesis. Herein, we present a library of anticancer peptoids that mimics the cationic, amphipathic structural feature of the host defense peptides and explore the relationships between the structure, anticancer activity and selectivity of these peptoids. Several peptoids are found to be potent against a broad range of cancer cell lines at low-micromolar concentrations including cancer cells with multidrug resistance (MDR). These peptoids interact with cells in a concentration-dependent manner. They can penetrate into cells, but their killing also involves plasma membrane perturbations. Furthermore, peptoid 1, the most potent peptoid synthesized, significantly inhibited tumor growth in a human breast cancer xenotransplantation model without any noticeable toxicity in mice. Taken together, these cationic, amphipathic peptoids are very attractive for anticancer applications due to their high solubility, stability against protease degradation, their broad, potent cytotoxicity against cancer cells and their ability to overcome multidrug resistance.

Chapter 3

**Cationic, amphipathic peptoid transporters:
structures determines their cellular uptake
mechanisms**

3.1 Introduction

Cell-penetrating peptides are a group of molecular transporters that can be used to facilitate the cellular uptake of drugs, drug candidates or probes (43-45, 48, 171, 172). The advantages of these peptides lie in their highly efficient translocation capacities in delivering cargos into a variety of cell lines *in vitro*, as well as their promising applications in delivering biologically active molecules into cells *in vivo* (48). They are typically short cationic sequences, either derived from natural sources or designed synthetically. They can be composed of just cationic residues, like polyarginines and other synthetic guanidinium-rich transporters (45), or have a cationic, amphipathic structure with a large fraction of basic residues and some hydrophobic residues, such as pAntp-(43-58) (*Drosophila antennapedia* transcription protein) (173), Transportan (174), and pVEC-(615-632) (murine vascular endothelial cadherin) (175). The mechanisms by which cell penetrating peptides enter mammalian cells have not been elucidated in all the specific details (44, 45). It is generally accepted that both energy-dependent endocytosis and energy-independent direct translocations are involved. It has also been suggested that structural properties of peptides as well as properties of the associated cargos can have a significant impact on the adopted internalization mechanisms, and that a single cell-penetrating peptide may use multiple modes of cellular entry (173, 176).

Cell-penetrating capabilities have also been observed with some host-defense peptides(61, 62, 177, 178), which can transport cargos into mammalian cells, such as KLA-derived α -helical model amphipathic peptides (MAPs)(56, 179), magainin 2(180), buforin 2(180), LL37(181), and *etc.* In general, host-defense peptides have a cationic, amphipathic structure and are strongly active on negatively charged membranes and/or

zwitterionic membranes. The barrel-stave, inverted micelle and carpet models have been proposed for their membrane perturbations(145, 159). The membrane-crossing host-defense peptides were found with extensive cellular uptake without a necessity for permanent membrane permeabilization at least during some low concentration ranges. This has led to interests of employing membrane-crossing host-defense peptides as transporters or as templates to design novel vectors(61, 62). A major concern is the cytotoxicity associated with cationic, amphipathic molecules. Even the frequently used cell-penetrating peptides, such as pAntp(43-58) and pVEC(615-632), also harm cell viability at higher concentrations (173, 182, 183). Therefore, to design effective cationic, amphipathic transporters, their toxic effects should be kept at a minimum when killing the cells is not the purpose.

One limitation of peptide-based transporters is their rapid degradation and clearance *in vivo*. Peptidomimetic scaffolds can be used to circumvent the proteolytic sensitivity. Poly-*N*-substituted glycines, or peptoids, in which side chains are attached to the backbone nitrogen rather than to the α -carbon as in peptides, is one of the most popular class of protease-resistant peptidomimetics to design molecular transporters (29, 88, 152). Polyguanidine peptoids(88), lysine-like polypeptoids(91, 92, 184), α -peptide/ β -peptoid chimeras (93) have been developed as efficient transporters. Tan and coworkers also demonstrated that peptoids were anywhere from 3- to 30-fold more permeable than the analogous peptide in their library depending on the structures, and they explained the enhanced permeability was from the stability of peptoids(185).

Peptoids can be easily synthesized on solid phase on an automated peptide synthesizer (152). Virtually any desired chemical functionality available as a primary

amine can be incorporated into peptoids via a two-step submonomer method (Figure 1A), which provides great chemical “design diversity” in peptoid sequences. Peptoids can form helical structures with a periodic incorporation of α -chiral side chains, giving rise to polyproline type-I-like helices with approximately three-residue per turn and a helical pitch of 6.0 – 6.7 Å (40). This threefold periodicity of peptoids allows the cationic, amphipathic structure to be easily recapitulated in three-faced helices, simply using peptoids comprising trimer repeats, (X-Y-Z)_n (30, 40). Accordingly, our group has successfully developed several cationic, amphipathic peptoids for antimicrobial and anticancer applications (30, 40, 186). In the work we present here, we designed a library of cationic, amphipathic peptoids to be used as molecular transporters and investigated how the peptoid structures may influence their cellular uptake as well as their toxicity on mammalian cells.

3.2 Materials and methods

3.2.1 Peptoid synthesis and purification

Peptoids were synthesized using an ABI 433A peptide synthesizer (Applied Biosystems, Inc.) on Rink amide MBHA resin (EMD Biosciences, Gibbstown, NJ) using the submonomer protocol. Briefly, the amine on the nascent chain is bromoacetylated followed by S_N2 displacement of bromide by a primary amine to form a side chain(17, 30). These two steps were iterated with appropriate amines until the desired peptoid sequence was obtained. Resin-bound peptoids were then exposed to a mixture of trifluoroacetic acid (TFA): triisopropylsilane: water (95:2.5:2.5, volume ratio) for 10 minutes to cleave peptoids from the resin. Crude peptoids were purified by reverse-phase high performance liquid chromatography (RP-HPLC) (Waters Corporation) using a C18

column and a linear acetonitrile/water gradient. A final purity > 95% as measured by analytical RP-HPLC (Waters Corporation) was achieved, and the identity of each peptoid was confirmed using electrospray ionization mass spectrometry (ESI/MS). Unless indicated otherwise, all reagents were purchased from Sigma Aldrich (St. Louis, MO). Among the submonomers used, *Nspe* was derived from (*S*)-*N*-(1-phenylethyl)amine; *Npm* from benzylamine (1-phenylethylamine); *Nssb* from (*S*)-(*sec*-butyl)amine; *NLeu* from isobutylamine; *NLys* from *N*-*tert*-butoxycarbonyl-1,4-butanediamine (CNH Technologies, MA). Guanidinylation of *NLys* was carried out according to the reported procedure(88). 5(6)-Carboxyfluorescein was used to label the N-terminus of all the constructs on solid-phase (154) .

RP-HPLC traces and ESI data of construct **1-11** are summarized in Figure S1 and S2.

3.2.2 Cell cultures

MCF-7 cells were kindly provided by Professor Branimir Sikic's lab at Stanford University, and were cultured in McCoy's 5A media (GIBCO) with 10% FBS (Hyclone, US sources) and antibiotics(Sigma). Cells were cultured in cell incubators (Thermo Scientific) at 37°C with 5% CO₂.

3.2.3 The MTS assays

Aliquots of 100μl media containing 1×10^4 cells were distributed into each well of a 96-well plate (BD Falcon). The following day, when cell density reaches about ~40% confluency, the cell media were removed and replaced with serial dilutions of peptoid stocks in culturing media. For peptoid dilutions, peptoid stocks were initially diluted in media at 100μM and then diluted by half in series using a multichannel pipette, and maintained in 100μl media for each concentration with triple repeats. After peptoid

solutions were transferred onto cells in 96-well plates, cells were incubated at 37°C for certain time periods. Then 20µl of the CellTiter 96[®] Aqueous Non-Radioactive cell proliferation assay (Promega) reagent which contains a tetrazolium compound, [3-(4,5-dimethylthiazol-2-yl)-5-(3-carboxymethoxyphenyl)-2-(4-sulfophenyl)-2H-tetrazolium, inner salt; MTS(a)], was added to each well and cells were further incubated for 2h to metabolize. The absorbance of formazan products were measured at 490nm in microplate reader (Molecular Devices). Percentage of cell viability = $(A - A_{\text{testblank}})/(A_{\text{control}} - A_{\text{blank}}) \times 100$, where A is the absorbance of the test well and A_{control} the average absorbance of wells with cells not treated with peptoids. $A_{\text{testblank}}$ (media, MTS, and diluted peptoids) and A_{blank} (media and MTS) were background absorbances measured in the absence of cells (30, 187).

3.2.4 The Guava flow cytometry cellular uptake assay

All the fluorescently labeled constructs were dissolved in water as stock solutions, and their concentrations were further matched by their absorbances of fluorescein at 490nm. Cells were plated as described in MTS assays. Constructs were diluted in McCoy's 5A media with 10% FBS at desired concentrations. MCF-7 cells were incubated with constructs for certain time period. Cells incubated with 5(6)-Carboxyfluorescein dissolved in DMSO was measured as the control. After the supernatant was removed, cells were washed with PBS for three times, each wash incubated at 37°C at 5min. Cells were then trypsinized and suspended in fresh media. Cellular fluorescent signals were quantified in Guava EasyCyte Plus[®] flow cytometry system (Millipore). Each sample was repeated three times, and the data presented are the mean fluorescent signal for 1500 cells.

3.2.5 The temperature study

The uptake assays were performed as described above with the exception that for the 4°C experiment, the cells and all the reagents were pre-incubated at 4°C for 30min. Cells were incubated with constructs for 1h at either 37°C or 4°C before flow cytometry analysis (180).

3.3 Results and discussion

3.3.1 Library design and quantifications of cellular uptake and cytotoxicity

We recently found that a peptoid (construct **4** in Table 3.1) we designed for antimicrobial and anticancer applications, [H-(NLys-Nspe-Nspe)₄-NH₂], composed of NLys (the peptoid analogue of lysine) and Nspe (the peptoid analogue of phenylalanine, Figure 3.1), can translocate into mammalian cells efficiently at low concentrations without causing observable cytotoxicity (186). As shown in Table 3.1, several cationic, amphipathic peptoids (construct **3-11**) were designed with different lengths, charges and hydrophobicity. Polyarginines, Arg₈ (construct **1**) and a polyguanidine peptoid (Nbtg₈, construct **2**) were tested as polyguanidine comparators with their charge-to-length ratio (CTRLR) at 1. To quantify their cellular uptake, all constructs were labeled with 5(6)-Carboxyfluorescein (CF) on the N-terminus without a spacer (Figure 3.1C). Therefore, all the studies and comparison described herein were about the cellular uptake of transporters conjugated to small molecules (MW ca. <3000), and the results do not preclude different phenomena in delivering larger cargos.

The sequences of the fluorescently labeled variants, the solvent composition at RP-HPLC elution as a relative measure of molecular hydrophobicity, molecular net charges, the CTRLR and the lethal concentration (LC₅₀) in MCF-7 (human breast cancer) were

summarized in Table 3.1. Peptoid monomers were illustrated in Figure 3.1B. Their purity and molecular weight characterizations were summarized in Figure 3.2 and Figure 3.3. LC₅₀ of the construct in MCF-7 cells was estimated from the cell viability curve for 24 h treatment and cell viability was measured with the MTS assays. Cellular uptakes of these constructs were quantified using Guava Easycyte Plus® flow cytometry system (Millipore). MCF-7 cells were incubated with 8 μ M of each construct in culturing media (10% FBS) for 1 h at 37 °C. Cells were washed thoroughly and trypsinized before the flow cytometry analysis. Cellular uptakes of construct **1-11** were compared in Figure 3.4.

	Construct	Sequence	MW (Da)	HPLC elution (% ACN)	Net Charge	CTLR	LC ₅₀ (μ M) (MCF-7)
Polyguanidine comparators	1	CF-(Arg) ₈ -NH ₂	1624	43.8	+8	1	80
	2	CF-(Nbtg) ₈ -NH ₂	1737	44.9	+8	1	>100
NLys(1/3) variants	3*	CF-(NLys-Nspe-Nspe) ₃ -NH ₂	1727	73.2	+3	0.33	67
	4*	CF-(NLys-Nspe-Nspe) ₄ -NH ₂	2177	76.0	+4	0.33	12
	5*	CF-(NLys-Npm-Npm) ₄ -NH ₂	2065	69.2	+4	0.33	38
	6*	CF-(NLys-Nssb-Nssb) ₄ -NH ₂	1905	73.0	+4	0.33	>100
NLys(2/3) variants	7	CF-(NLys-NLys-Nspe) ₄ -NH ₂	2045	52.5	+8	0.66	>100
	8	CF-(NLys) ₈ -(Nspe) ₄ -NH ₂	2045	60.0	+8	0.66	48
	9	CF-(NLys) ₈ -(NLeu) ₄ -NH ₂	1853	53.5	+8	0.66	>100
Guanidinylated variants	10*	CF-(Nbtg-Nspe-Nspe) ₄ -NH ₂	2345	75.5	+4	0.33	11
	11	CF-(Nbtg-Nbtg-Nspe) ₄ -NH ₂	2381	54.8	+8	0.66	50

Table 3.1 Sequences and molecular properties of fluorescently labeled cell-penetrating constructs. See Figure 3.1 for the structures of the peptoid monomers indicated in each sequence. HPLC elution is reported as the percentage of compound elution in the water/acetonitrile (0.1% trifluoroacetic acid) solvent in analytic HPLC. A linear water/acetonitrile (0.1% trifluoroacetic acid) gradient of 5%-95% acetonitrile over 30min was run on a C18 column. Net charge indicates molecular charges at neutral pH.

CTLR stands for charge-to-length ratio, which is defined as the ratio of the total number of cationic residues to the total number of residues in each sequence. LC₅₀ means lethal concentrations causing 50% death of the MCF-7 cells. Cell viability was measured via MTS assay in cells treated with compounds for 24h. Peptoids labeled with asterisk have been reported previously in our antimicrobial and anticancer activities(30, 40, 42, 186).

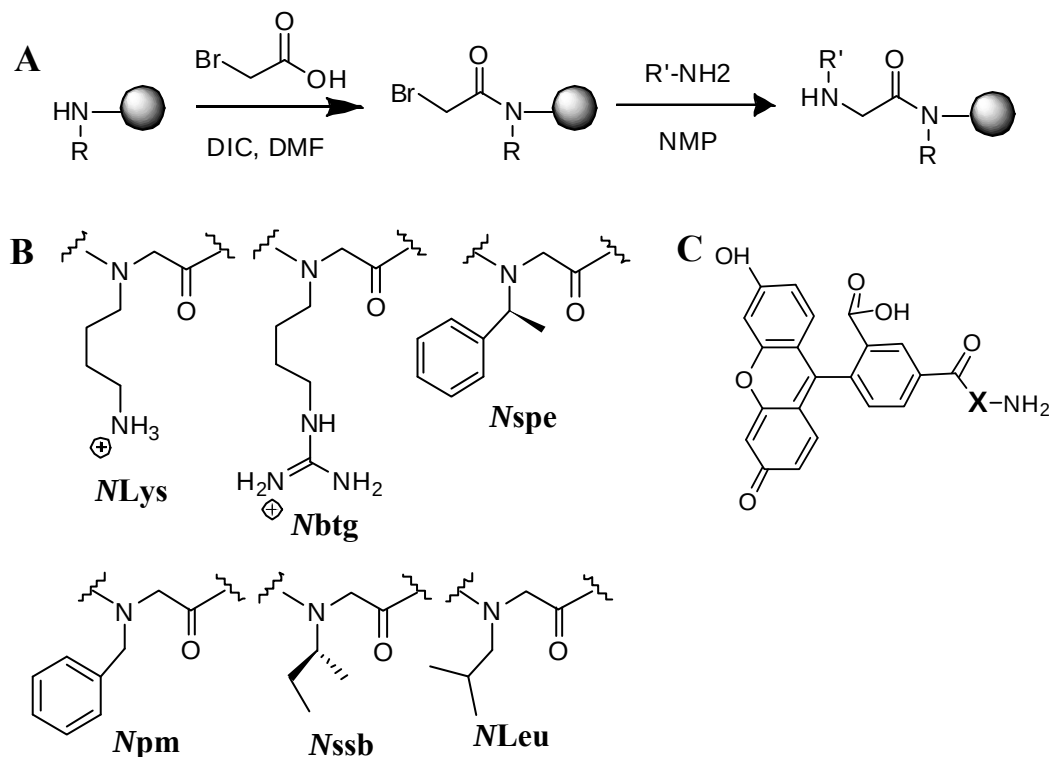
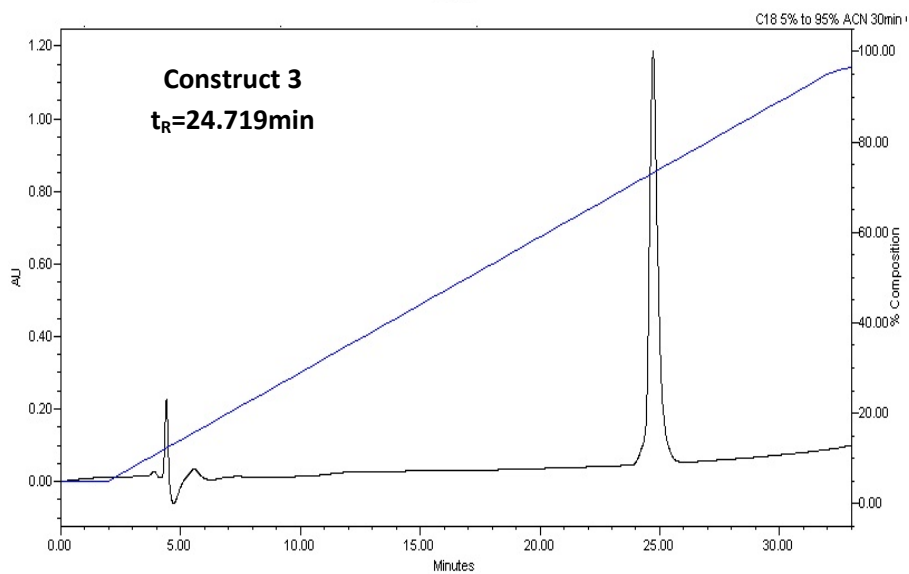
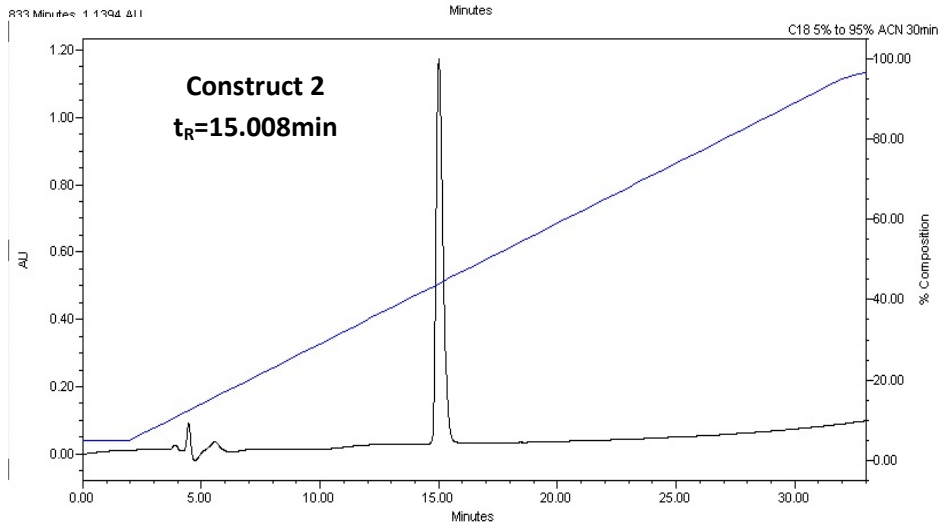
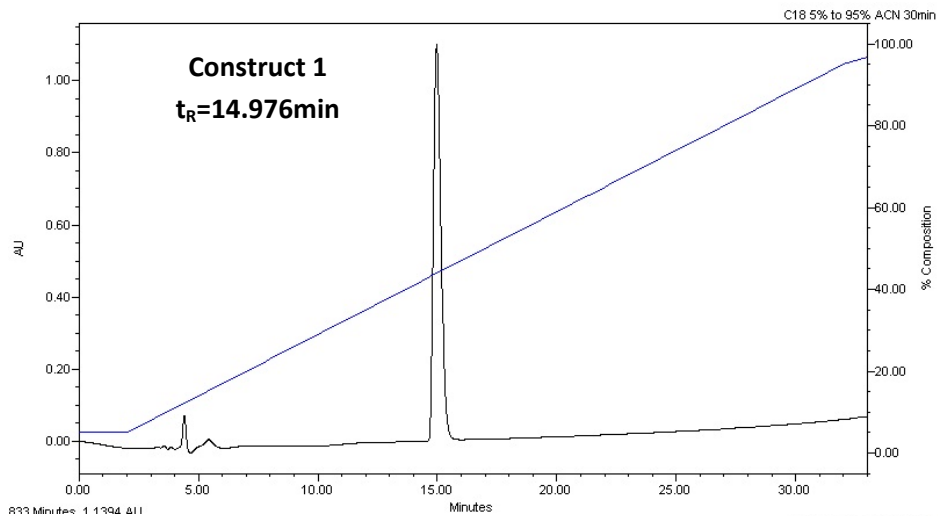
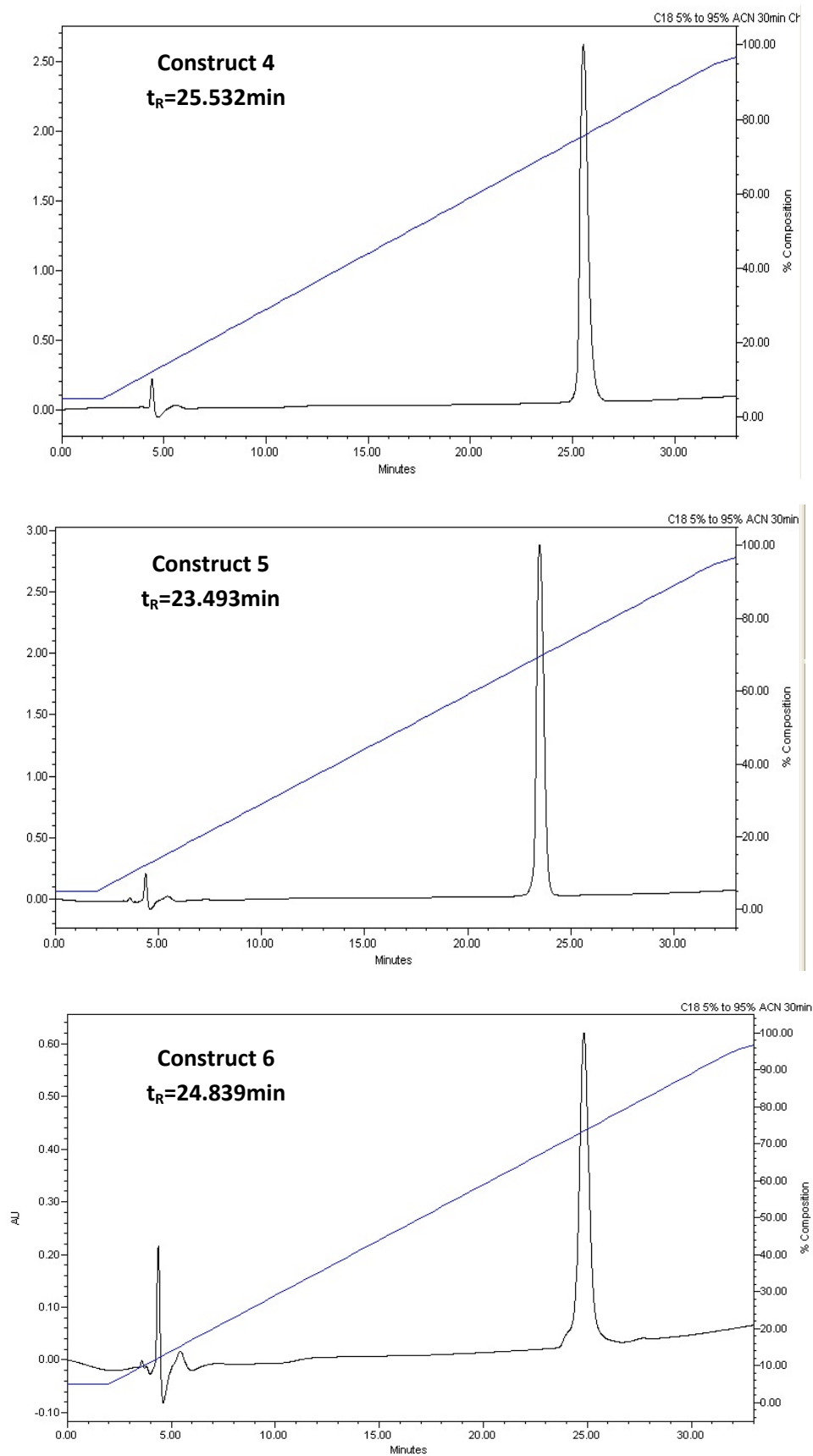
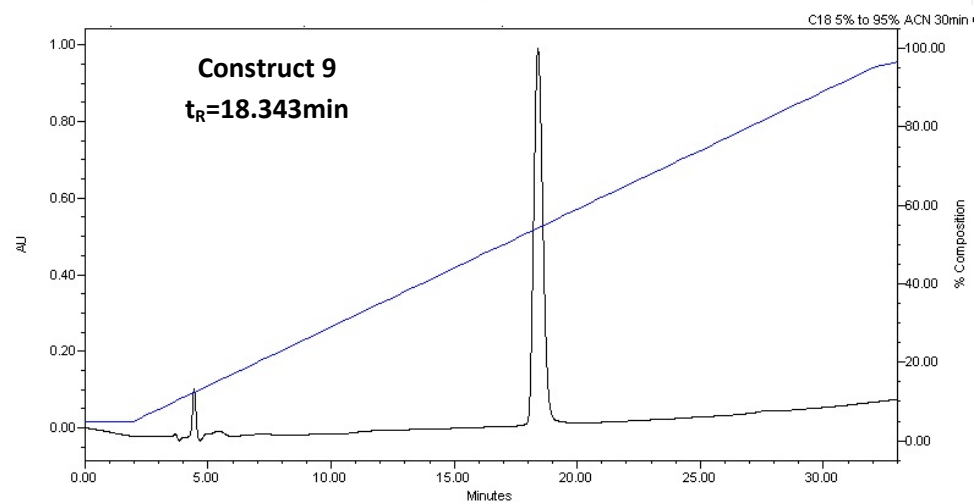
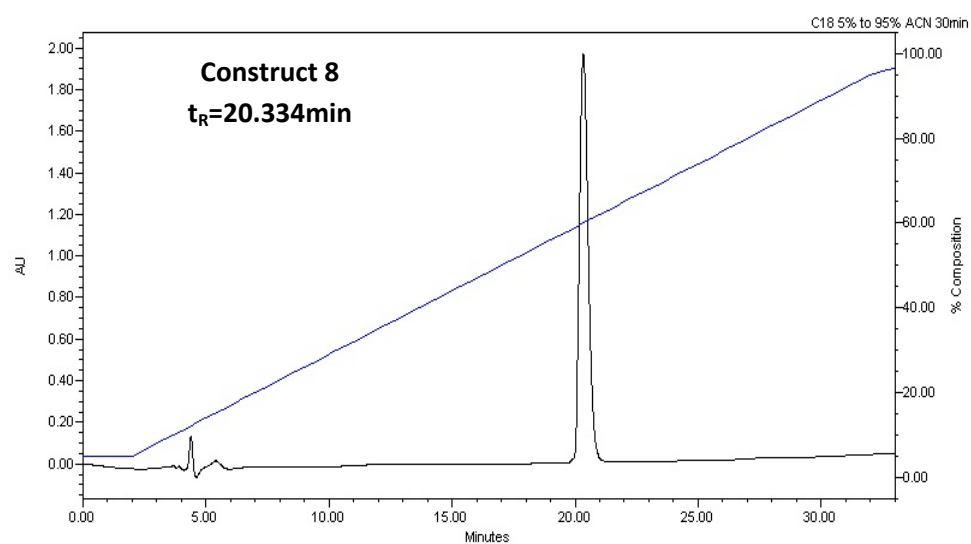
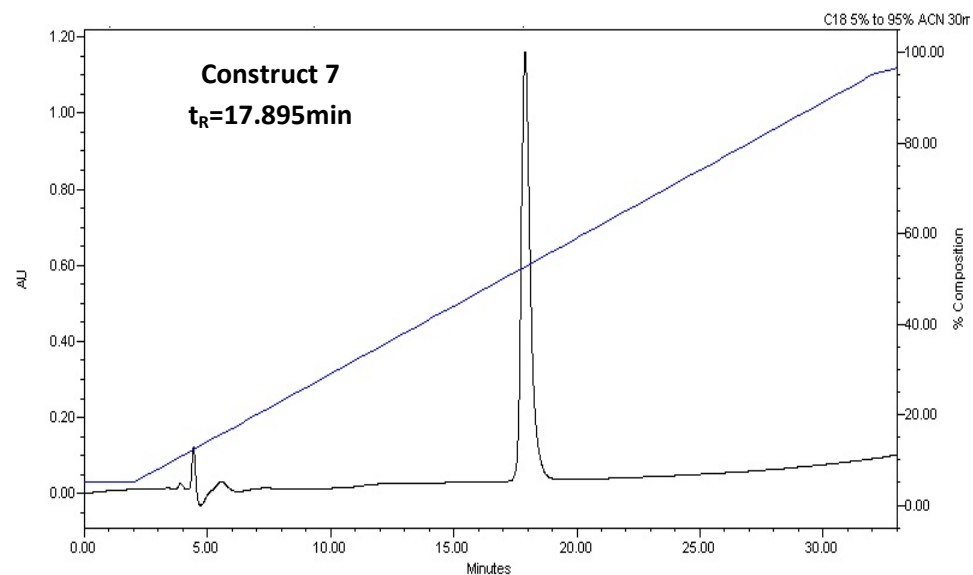


Figure 3.1 The peptoid synthesis and construct structures. A, the submonomer synthesis scheme of peptoids(17); B, peptoid monomer side-chain structures with shorthand names; C, the chemical structure of cell penetrating peptide or peptoids (X) labeled with 5(6)-Carboxyfluorescein (CF) on the N-terminus.







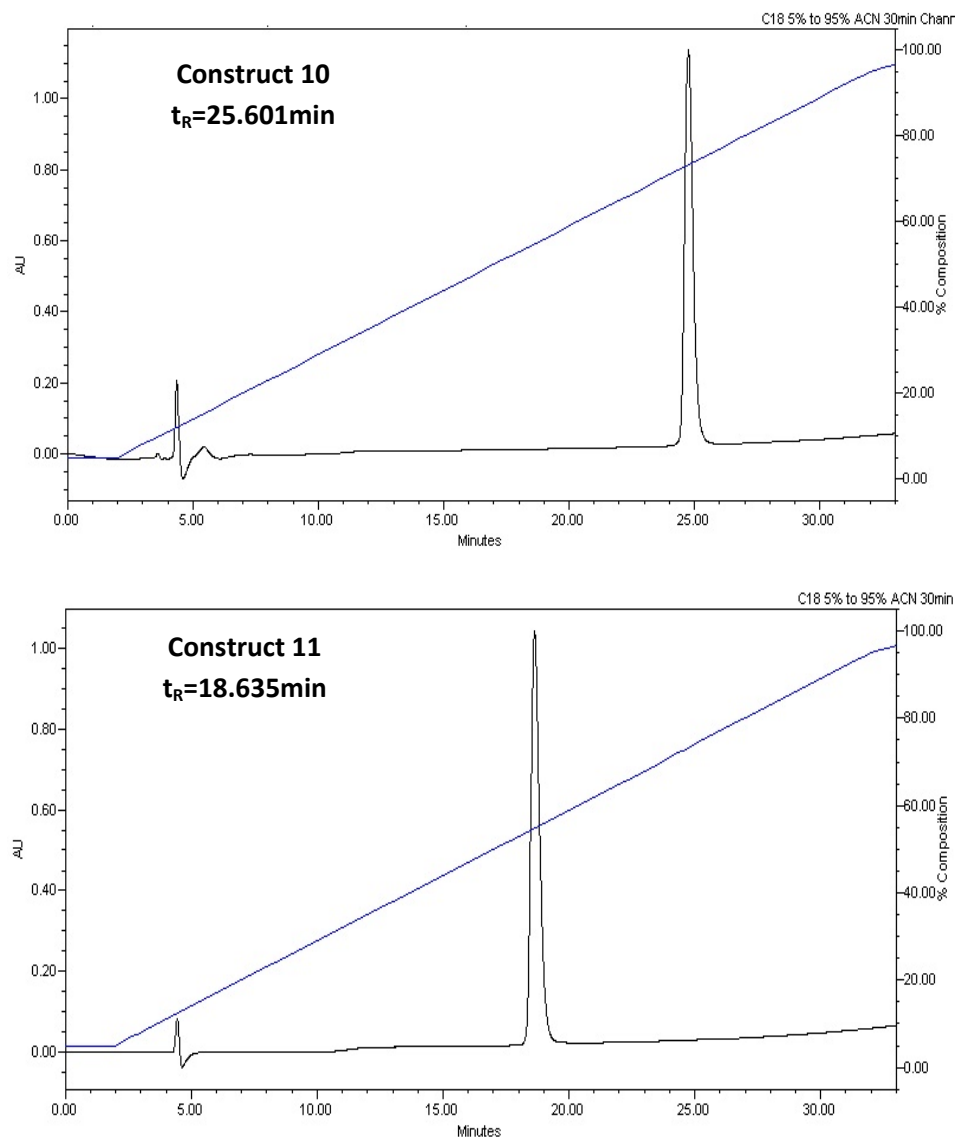
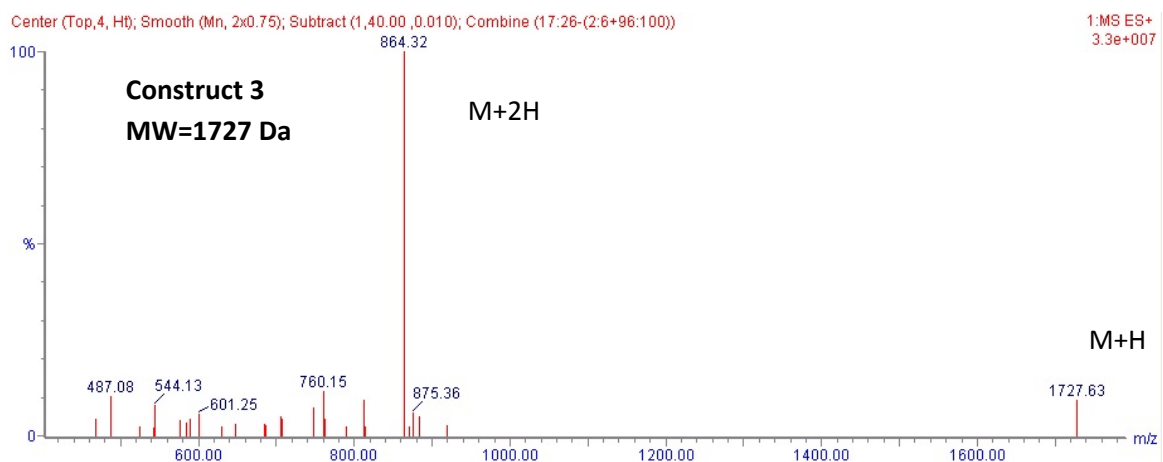
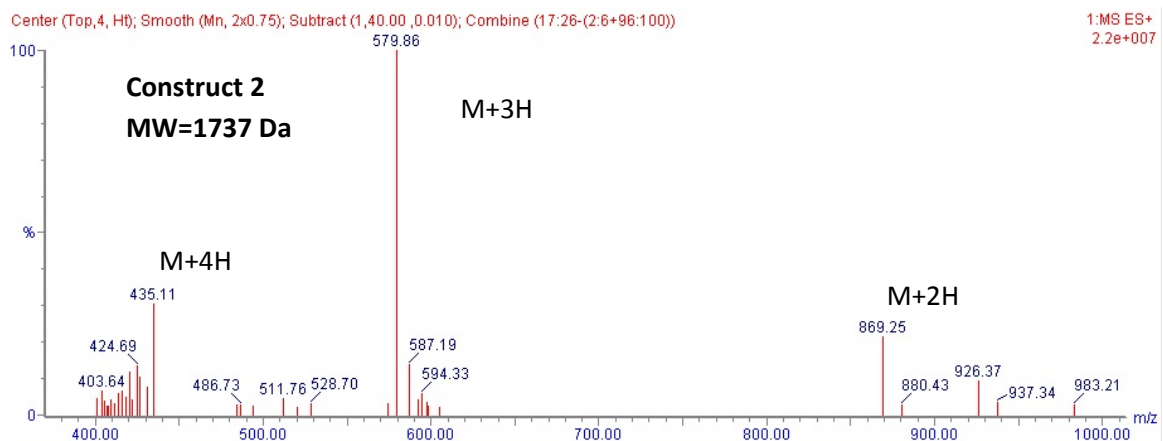
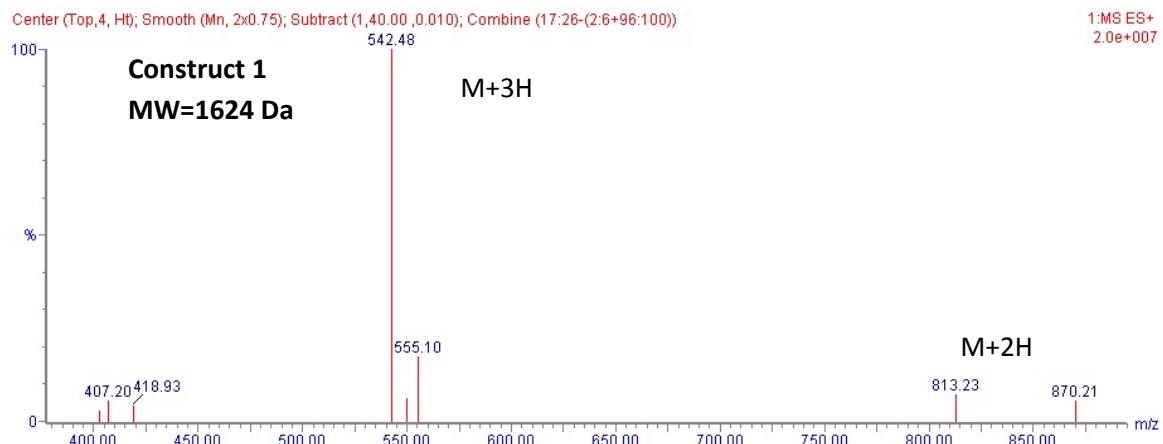
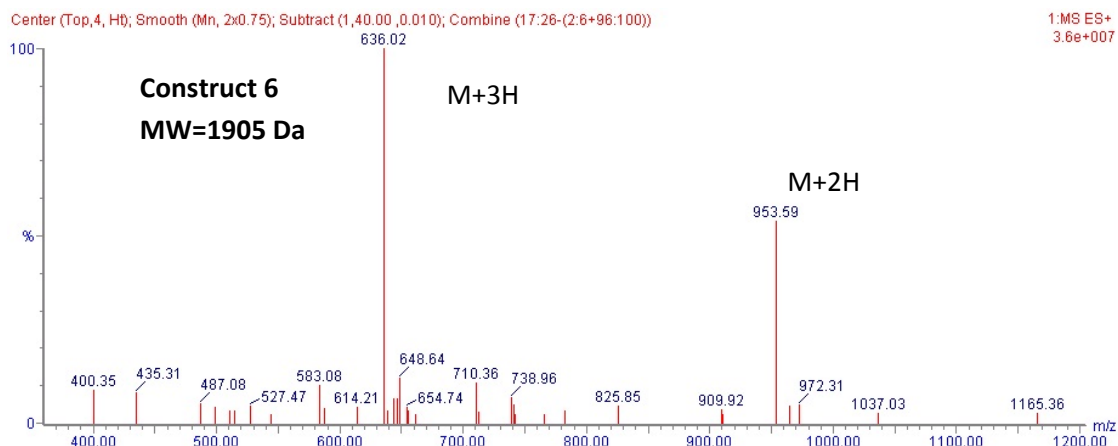
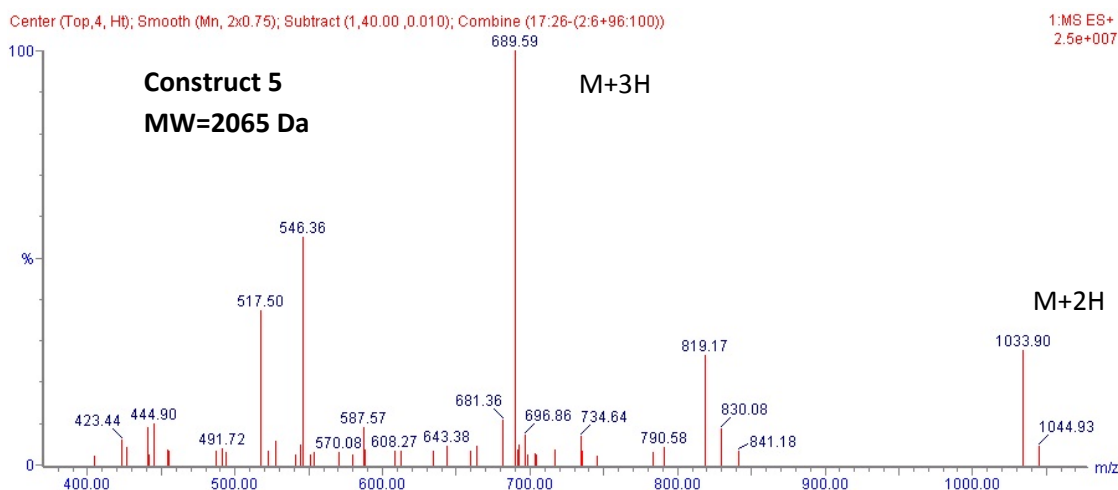
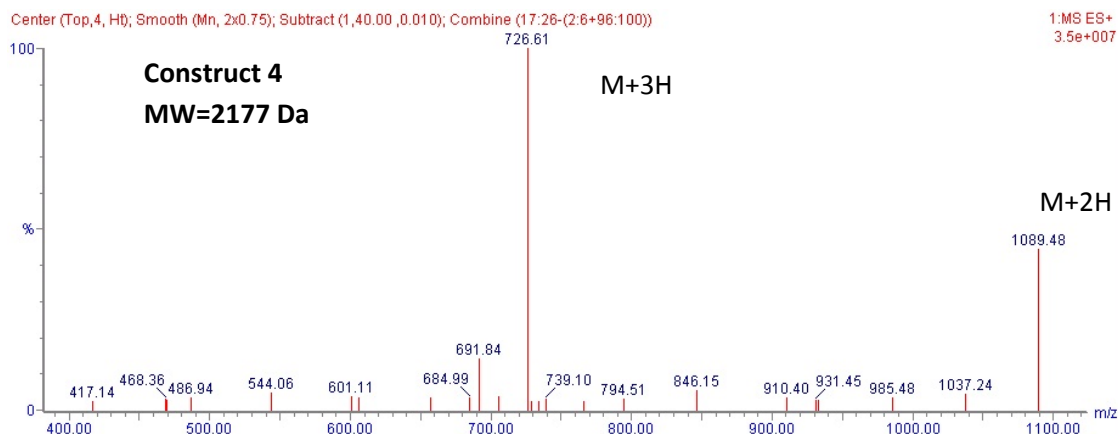
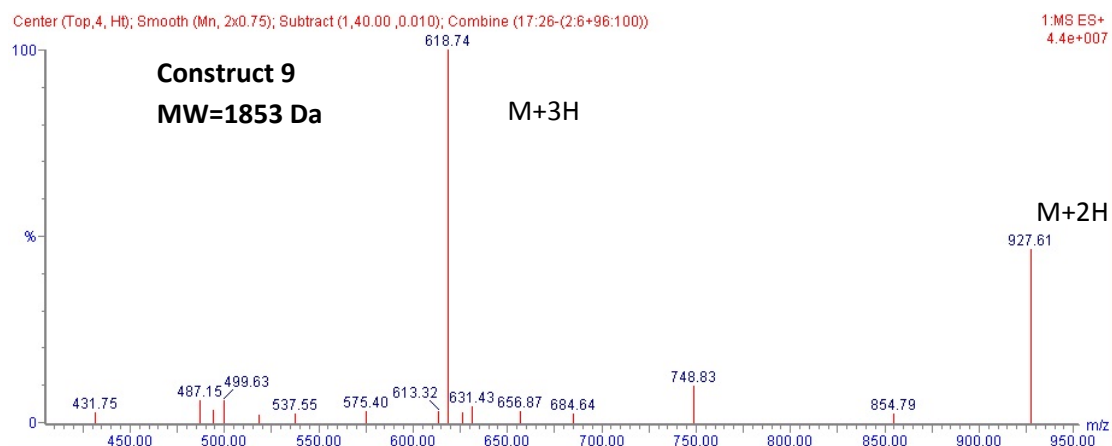
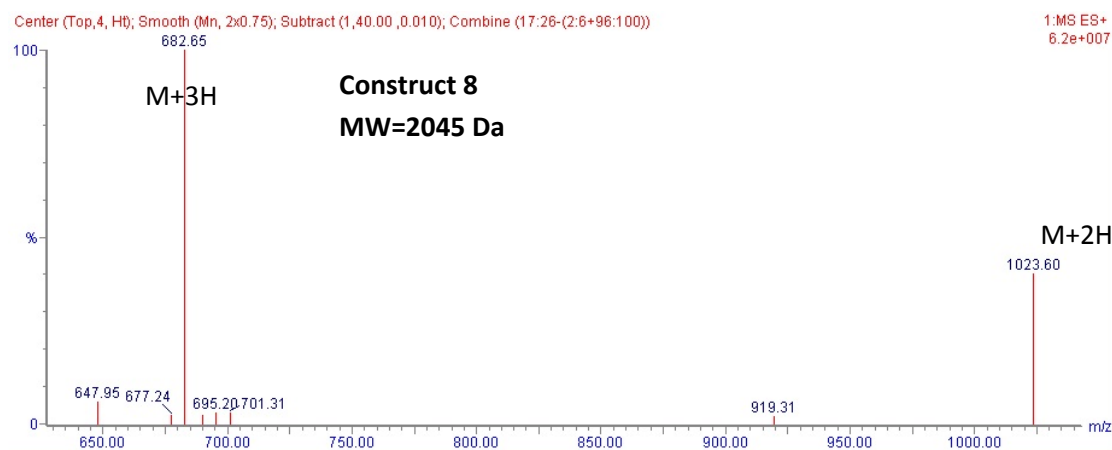
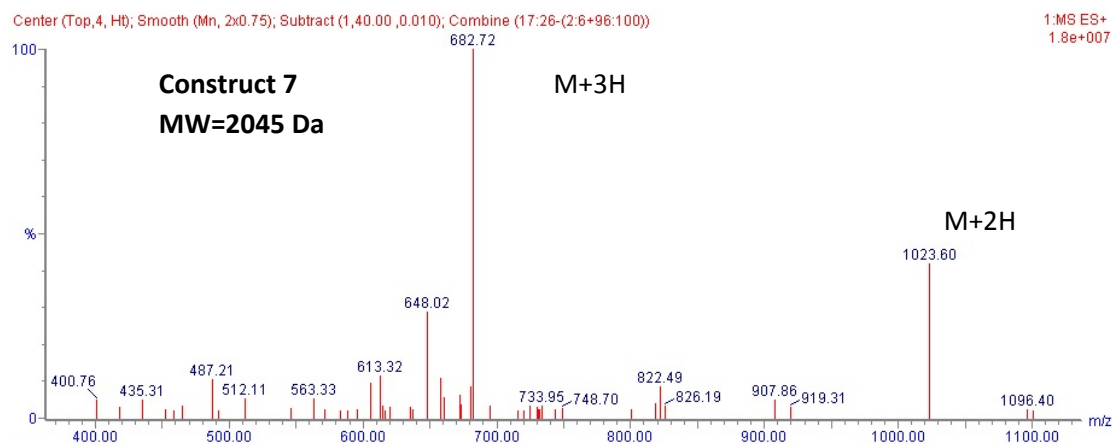


Figure 3.2 Analytical RP-HPLC traces of construct 1-11, measured at 220nm.
The retention time (t_R) of each construct was labeled in the chromatogram.







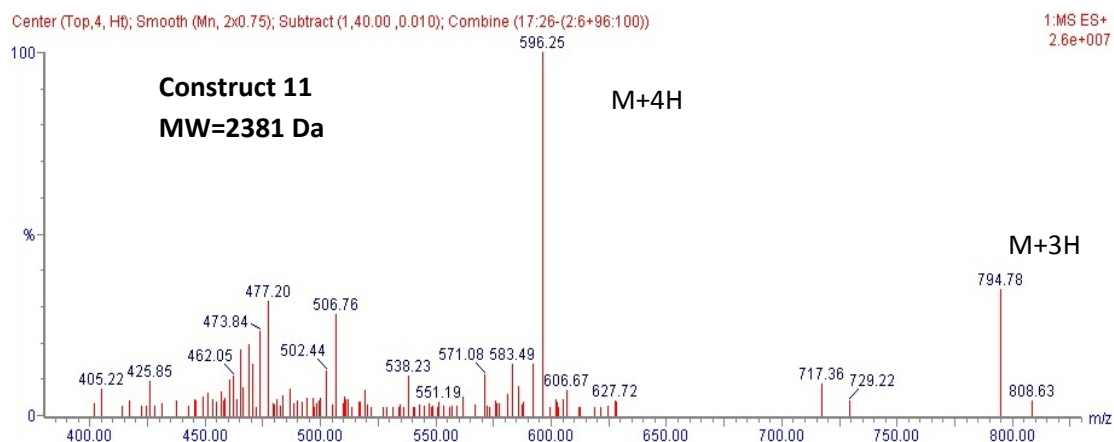
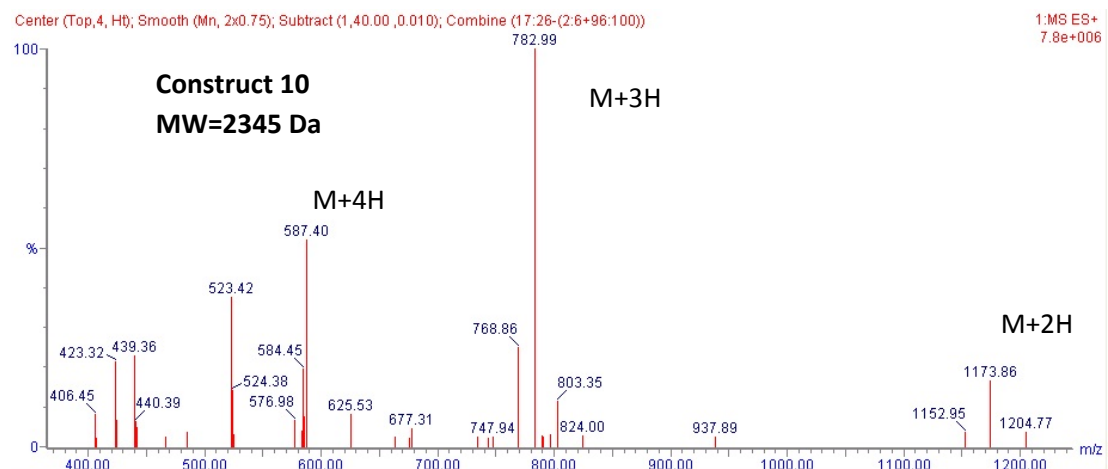


Figure 3.3 ESI data of construct 1-11. Construct molecular weight (MW) and the corresponding peaks were indicated in the mass spectrum.

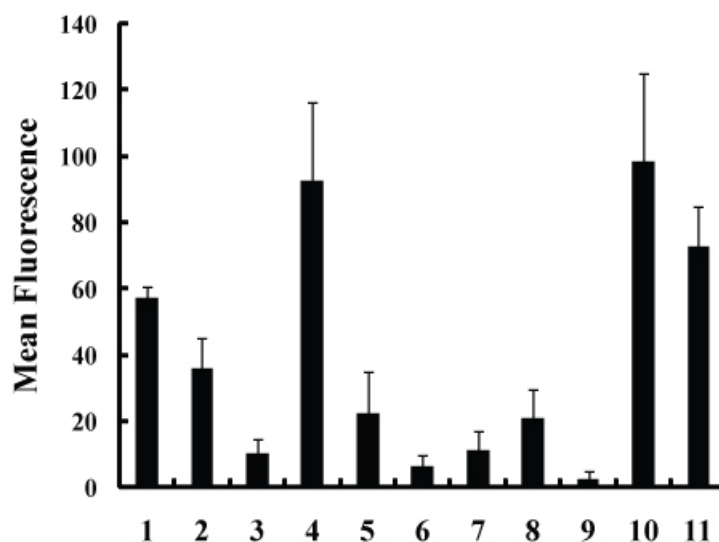


Figure 3.4 The flow cytometry cellular uptake assay of construct 1 to 11. MCF-7 cells were incubated with 8 μ M of each construct diluted in culturing media (10% FBS) for 1 h at 37 °C. Cells were washed thoroughly with PBS and trypsinized, and cellular fluorescent signals were quantified on Guava Easycyte Plus® flow cytometry system (Millipore).

3.3.2 Structure-Activity Relationship Studies

Polyguanidine comparators (construct **1** and **2**) were observed with efficient uptakes (Figure 3.4) and minor cytotoxicity (Table 3.1). The *N*Lys(1/3) variants, construct **3-6**, which have one *N*Lys every three residues with CTLR at 0.33, have been reported previously in our antimicrobial and anticancer studies. Consistent with previous findings, the cytotoxicity of construct **4** comprising *N*spe as the hydrophobic residues, **5** (with *N*pm) and **6** (with *N*ssb) decreased as the bulkiness and hydrophobicity of their hydrophobic residues decreased, indicated with their increased LC_{50} . Moreover, the cytotoxicity of construct **3** (a 9mer) was much lower than the 12mer analogue construct **4** (Table 3.1). Interestingly, the cytotoxicity of construct **3-6** decreased as their uptake efficiencies

decreased (Figure 3.4). Construct **4** with the most efficient uptake was found with the highest cytotoxicity.

The *N*Lys(2/3) variants, construct **7-9**, have 8 *N*Lys per molecule with CTLR at 0.66. Compared to construct **4**, construct **7** which has an increased number of *N*Lys and thus reduced molecular hydrophobicity displayed dramatically decreased cellular uptakes and cytotoxicity. Interestingly, construct **8** with the same composition as construct **7** but with segregated *N*Lys's and *N*spe's were found with higher cytotoxicity and higher cellular uptakes. Construct **9** with nonaromatic residue *N*Leu were basically found to be inactive.

Guanidinium heads have been reported to enhance the membrane interactions of cell penetrating peptides through hydrogen bonding. Construct **10** is the guanidinylated analogue of construct **4**, and construct **11** is the guanidinylated analogue of construct **7**. We found that with CTLR at 0.33, guanidylation did not have obvious improving effects, suggested by the similar cellular uptakes and cytotoxicity of construct **4** and **10**. However, for peptoids with CTLR at 0.66, guanidylation dramatically increased cellular uptakes. As shown in Figure 3.4, the uptake efficiency of construct **11** was much higher than construct **7**'s, so was the cytotoxicity, which indicated stronger interactions with membranes contributed by guanidinium heads. In our structure-activity studies, both guanidinium heads and bulky, aromatic hydrophobic residues can make the cationic, amphipathic constructs more permeable. Construct **4** with CTLR at 0.33 was found with the most efficient uptake, good water solubility and ease of synthesis but also the highest cytotoxicity. Construct **11** which had multiple guanidinium heads with CTLR at 0.66 also had a strong uptake but was much safer to the cells.

3.3.3 Internalization mechanism Studies

After the efficiency and cytotoxicity screening of cationic, amphipathic constructs, we wanted to compare the uptakes of construct **4** and **11** with polyguanidine comparators **1** and **2**. The time dependence of the construct uptake was shown in Figure 3.5. We found that the longer incubation gradually increased the uptake of peptoid constructs, **4**, **11** and **2**, up to 18 h we tested. The time course of **1** (Arg₈) was different: its initial uptake was slower than the peptoid constructs, but it reached the maximum around 4 h, and then gradually decreased. The decrease is likely to be caused by peptide degradation. Their intracellular distributions were further compared with live cell confocal laser microscopy. The distributions of these two groups were distinct from each other. Cationic, amphipathic peptoids **4** and **11** had a punctuated distribution. Construct **4** was mainly present in the cytoplasm (Figure 3.6A), and **11** also displayed some nuclear distributions (Figure 3.6B). In contrast, **1** (Figure 3.6C) and **2** (Figure 3.6D) had a diffusive distribution throughout the cells. In the temperature studies, the cellular uptake of **4** was significantly reduced at 4 °C compared to 37 °C (Figure 3.7). The uptake of **11** was greatly reduced at 4 °C as well but it still displayed a certain degree of uptake. The data suggested that the translocation of **4** and **11** were either energy-dependent or very sensitive to the membrane fluidity which would have greatly changed at 4 °C. In contrast, at 4 °C, the uptake of **1** and **2** was not reduced, indicating an energy-independent internalization mechanism for polyguanidine comparators. The uptake of **1**, R₈, was even increased at 4°C. The intracellular distribution and temperature studies indicated the distinct mechanisms employed by polyguanidine constructs and cationic, amphipathic constructs.

In summary, we demonstrated the structures of cationic, amphipathic peptoids influence greatly on their cellular translocation and cytotoxicity. Two constructs (**4** and **11**) were found to have higher cellular uptakes than Arg₈ and the peptoid analogue of Arg₈. Different intracellular distribution patterns and different uptake behaviors in the temperature study suggested that distinct internalization mechanisms were employed by cationic, amphipathic constructs and polyguanidine constructs. Our findings can help understand the structural influence on the internalizations of cell-penetrating peptides and peptidomimetics, and should provide guidance in the design of new cell-penetrating transporters.

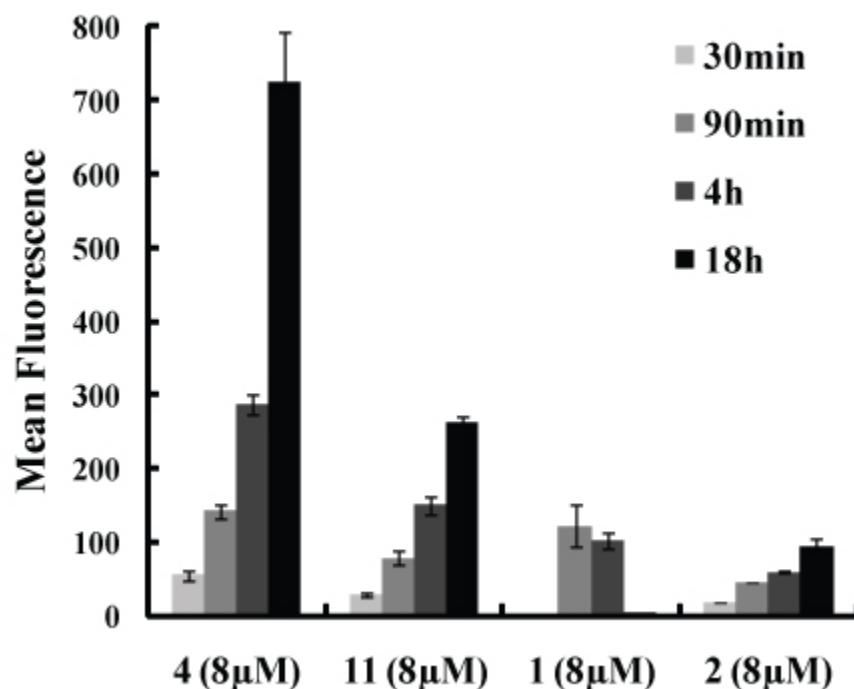


Figure 3.5 The time-dependent cellular uptake of construct 4, 11, 1 and 2. MCF-7 cells were incubated at 37°C with 8μM of each compound diluted in culturing media

(10% FBS) for 30min, 90min, 4h and 18h, before the cellular uptake analysis on Guava Easycyte Plus® flow cytometry system.

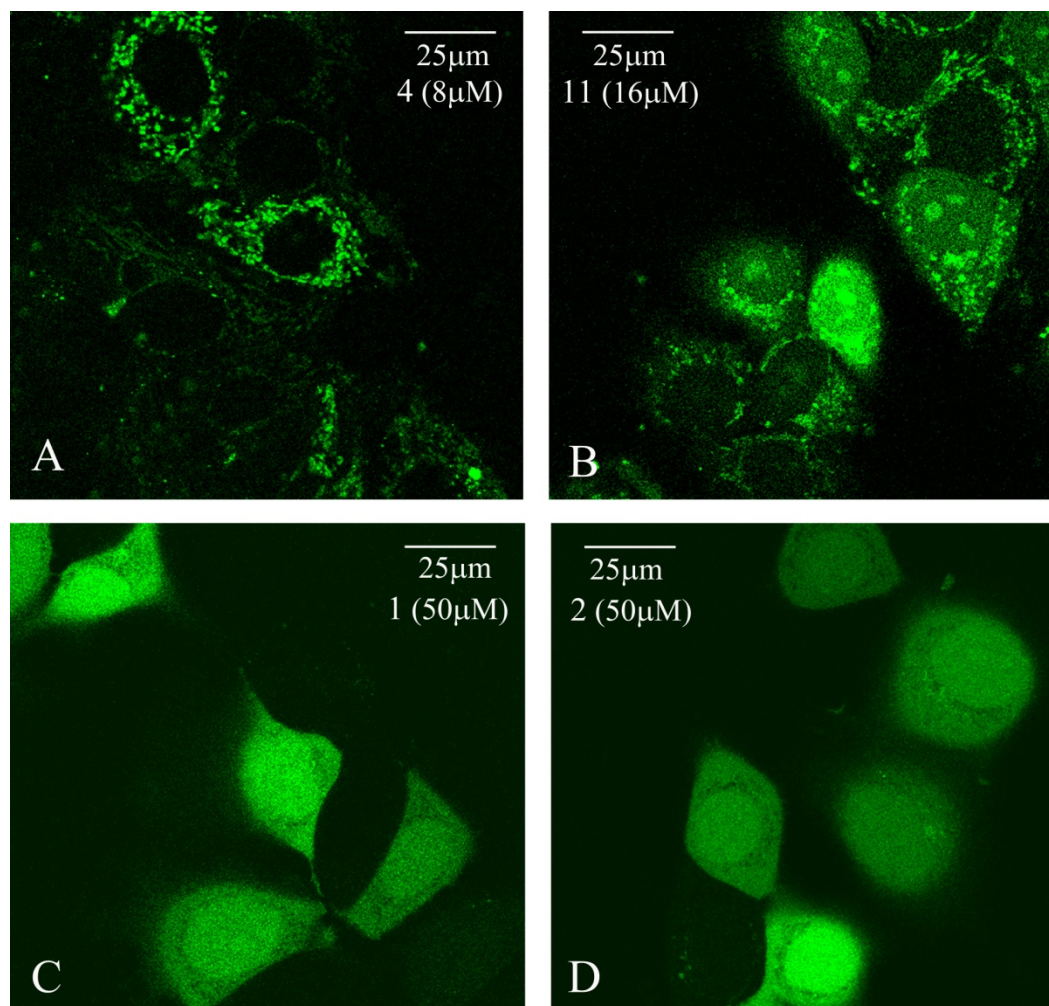


Figure 3.6 The intracellular distribution of construct 4, 11, 1 and 2 as examined by live cell confocal laser microscopy. MCF 7 cells were incubated at 37 °C with 8 μM of compound 4 (A)(186), 16 μM of compound 11 (B), 50 μM of compound 1 (C) and compound 2 (D) in culturing media (10% FBS) for 1h. After removing supernatants, cells were washed with PBS three times, and were imaged on Leica SP2 AOBS confocal laser scanning microscope using ×63 oil lens.

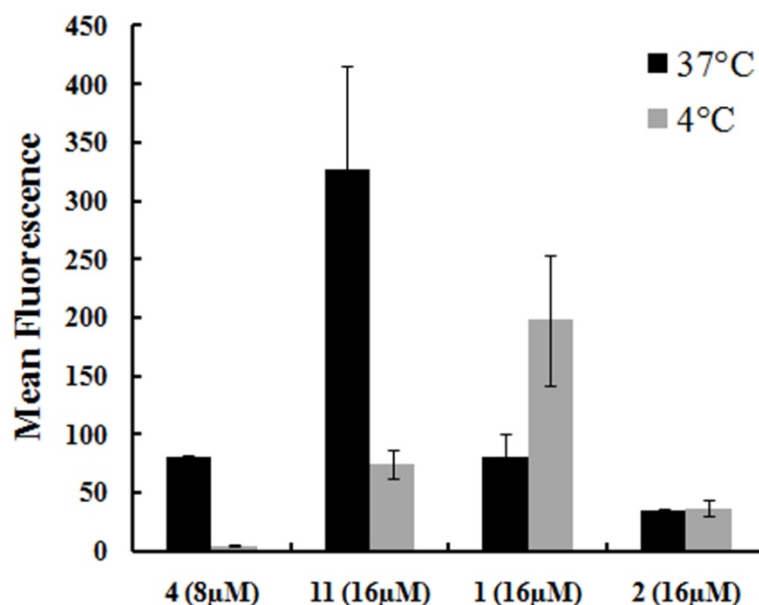


Figure 3.7 The cellular uptake of construct 4, 11, 1 and 2 at 37 °C and 4 °C. MCF-7 cells were incubated with 8 μM of 4, 16 μM of 11, 1, and 2 in culturing media (10% FBS) for 1 h at 37 °C and 4 °C. For 4 °C experiments, MCF-7 cells and all the reagents were pre-cooled at 4 °C for 30 min. Cellular fluorescent signals were quantified on Guava EasyCyte Plus® flow cytometry system.

3.4 Conclusions

Cell-penetrating peptides have found numerous applications in biology and medicine as molecular transporters. The search for highly efficient, stable transporters and the understanding of their internalization mechanisms are two areas under extensive investigation. Herein, we developed a library of cationic, amphipathic peptoids (poly-*N*-substituted glycines) as a new class of transporters and investigated the relationships between their structure, uptake efficiency and the associated cytotoxicity. Both

guanidinium heads and bulky, aromatic hydrophobic residues were found to render the cationic, amphipathic constructs more permeable. Constructs **4** and **11** were found to be promising molecular transporters with excellent cellular uptake efficiency. Moreover, the internalization mechanisms of these cationic, amphipathic peptoid transporters were distinct from those of polyguanidine constructs (Arg_8 and Nbtg_8) as indicated by their different intracellular distribution patterns and different uptake behaviors at 4°C.

Chapter 4

A novel PEG platform for multivalent Bombesin peptide delivery to GRPR/BB2 in prostate cancer

4.1 Introduction

Bombesin (BBN) peptides, an amphibian homologue of mammalian gastrin-releasing peptide (GRP), can bind with high affinity and specificity to the GRP receptors (GRPR/BB₂) which are upregulated in various types of cancers including small cell lung cancer (SCLC), prostate cancer, breast cancer, gastric cancer, and pancreatic cancer, and has been regarded as an important marker for malignant transformation (188, 189). Especially in prostate cancer, Markwalder *et al* reported the high expression of GRPR/BB₂ in 30 of the 30 tested invasive prostatic carcinoma (190). Hence, BBN peptides, especially the truncated form BBN(7-14) (Q-W-A-V-G-H-L-M), provide a promising basis for developing radiometallated diagnostic or therapeutic radiopharmaceuticals to target GRPR/BB₂/BB₂ positive prostate cancers(188, 189, 191).

A variety of BBN(7-14) derived radiopeptides have been developed (192), and the spacers between BBN(7-14) and the chelators have been found to influence the conjugates' affinity to GRPR/BB₂ and their *in vivo* pharmacokinetics (193-195). Smith *et al* reported that a linear hydrocarbon spacer group with 3 to 8 carbon atoms in length wouldn't compromise the conjugates' binding affinity to GRPR/BB₂(189, 194, 195). Spacers with aromatic groups, such as *p*-aminobenzoic acid (AMBA) were reported with even higher tumor retentions than radioconjugates with ether linker moieties (191, 193).

Despite the variety of derivatives developed, small peptide-based reagents still suffer from a fast renal clearance from the blood circulation and thus a short tumor retention for *in vivo* application (196). PEGylation, covalently attaching polyethylene glycol (PEG) to therapeutic molecules, has been used to prolong the half-life of low-molecular weight drugs or peptides and enhance their stability in the plasma. Small molecules can diffuse

freely in and out of the tumor blood vessels and their effective concentration in the tumor can diminish quickly when their concentrations in the blood circulation are low. PEGylation of low-molecular weight compounds can improve the compound accumulations in tumor tissues with time by the enhanced permeability and retention (EPR) effect, which provides a passive tumor targeting (196, 197). An anticancer drug delivery system with PEG (MW $\approx$ 5,000) as a carrier and LHRH peptides as the tumor targeting moieties was found with significantly enhanced selectivity (166).

Herein, we report a design of a 4-arm PEG-based platform with multivalent BBN(7-14) for targeted delivery to the GRPR/BB₂ positive prostate cancer. Compared to free BBN(7-14)-based radiopeptides, the target specificity of this conjugate is achieved via a combination of multivalent targeting moieties with BBN(7-14) and the benefits provided by the PEG carrier.

This work is being prepared as a manuscript titled as “A novel StarPEG platform for Bombesin peptide delivery to GRPR/BB₂ in prostate cancer”, with the authors: Yingding Xu[&], Wei Huang[&], Ren Gang[&], Shibo Qi, Han Jiang, Zheng Miao, Hongguang Liu, Ermelinda Lucente, Annelise Barron and Zhen Cheng (authors labeled with [&] contributed equally). The material synthesis part was done by Wei Huang, while the characterization work presented here was mainly contributed by Yingding Xu and Ren Gang.

4.2 Materials and methods

4.2.1 General experimental procedures

All commercially obtained chemicals were of analytical grade and used without further purification. Chelex 100 resin (50–100 mesh) was purchased from Sigma-Aldrich (St. Louis, MO). Water and all buffers were passed through Chelex 100 column before

use in ^{64}Cu labeling procedures to ensure that the aqueous buffer is heavy metal-free. The syringe filter, polyethersulfone membranes (pore size, 0.22 μm ; diameter, 13 mm) were obtained from Nalge Nunc International (Rochester, NY). ^{125}I -[Tyr4]BBN (2,000 Ci/mmol) was purchased from GE Healthcare (Piscataway, NJ). ^{64}Cu was obtained from University of Wisconsin (Madison, WI). The reversed-phase high-performance liquid chromatography (HPLC) system was the same as previously reported (198). For analytical HPLC and ^{64}Cu labeling purification, a Vydac 218TP54 column (5 μm , 250Å~4.6 mm) was used with a flow rate of 1 mL/min. The mobile phase was changed from 95% solvent A (0.1% trifluoroacetic acid [TFA] in water) and 5% solvent B (0.1% TFA in acetonitrile, [ACN]) (0–2 min) to 35% solvent A and 65% solvent B at 32 min. The UV absorbance was monitored at 218 nm, and the identification of the peptides was confirmed based on the UV spectrum acquired using a photodiode array (PDA) detector. The radioactivity was detected by model 105S single-channel radiation detector (Carroll and Ramsey Associates).

4.2.2 Solid-phase DOTA-peptides Synthesis

Peptides were synthesized using an ABI 433A peptide synthesizer (Applied Biosystems, Inc.) on Rink amide MBHA resin (EMD Biosciences, Gibbstown, NJ) using the standard Fmoc chemistry. Briefly, the resin was deprotected using piperidine to give rise to a primary amine for further peptide assembly. The Fmoc protected amino acids with appropriate orthogonal protection were activated using HCTU as sequentially added to the resin. Fmoc-8-Aoc-OH (Creosalus, Louisville, KY) was used as the linker for DOTA-G-C-8-Aoc-BBN(7-14). Fmoc-PABA-OH was used as the linker for DOTA-G-AMBA-BBN(7-14)(193). DOTA-tris (t-Bu ester) (Macrocylics, Inc.) was added to

the N-terminal of peptides via standard HCTU coupling in solid phase. The resin-bound conjugates were then exposed to a mixture of trifluoroacetic acid (TFA): triisopropylsilane: water (95:2.5:2.5, volume ratio) for ~3h to cleave conjugates from the resin and remove all the protecting groups. Crude peptides were purified by reverse-phase high performance liquid chromatography (RP-HPLC) (Waters Corporation) using a C18 column and a linear acetonitrile/water gradient. A final purity > 93% as measured by analytical RP-HPLC (Waters Corporation) was achieved, and the identity of each compound was confirmed using electrospray ionization mass spectrometry (ESI/MS). Unless indicated otherwise, all amino acids were purchased from Novabiochem, EMD, Inc., and all reagents were purchased from Sigma Aldrich (St. Louis, MO).

4.2.3 Bioconjugation of DOTA-peptides to the 4-arm PEG

The 4-arm PEG-MAL, 10K Da was purchased from Creative PEGWorks, NC. DOTA-G-C-8-Aoc-BBN(7-14) reacted with the maleimide groups at the end of 4-arm PEG-MAL via its thiol in Cysteine. Briefly for the conjugation, peptides and the 4-arm PEG-MAL were dissolved in PBS, pH 7.4 at 10mg/ml, and they were mixed together to make the ratio of thiols : maleimides equal to 3:1. The mixed solution was stirred at room temperature for 1h and was protected from light. Dithiothreitol was added to the reaction at half of the thiols' molar amounts and reacted for half an hour to remove potentially formed disulfide bonds. The PEG-peptide conjugation was first confirmed via analytic RP-HPLC (Figure S2) and then the PEG-BBN conjugates were purified via RP-HPLC on a C18 column. The molecular weight of the final conjugates was analyzed via matrix assisted laser desorption ionization time of flight mass spectrometry (MALDI-

TOF MS) using sinapinic acid matrix at Stanford University's PAN Facility Center to derived the functionalized ratio of DOTA-G-C-8-Aoc-BBN(7-14) to PEG.

4.2.4 The cell uptake study

The cell uptake studies were performed as previously described with some modifications (199). PC3 cells were seeded into 12-well plates at a density of 5×10^5 cells per well for overnight incubation. Cells were rinsed 3 times with phosphate buffered saline (PBS), followed by the addition of ^{64}Cu -DOTA-StarPEG-BBN (~ 18 kBq/well) at 37°C or 4°C for 30, 60, 120, and 300 min. For blocking study cells were incubated with both ^{64}Cu -DOTA-StarPEG-BBN (~ 18 kBq/well) and unlabeled 8-AOC-BBN monomer at 37°C for the aforementioned time points. Tumor cells were then washed three times with chilled PBS and harvested by trypsinization with 0.25% trypsin/0.02% EDTA (Invitrogen, Carlsbad, CA). The cells suspensions were collected and measured in a γ counter (Packard, Meriden, CT). The cell uptake was expressed as the percent added dose (%AD) after decay correction. Experiments were performed with triplicate wells.

4.2.5 MicroPET Imaging

For microPET imaging studies, nude mice were implanted with PC3 cells in the right shoulder and divided into 2 groups (n=3 each). The first group was injected with about 3.7 MBq (100 μCi) of ^{64}Cu -StarPEG-BBN; the second group was injected with 250 μg of unlabeled 8-AOC-BBN along with ^{64}Cu -StarPEG-BBN. Mice were then imaged with PET at 1 h, 4 h, and 24 h post injection (p.i.). Mice were anesthetized with isoflurane (5% for induction and 2% for maintenance in 100% O_2) using a knock-down box. With the help of a laser beam attached to the scanner, the mice were placed in the prone position near the center of the field of view of the scanner. The 5-min static scans were then

obtained. Images were reconstructed by using a two-dimensional ordered-subset expectation maximization (OSEM) algorithm. No background correction was performed. Regions of interest (ROIs; 5 pixels for coronal and transaxial slices) were drawn over the tumor on decay-corrected whole-body coronal images. The maximum counts per pixel per minute were obtained from the ROI and converted to counts per milliliter per minute by using a calibration constant. With the assumption of a tissue density of 1 g/ml, the ROIs were converted to counts per gram per min. Image ROI-derived %ID/g values were determined by dividing counts per gram per minute by injected dose. No attenuation correction was performed.

4.2.6 Animal Biodistribution Studies

Animal procedures were performed according to a protocol approved by the Stanford University Institutional Animal Care and Use Committee. Approximately 5×10^6 cultured PC3 cells were suspended in PBS and subcutaneously implanted in the right shoulders of nude mice purchased from *Charles River Laboratory* (Wilmington, MA). Tumors were allowed to grow to a size of 0.5-1 cm in diameter. For biodistribution studies, the nude mice bearing PC3 xenografts ($n = 3$) were injected with about 3.7 MBq (100 μ Ci) of ^{64}Cu -StarPEG-BBN through the tail vein and sacrificed at 24 h post injection (p.i.). The blocking study of ^{64}Cu -StarPEG-BBN was performed by co-injection of the probe with 250 μ g of unlabeled 8-AOC-BBN through the tail vein. The tumor and normal organs of interest were removed and weighed, and their radioactivity was measured in a gamma-counter. The radioactivity uptakes in the tumor and normal organs were expressed as a percentage of the injected radioactive dose per gram of tissue (% ID/g).

4.3 Results and discussion

4.3.1 Synthesis of DOTA-peptides and Conjugation to the 4-arm PEG

The 4-arm PEG with MW ~10 KDa was chosen as the carrier to multidisplay BBN peptides, and the thiol-maleimide chemistry was adopted for a specific and easy peptide-PEG conjugation. The 8-aminooctanoic acid (8-Aoc) has been reported as an effective spacer between the chelator and the BBN peptide with the ease of synthesis (189, 193). A Cysteine was further designed in the spacer part to provide a thiol for further attachment to the PEG carrier. Taken together, the 4-arm PEG-MAL (MW~10KDa) and DOTA-G-C-8-Aoc-BBN(7-14) were designed to synthesize a PEG-based platform with multivalent BBN peptides (Figure 4.1).

Spacers or chelators added onto the C-terminus of BBN(7-14) significantly reduced the conjugate's affinity to GRPR/BB₂ (data not shown). Hence, all the functional groups were added on the N-terminus of BBN(7-14). DOTA-G-C-8-Aoc-BBN(7-14) was characterized as shown in Figure 4.2, and then reacted with the 4-arm PEG-MAL with a thiol-to-maleimide ratio at 3:1 (Figure 4.1). The PEG-BBN conjugates were further characterized and purified with RP-HPLC (Figure 4.3). By analyzing the molecular weight of the resulted conjugates via MALDI (Figure 4.4), the functionalizing ratio of peptides to PEG is ~4, with the 4 arms of each PEG all functionalized with BBN peptides.

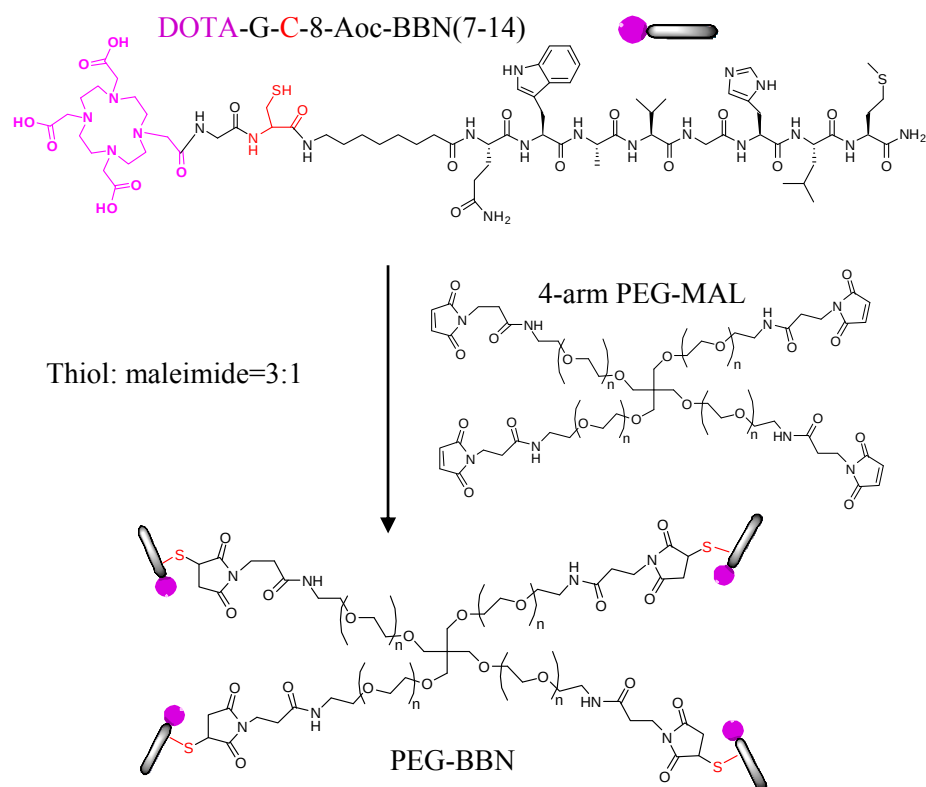


Figure 4.1 DOTA-peptides and PEG-peptide conjugation. The chemical structures of DOTA-G-C-8-Aoc-BBN(7-14) and the 4-arm PEG-MAL were shown, and the coupling of PEG-BBN conjugates was illustrated.

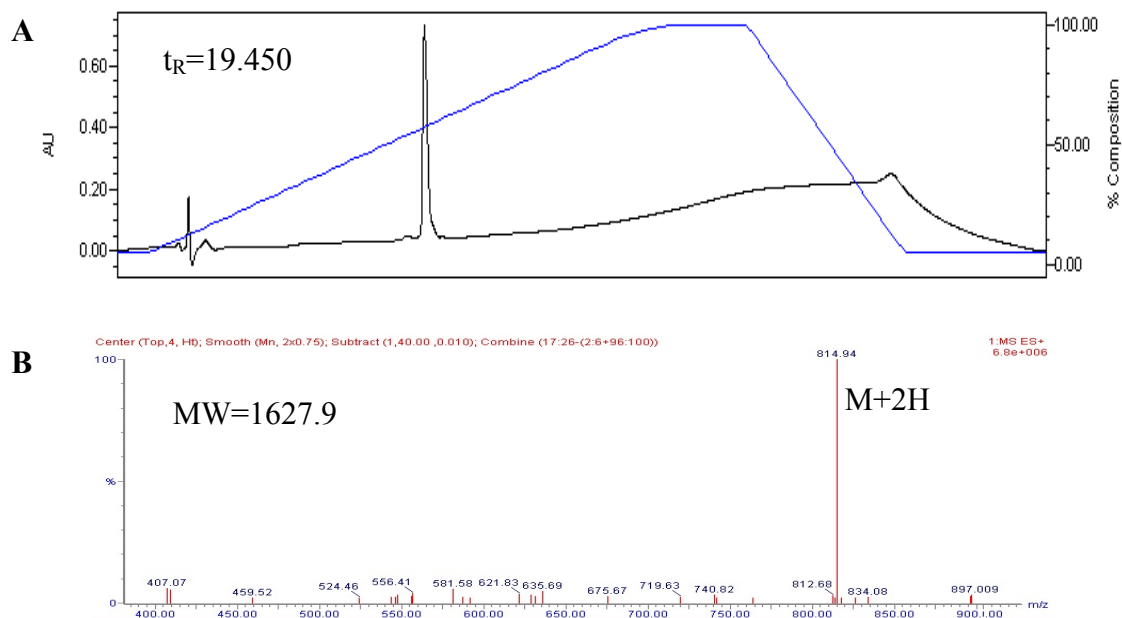


Figure 4.2 Characterization of DOTA-peptides. Analytical RP-HPLC traces of DOTA-G-C-8-Aoc-BBN(7-14) (A), were measured at 220nm, with a linear gradient of 5-95% acetonitrile in water (with 0.1% trifluoroacetic acid) over 30 min at the flow rate of 15 ml/min followed by column washing with 99% acetonitrile. The retention time (t_R) of each construct was labeled in the chromatogram. The molecular weight (MW) of DOTA-G-C-8-Aoc-BBN(7-14) (B) was measured via electrospray ionization mass spectrometry (ESI/MS). The construct MW and the corresponding peaks were indicated in the mass spectrum.

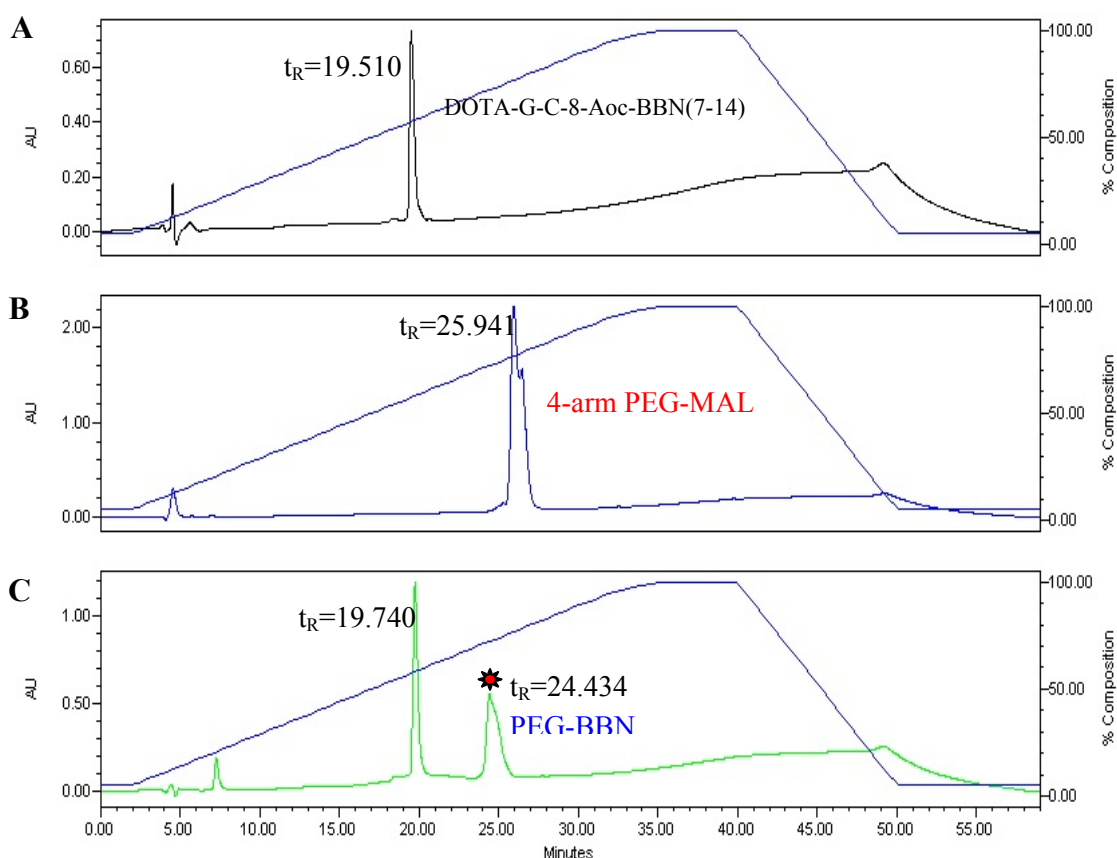


Figure 4.3 Characterization of the PEG and peptide conjugation via RP-HPLC. Analytical RP-HPLC traces of DOTA-G-C-8-Aoc-BBN(7-14) (A), the purchased 4-arm PEG-MAL (B), and the mixture after conjugation before purification (C) were measured at 220nm, with a linear gradient of 5-95% acetonitrile in water (with 0.1% trifluoroacetic acid) over 30 min at the flow rate of 15 ml/min followed by column washing with 99% acetonitrile. The retention time (t_R) of each peak was labeled in the chromatogram. After conjugation, in C, we observed the extra DOTA-G-C-8-Aoc-BBN(7-14) peptide peak and

the conjugated PEG-BBN peak (indicated with a star, the final product we intended to synthesize).

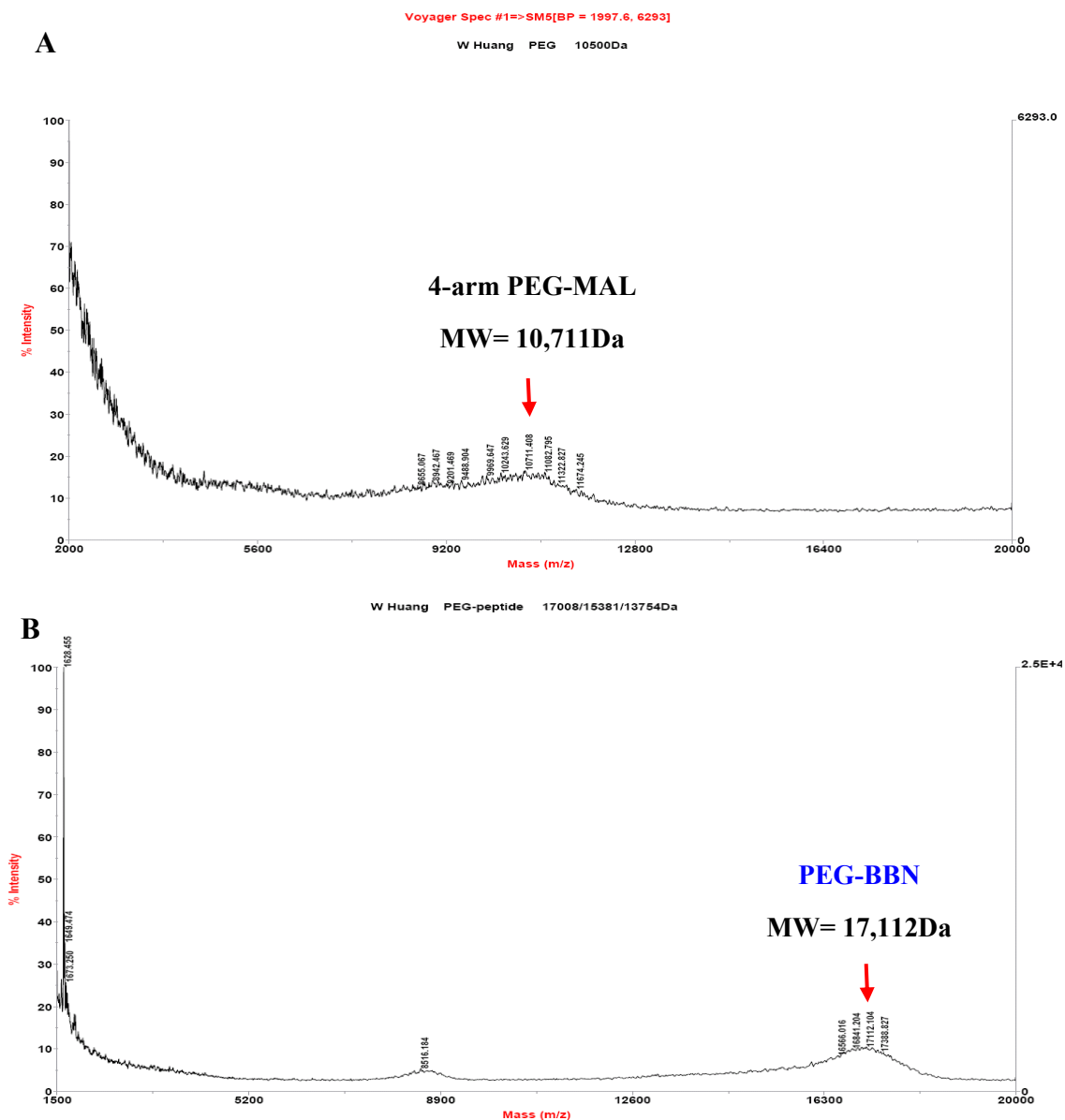


Figure 4.4 Molecular weights (MWs) of the 4-arm PEG-MAL and the PEG-BBN conjugate. MWs were determined by matrix assisted laser desorption ionization time of flight mass spectrometry (MALDI-TOF MS) using sinapinic acid matrix at Stanford University's PAN Facility Center. PEG-based samples don't fly very well in MALDI. The MW of the 4-arm PEG-MAL was 10,711Da, and the MW of PEG-BBN

was 17,112Da. The MW of DOTA-G-C-8-Aoc-BBN(7-14) is 1627.9. Thus, the functionalizing ratio of peptide to PEG is $(17112-10711)/1627.9 \approx 4$.

4.3.2 The cell binding assay

This assay was performed by Yingding Xu in Zhen Cheng lab. The competitive cell-binding assay was used to determine the receptor-binding affinity of the PEG-BBN. The PEG-BBN inhibited the binding of ^{125}I -[Tyr4]BBN to PC3 cells in a concentration dependent manner (Figure 4.5). The IC_{50} value of PEG-BBN was determined to be 38.3nM. This IC_{50} value suggests that the conjugation of the BBN peptides to the 4-arm PEG results in a good receptor binding affinity.

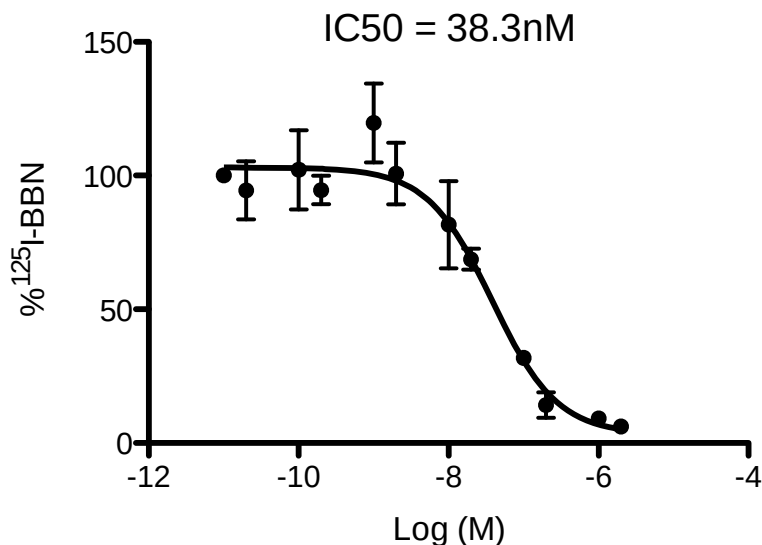


Figure 4.5 The cell binding affinity assay of the PEG-BBN

4.3.3 The cell uptake study

This assay was performed by Yingding Xu. The cell uptake studies of ^{64}Cu -DOTA-PEG-BBN and ^{64}Cu -DOTA-PEG-BBN with blocking were performed on PC3 cells. As

shown in Figure 4.6, the radiotracers exhibited an increased uptake with time. Most of the cellular uptake occurred at 37 °C. Comparing the blocking and non-blocking part of the study, the uptake of ^{64}Cu -DOTA-PEG-BBN was significantly lower when the cells were incubated with unlabeled 8-AOC-BBN peptide at all time points ($P < 0.05$). Therefore the cell uptake of ^{64}Cu -DOTA-PEG-BBN by PC3 cells was mostly specific, as indicated by this uptake assay.

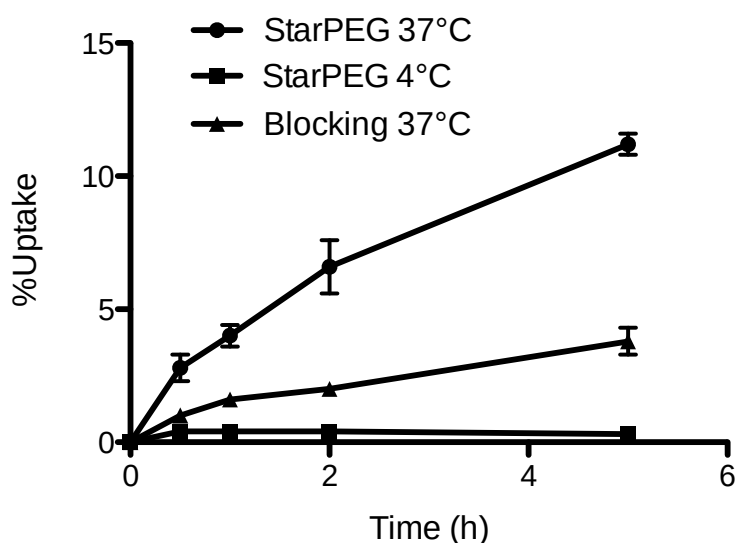
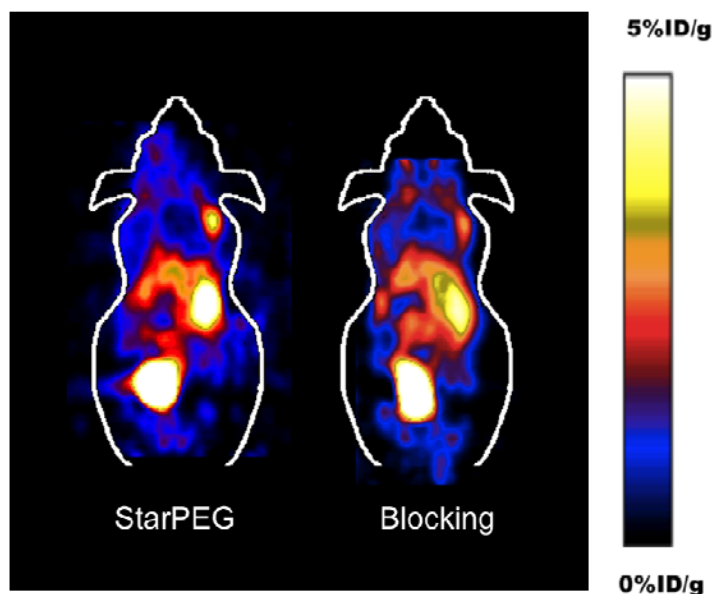


Figure 4.6 Cellular uptake studies of PEG-BBN. The cellular uptake of ^{64}Cu -PEG-BBN was quantified at 37°C and 4°C. For the blocking assay, cells were pre-incubated with unlabeled 8-AOC-BBN at 37°C before the incubation with ^{64}Cu -StarPEG-BBN ($n=3$, mean $\pm$ SD).

4.3.4 MicroPET Imaging Studies

This study was contributed by Yingding. Representative decay-corrected coronal images of static scans at 4 h p.i. were shown in Figure 4.7. The PC3 tumors were clearly visualized with good tumor-to-background contrast for ^{64}Cu -StarPEG-BBN (Figure 4.7A). The tumor uptakes of the ^{64}Cu -labelled conjugates were measured to be 3.2 ± 0.4 ,

4.4 ± 0.2 and 4.4 ± 0.3 %ID/g at 1, 4, and 24 h p.i., respectively ($n = 3$, Figure 4.7B). As shown in Figure 4.7A, the targeting of ^{64}Cu -PEG-BBN was significantly blocked by co-injection of unlabeled 8-AOC-BBN, particularly at 4 h and 24 h p.i.. This reduction was further verified by quantification, which yielded tumor uptakes of 3.1 ± 0.5 , 3.0 ± 0.3 and 2.9 ± 0.2 %ID/g at 1, 4, and 24 h p.i., respectively, when blocked by unlabeled BBN monomers (Figure 4.7B). This finding confirms the GRPR/BB₂-specific targeting of StarPEG-BBN in the GRPR positive PC3 tumor.



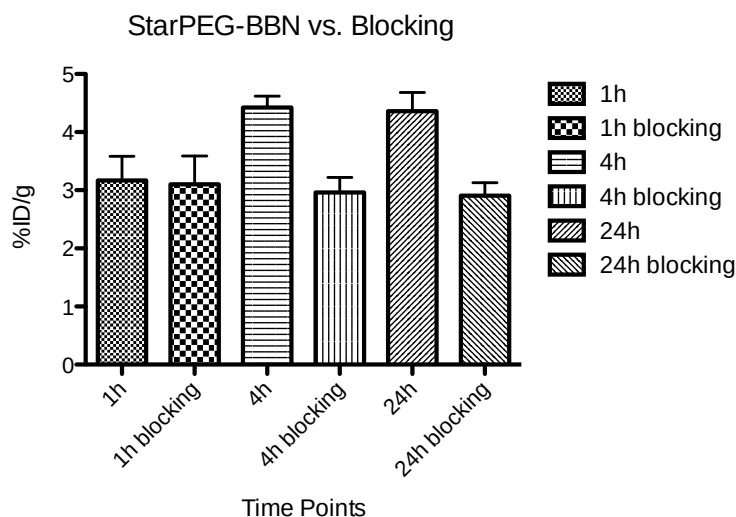


Figure 4.7 In vivo imaging of PC3 xenograft mouse models. (A) From left to right, representative PET images of PC3 tumor mice injected with ^{64}Cu -PEG-BBN and ^{64}Cu -PEG-BBN with 250 μg of unlabeled 8-AOC-BBN, respectively, at 4 h p.i. (B) Quantifications of the tumor uptake of ^{64}Cu -PEG-BBN and ^{64}Cu -PEG-BBN with blocking at 1h, 4h and 24h p.i.

4.3.5 Biodistribution Study

This study was contributed by Yingding. To validate microPET quantifications, we performed a biodistribution experiment by directly sampling tumors and normal organs of PC3 tumor bearing mice 24 h after tail-vein injection of ^{64}Cu -DOTA-PEG-BBN alone and ^{64}Cu -DOTA-PEG-BBN with unlabeled AOC-BBN as the blocking agent ($\sim 100 \mu\text{Ci}/\text{mouse}$). The results were shown in Figure 4.8. The tumor uptakes for ^{64}Cu -DOTA-PEG-BBN and ^{64}Cu -DOTA-StarPEG-BBN blocked with unlabeled 8-AOC-BBN were 5.3 ± 0.2 and 3.1 ± 0.6 %ID/g, respectively ($p = 0.003$). The liver and kidney uptakes were not statistically significant comparing the blocked and the non-blocked radiotracers. However, in pancreas a significantly reduced uptake was observed when ^{64}Cu -DOTA-

PEG-BBN was blocked with unlabeled 8-AOC-BBN, which is consistent with the fact that murine pancreatic cells express a very high level of GRPR.

4.4 Conclusions

Gastrin-releasing peptide receptor (GRPR/BB₂) is up-regulated in numerous cancers including small cell lung cancer (SCLC), prostate cancer, breast cancer, gastric cancer, and pancreatic cancer, and has been regarded as an important marker for malignant transformation. Bombesin (BBN) is a homolog of gastrin-releasing peptide and its analogs have been widely used for developing molecular probes. *In vivo* pharmacokinetics of radiolabeled BBN analogues can be improved by modifying the linker between BBN and its chelating agent. A number of linkers including 8-aminooctanoic acid (8-AOC) have been developed without influencing the binding affinity of the chelated BBN analogues. In this study we designed and synthesized a novel BBN positron emission tomography (PET) probe featuring 4 copies of 8-AOC-BBN peptides displayed on the 4-arm poly(ethylene glycol) (PEG).

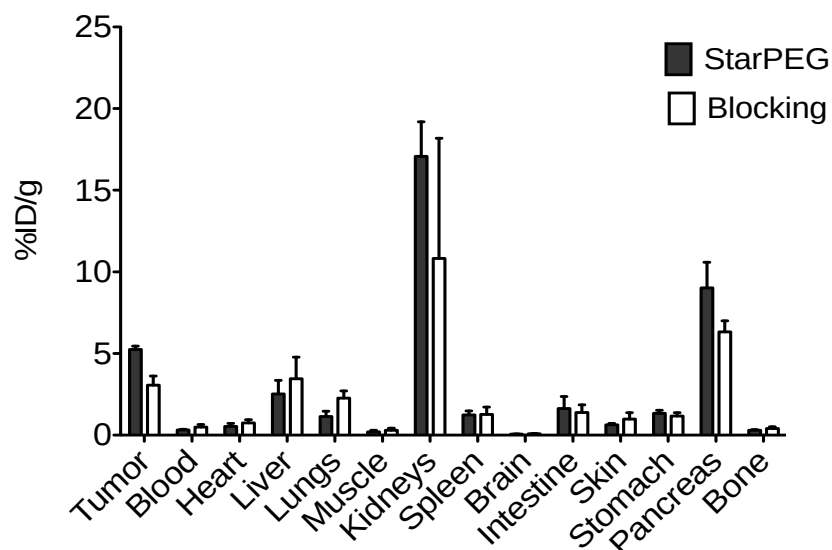


Figure 4.8 Biodistribution studies of ^{64}Cu -PEG-BBN and ^{64}Cu -PEG-BBN blocked by unlabeled 8-AOC-BBN in PC3 xenograft-bearing athymic nude mice at 24 h p.i. Data are expressed as normalized accumulation of activity in %ID/g $\pm$ SD (n=3)

Briefly, DOTA-GC-8-AOC-BBN(7-14) NH_2 was synthesized on solid-phase using conventional Fmoc chemistry. The monomers were attached to the 4-arm PEG-Maleimide via the thiol groups of the cysteines in the sequences, which gave rise to the 4-arm full functionalized PEG-BBN as confirmed by the molecular weight measurement. Cell binding studies and the cell uptake studies with blocking showed that PEG-BBN had good and specific binding affinity to PC3 cells (IC_{50} =39.12 nM).

For *in vivo* studies, athymic nude mice were implanted with PC3 cells in the right shoulder to set up the PC3 xenograft-bearing mouse models. *In vivo* images, quantifications, and biodistribution studies revealed that ^{64}Cu -PEG-BBN had good, specific tumor uptakes.

Chapter 5

**Creation of protein polymers with multiple matrix
metalloproteinase degradation sites for hydrogel
development**

5.1 Introduction

In situ forming hydrogels have emerged as a leading trend in biomedical hydrogel research. The precursors can be delivered via minimally invasive procedures prior to solidification and gelation occurs at the site of injection (200). In tissue engineering, biomaterials are needed as a scaffold to provide a three-dimensional support, interact with cells and guide the processes of tissue formation and regeneration, functioning like the natural extracellular matrix (ECM). The ECM is a network of different types of proteins, protein fibrils, glycosaminoglycans (GAGs) and soluble cues. Hydrogels, networks of hydrophilic polymer chains, provide a good mimic of natural ECM and have been explored extensively for tissue engineering applications (109, 110).

A variety of synthetic and naturally derived polymers have been used to form hydrogels. Materials purified from natural sources are advantageous because of their inherent biological activities, but they suffer from several drawbacks including batch-to-batch variability, limited sources, limited control of material structure and properties, and possible pathogen transmission and immunogenicity. Synthetic polymers allow more specific control over the material properties and are easy to scale up in synthesis (109, 201, 202). However, chemically synthesized polymers themselves are inactive, and may be toxic due to crosslinkers or initiators needed for their gelation. In addition, bioactive peptides usually need to be incorporated to allow for cell responsiveness.

Genetically engineered protein polymers provide a platform to design monodisperse protein polymers with facile incorporation of bioactivity. As discussed in Chapter 1.4.2, several biomimetic recombinant protein polymers have been designed. Some of them are

designed to self-assemble into hydrogels via physical crosslinking, and some can be chemically crosslinked to achieve more rigid and stable hydrogels (203, 204).

The Barron group has utilized tissue transglutaminase (TG) as an alternative to chemical crosslinking to develop a protein-based hydrogel platform with a mild and biocompatible enzymatic crosslinking (134, 135). Tissue transglutaminase catalyzes the formation of a covalent bond between a lysine and a glutamine residue in a calcium dependent reaction (Figure 5.1A). This crosslinking mechanism naturally occurs *in vivo* during the processes of wound healing and the stabilization and organization of the extracellular matrix (205), and has been widely employed in food processing to bond food together (206). Transglutaminases have been used to create hydrogels made of poly(ethylene glycol) (PEG) (207, 208) and elastin-like protein polymers (116).

In the enzyme-crosslinking hydrogel system developed previously in the Barron group, a family of high-molecular weight protein polymers was developed by repeating two modules, the lysine-containing substrate, and the glutamine-containing substrate, to act as substrates for TG crosslinking into hydrogels (Figure 5.1B)(209). For example, BQ₆ (containing multiple glutamines, Q) and K8₃₀ (containing multiple lysines, K) were found to be efficient substrates for the enzyme transglutaminase (TG, which covalently crosslinks K and Q) and thus can form purely protein-based hydrogels, the physical properties of which can be controlled by the formulation of the hydrogels. These recombinant protein polymers are macromolecular, highly water-soluble, and random coil.

We aimed to introduce bioactive motifs into this hydrogel system to interact better with cells and to guide cell behavior. Herein, the matrix metalloprotease (MMP)

degradable sites were engineered into the lysine-containing substrate protein polymers via the “controlled-cloning” method, and the functions of the MMP degradable sites were further evaluated.

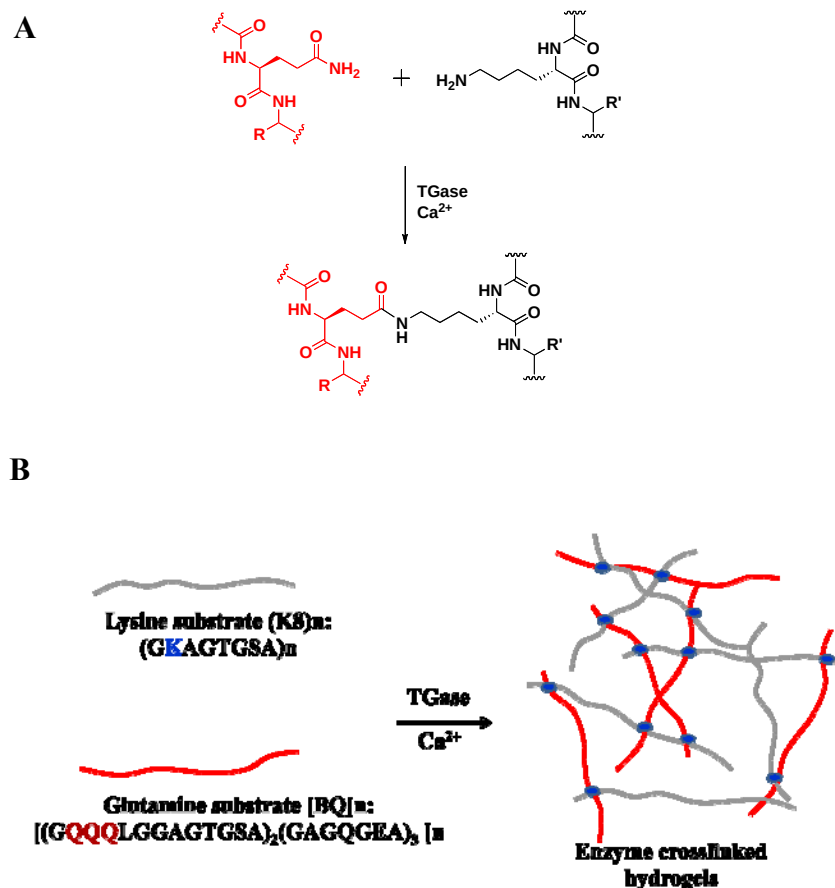


Figure 5.1 Illustrations of the reactions catalyzed by transglutaminase (TGase). (A) Transglutaminase can catalyze covalent bonding between a lysine and glutamine residue in a calcium-dependent manner; (B) A gelation scheme of the enzyme-crosslinking protein hydrogel. Two types of protein polymers are needed in the system, one as the lysine substrate and the other as the glutamine substrate.

5.2 Materials and methods

5.2.1 Plasmid constructions

MMP: 5'-
GCTAGCCATATGCTCTTCAGGTGCGGGCCAAGGAGAAGCAGGTGCTGGACAG
GGCGAGGCAGGTCCGCAAGGCATCTGGGGACAGGGTTGAAGAGGGATCCAC
TAGT-3', the DNA oligonucleotides (Integrated DNA Technologies) were amplified by
PCR with the primers 5'-GCT AGC CAT ATG CTC TTC AGG-3' and 5'-ACT AGT
GGA TCC CTC TTC AAC-3' with high-fidelity *Pfu* polymerase, separated by
electrophoresis on a 2% agarose gel, and purified using the QIAquick Gel Extraction Kit
(Qiagen). The amplified DNA was then digested with *EarI* and the monomers were self-
ligated utilizing T4 ligase to form DNA multimers. The multimers were ligated into the
cloning vector pUC18 (Life Technologies) and transformed into *E. coli* NovaBlue cells
(Novagen) by heatshock. Transformed colonies were screened for the presence of
desired DNA inserts using colony PCR, with *Taq* polymerase and primers 5'-TTA ATG
AAT CGG CCA ACG CGC-3' and 5'-GGA TAA CCG TAT TAC CGC CTT-3,' which
flank the insert region in the pUC18 vector.

For the “controlled cloning”, DNA can be excised from minipreped pUC18 plasmids.
All restriction enzymes including calf intestinal phosphatase (CIP), T4 polynucleotide
kinase (PNK), and T4 DNA ligase were used according to standard protocols. PCR was
used to amplify the gene inserted in the pUC18 vector, which contains a *Sap* I site on
both ends of the inserted gene, with the forward primer, 5'-TTA ATG AAT CGG CCA
ACG CGC-3', which is perfectly complementary to the vector, and the reverse primer,
5'-TGA GCG AGG AAC TCT TCA GGT-3', which contains one mismatched base
which changes one *Sap* I site to an *Ear* I site.

The expression vector, pET-19b (Novagen) was modified to remove the internal *Sap* I site using QuickChange Site-Directed Mutagenesis (Stratagene, La Jolla, CA) to change a single base in the restriction site (111, 210). The modified vector was then amplified by PCR with primers containing *Ear* I recognition sites as well as a stop codon [forward primer: 5'-AGTTACTCTTCAGGTTAAGGATCCGGCTGCTAACAAAG-3'; reverse primer: 5'-AGTTACTCTTCAACCCATATGCTTGTCGTCGTCGTC-3'] (111). The amplified linear plasmid was digested with *Ear* I for ligation to protein polymer genes digested from pUC18. To screen for the correct insertion, these plasmids were transformed into NovaBlue cells, and correct colonies were verified by digestion with *Bam*HI and *Nde* I restriction enzymes and through DNA sequencing (SeqWright). The final proteins expressed in the modified pET-19 are fusion proteins with an *N*-terminal histidine tag (tag amino acid sequence: GH₁₀SSGHIDDDDKHM).

5.2.2 Expression and purification of protein polymers

The modified pET-19 with correct encoding sequences was transformed into *E. coli* expression strain BLR (DE3) cells (Novagen). A small-scale expression test was carried out first to identify colonies with good protein expression (211). For large-scale protein expression, an overnight culture in LB Broth was first grown and then used to inoculate cultures of Terrific Broth, both supplemented with 200 µg/mL ampicillin and 12.5µg/mL tetracycline. The culture was induced with 0.5 mM isopropyl thiogalactoside (IPTG) (US Biologicals) when the OD₆₀₀ of the culture reached between 0.6 and 0.8. The cells were harvested 4 h after the IPTG addition by centrifugation and the cell pellet was resuspended in denaturing lysis buffer (6 M guanidine hydrochloride, 20 mM sodium phosphate, 500 mM NaCl, pH 7.8). The cells were lysed by three successive freeze (–

80°C)-thaw (37°C) cycles followed by sonication. Cellular debris was removed by centrifugation for 30 min at 15,000 g at 4°C. The lysate supernatant was bound to Chelating Sepharose Fast Flow nickel-charged resin (GE Healthcare) under denaturing conditions, washed, and then eluted with imidazole based competitive elution buffers. The purification process was checked via sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE). Purified proteins were dialyzed against deionized water thoroughly and then lyophilized. They were further purified via RP-HPLC to remove the associated endotoxins.

5.2.3 MALDI analysis

The molecular weights of expressed proteins were measured using matrix-assisted laser desorption/ionization-time-of-flight (MALDI-TOF) mass spectrometry on a Voyager-DE PRO instrument (the Protein and Nucleic Acid Facility, Stanford University). The protein was dissolved in a 50% water/50% acetonitrile with a 0.1% trifluoroacetic acid solution of sinapinic acid and allowed to dry on the surface of the metal target plate. A laser hits the sample under vacuum. The matrix absorbs most of the energy, preventing unwanted fragmentations of the protein sample. The protein becomes ionized and is accelerated in an electric field. The time-of-flight analyzer separates the ions according to mass-to-charge ratios (m/z).

5.2.4 Protease degradation of the protein polymers

The recombinant protein polymers were characterized in terms of their degradability by human plasmin (Roche) and recombinant human MMP-2 (matrix metalloproteinase 2, Calbiochem). Lyophilized protein polymers were dissolved in enzyme buffers at concentrations of 1.6 mg/mL (plasmin buffer: 20 mM Tris, 150 mM NaCl, pH 7.6; MMP

buffers: 50 mM Tricine, 50 mM NaCl, 10 mM CaCl₂, pH 7.5) . Plasmin (0.5 U/mL) and MMP-2 (50 nM) were incubated with the protein polymer solutions at 37 °C for 24h. Degradation of proteins was visualized on Coomassie blue-stained 12% polyacrylamide gel via SDS-PAGE.

5.2.5 The hydrogel invasion assay with transwells

24-well transwell permeable supports with 8 µm pore size were purchased from Corning. MRC-5 fibroblast cells were purchased from ATCC and were cultured in suggested media supplemented with 10% FBS and antibiotics. 40 µl of protein solution comprising K8₃₀+BQ or (MMP-K8₆)₅+BQ was enzymatically crosslinked by TG from guinea pig liver (Sigma-Aldrich) into a ~1 mm-thick hydrogel layer on top of the membrane in each transwell support. 1×10⁵ of MRC5 cells suspended in 200 µl of culturing media were seeded on top of the hydrogel, with 800 µl of culturing media in the lower chamber of the transwell. Culturing media in the upper and lower chamber of the transwell were changed with fresh media every day, and cells were allowed to grow and migrate for 5 days. Then, the membrane was cut from the transwell support, with the gel side wiped with Q-tips, fixed in methanol for 1 h, and stained with Giemsa Stain solution (Sigma) for 1 h. Cells that have penetrated through the hydrogel and reached the bottom of the membrane would be stained. The number of migrated cells for each transwell was counted at 10X on a light microscope. Each gel composition was tested in triple repeats.

5.3 Results and discussion

5.3.1 Design of the block protein polymer sequences

5.3.1.1 An overview of recombinant protein expression

The overall process to generate the recombinant protein polymers consisted of four sequential steps: designing the sequence, cloning the gene, expressing it, and finally purifying the desired protein.

The selection of the expression system depends on the production costs and any modifications the protein may require for its biological activity. They can be expressed in many production systems, including bacteria, yeast, insect cells, mammalian cells, and transgenic animals and plants. The Gram-negative bacterium *E.coli* is a widely used expression host for genetic engineering and recombinant protein production. Our knowledge of its growth, genetics, molecular biology, and genome has enabled efficient manipulation of *E. coli*, and it has become the standard organism for production of recombinant proteins which do not require complicated post-translation modifications or unusual cofactors. To employ *E. coli* as the expression strain, we need to pay attention to the codon bias of *E. coli*. When heterologous proteins are encoded with rare *E.coli* codons, their expression levels can be very low and may result in truncated proteins (212). Moreover, the protein over-expression in the cytoplasm of *E. coli* can lead to inclusion body formation. To minimize protein aggregation, strategies have been developed such as using lower expression temperature and the introduction of solubility tags (213).

After choosing *E. coli* as the expression system, the target gene with correct sequence is placed in an expression plasmid containing a bacteriophage T7 promoter,

coding regions for a specific antibiotic resistance, and an affinity tag sequence. This plasmid DNA is then “transformed” into *E. coli* cells which possess a chromosomal copy of the T7 RNA polymerase under *lacUV5* control. Therefore, upon the addition of a lactose analogue, isopropyl- β -D-thiogalactopyranoside (IPTG) to the bacterial culture, multiple copies of T7 RNA polymerase are produced which further transcribe the heterologous gene inserted in the plasmid containing the T7 promoter sequence. These transcripts are then translated into proteins by the ribosomes. The desired protein can constitute up to 50% of the total cellular protein after a few hours of induction (214).

To purify the expressed proteins, cells are lysed by sonication or other means and then the target protein is purified from cell lysate by affinity chromatography. For the protein polymers used in the hydrogel system, we are aiming to obtain extremely water-soluble macromolecules with good expression yields in *E. coli*. Though various motif sequences published in literature can give some guidance in the *de novo* design, it is difficult to predict beforehand whether a sequence can be well expressed, be highly soluble in water, or have desired properties.

5.3.1.2 The previous enzyme-crosslinking hydrogel system

In previously established hydrogel systems, two types of protein polymers were designed, one as the lysine substrate and the other as the glutamine substrate. The sequences of BQ₆ (containing multiple glutamines, Q) and K8₃₀ (containing multiple lysines, K) which have been proven to be efficient substrates for transglutaminase to form a hydrogel are shown in Figure 5.1B. To *de novo* design highly water-soluble, predominantly unstructured proteins with good expression yields and a minimal risk of immunogenicity, the choice of their amino acid sequences is critical. The amino acid

sequences other than the enzyme substrate sequences in these polymers originated from previous research of random-coil proteins (111, 210, 211). To achieve good solubility and avoid possible compact structures and aggregation, hydrophobic amino acids (*e.g.* F, L, I, M, V, W and Y) were excluded. Cysteine residues were also eliminated to prevent heterogeneous disulfide bond formation. Polar amino acids (S, T) were included to promote water solubility.

To design lysine substrate polymers, it has been shown that the lysine residue alone can be catalyzed by TG for bonding independent of its neighboring sequences, while the efficiency of the glutamine for TG usually depend on its nearby residues (135, 203). Three lysine substrate protein polymers were designed, with the lysine density to be one in every 4, 6, and 8 amino acids (K4, K6, K8, sequence shown in Table 5.1). The glutamine substrate module, BQ, is (GQQQLGGAGTGSA)₂(GAGQGGA)₃, comprising two different blocks. The underlined block has an efficient glutamine substrate for TG based on a reported optimized TG substrate (in bold) (209). The second block is included for water solubility (134), and the glutamine in this block has marginal TG recognition (209).

5.3.1.3 Design of block protein polymers with MMP degradable sites

Based on our previous unpublished data, the enzyme crosslinked hydrogels, though degradable by plasmin, are still too stable to enable fast proliferation of encapsulated cells. Therefore, introducing alternative protease degradation mechanisms into the system would be essential to better allow cell proliferation and migration. A peptide substrate sequence for MMP degradation, (GAGQGGA)₂GPQGIWGQ has been cloned. The underlined sequence is the MMP degradation site and a repeat of GAGQGGA was

included to enhance its solubility (215, 216). With the controlled cloning method, this MMP degradation block can be cloned as a block into lysine substrate proteins, which provides evenly spaced MMP degradation sites along the protein backbone (as illustrated in Figure 5.2, sequences of various modules summarized in Table 5.1).

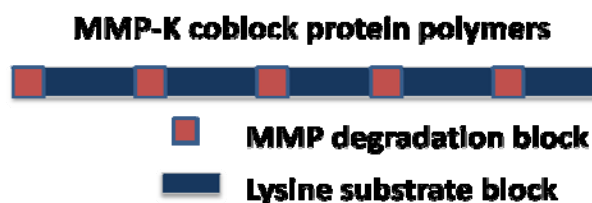


Figure 5.2 A cartoon illustrating the MMP-K block protein polymers

Module Name	Amino acid sequence
(K4) _n	(GKGSGKGA) _n
(K6) _n	(GKGTGA) _n
(K8) _n	(GKAGTGSA) _n
(BQ) _n	[(GQQQLGGAGTGSA) ₂ (GAGQGEA) ₃] _n
MMP	(GAGQGEA) ₂ GPQGIWGQ

Table 5.1 Amino acid design of the protein module sequences. The bioactive residues for each module were shown in bold.

5.3.2 Gene construction of the MMP-K block protein polymers

The whole cloning and expression protocol for these MMP-K block protein polymers are summarized in a flowchart in Figure 5.3.

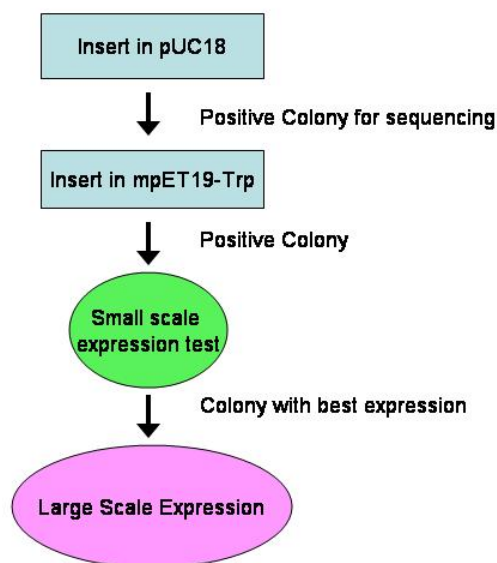


Figure 5.3 A flowchart summarizing the cloning and expression process of protein polymers in this study. Modified pUC18 vector is used for cloning, and mpET19-Trp vector is used for protein expression. Via the small scale expression screening, the colony with the best expression levels of desired protein is chosen for large scale protein production (111).

The synthetic oligonucleotides encoding the MMP degradation modules were first amplified via PCR with proper primers, and then cleaved with *Ear* I to create module monomers. These monomers were ligated into a modified pUC18 vector for “controlled cloning” (111). Figure 5.4A illustrates how to exactly double a DNA sequence in the controlled cloning. One part of the encoding gene in modified pUC18 vector was amplified by growing the plasmid in *E. coli* cell culture, extracted by plasmid mini-preparation, and then digested with *Sap* I followed by the removal of the phosphate groups at the two ends with calf intestinal phosphatase (CIP); for the other part, the encoding gene was amplified through PCR using primers that eliminate one of the *Sap* I restriction sites at the end of the DNA sequence, and thus the PCR-amplified DNA would only have one adhesive end after *Sap* I digestion. Hence, DNA segments derived from the above two ways can be ligated to

give the exact doubled sequence. After replacing the phosphate group with T4 nucleotide kinase and digesting the double gene with *Ea*RI to create double adhesive ends, the DNA was ligated into the pUC18 plasmid, which resulted in a double gene of the original DNA segment. This controlled cloning method is built on two restriction enzymes, *Ea*RI and *Sap*I, the restriction sites of which only differ by one nucleotide. Hence, *Ea*RI can digest the *Sap*I site, but not vice versa (Figure 5.4B).

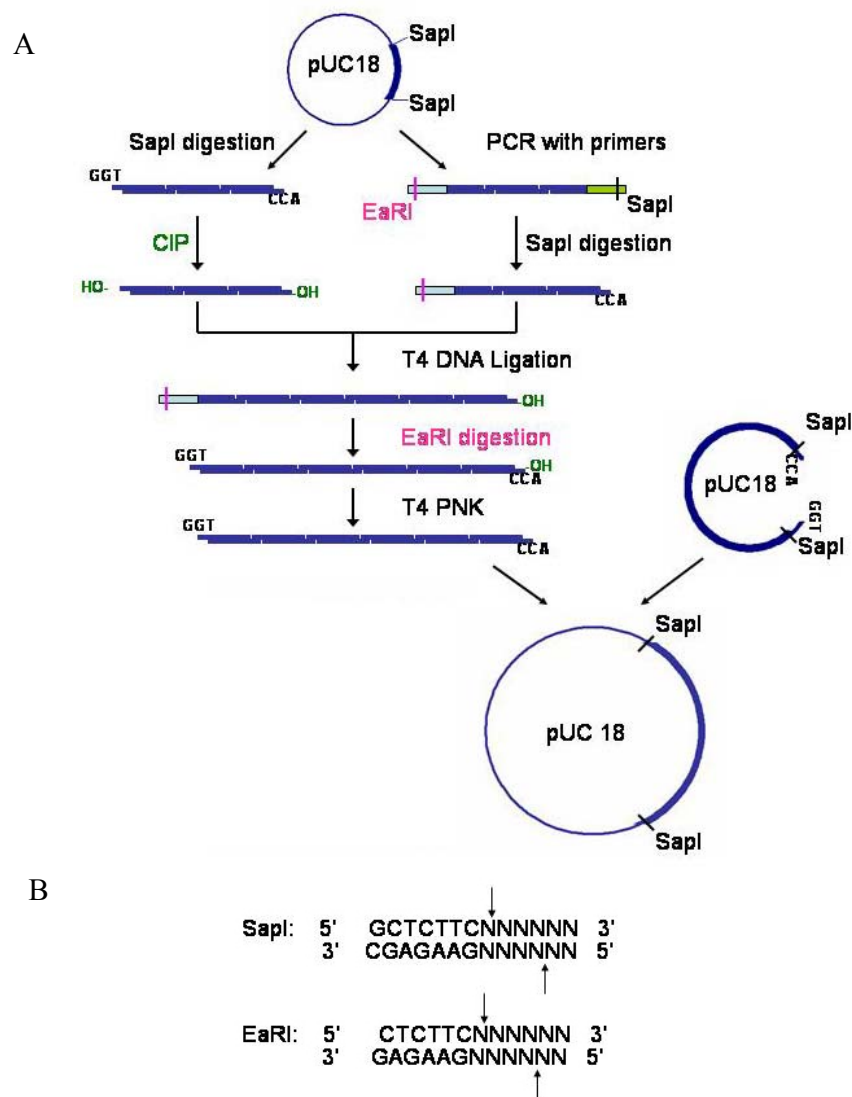


Figure 5.4 The controlled cloning method to double a DNA segment in a well-controlled manner. (A) Cloning details to generate a double repeat of a DNA insert in

the modified pUC18 vector; (B) Recognition and cleavage sequences of the restriction enzyme EarI and SapI (111) .

This method also allows ligation of two different DNA segments in a well-controlled manner, and thus was used to create monomers of the MMP-K block modules. We ligated the MMP modules with three K blocks, and cloned MMP-(K4)₁₂, MMP-(K6)₈ and MMP-(K8)₆ into the pUC18 vector (Table 5.2).

K blocks (in pUC18)	Monomers of MMP-K block modules	MMP-K block protein polymers
(K4) ₁₂ 1 K in 4aa	MMP-(K4) ₁₂ 118aa <u>GAGQGEAGAGQGEAGPQGIWGO</u> GKGSKGAGKGSKGAGKGSKGGA GKGSKGAGKGSKGAGKGSKGGA GKGSKGAGKGSKGAGKGSKGGA GKGSKGAGKGSKGAGKGSKGGA	[MMP-(K4) ₁₂] ₄ 464aa, ~1392bp,
(K6) ₈ 1 K in 6aa	MMP-(K6) ₈ 70aa <u>GAGQGEAGAGQGEAGPQGIWGO</u> GKGTGAGKGTGAGKGTGAGKGTGA GKGTGAGKGTGAGKGTGAGKGTGA	[MMP-(K6) ₈] ₅ 350aa, ~1050bp,
(K8) ₆ 1 K in 8aa	MMP-(K8) ₆ 70aa <u>GAGQGEAGAGQGEAGPQGIWGO</u> GKAGTGSAGKAGTGSAGKAGTGSA GKAGTGSAGKAGTGSAGKAGTGSA	[MMP-(K8) ₆] ₅ 350aa, ~1050bp

Table 5.2 Amino acid sequences of the MMP-K block protein polymers. The MMP degradable module (sequence underlined) was ligated with three K substrate modules via controlled cloning to generate three monomers of the MMP-K block modules (Figure 5.4A), which were further elongated via modified controlled cloning to give rise to the final MMP-K block protein polymers (Figure 5.2).

To derive longer repeats based on these MMP-K block monomers, a modified controlled cloning method was used based on self-ligation to select the DNA length of

interest to clone into pUC18. The elongated DNA segments were further confirmed via specific restriction enzyme digestion and further sequencing (Figure 5.5).

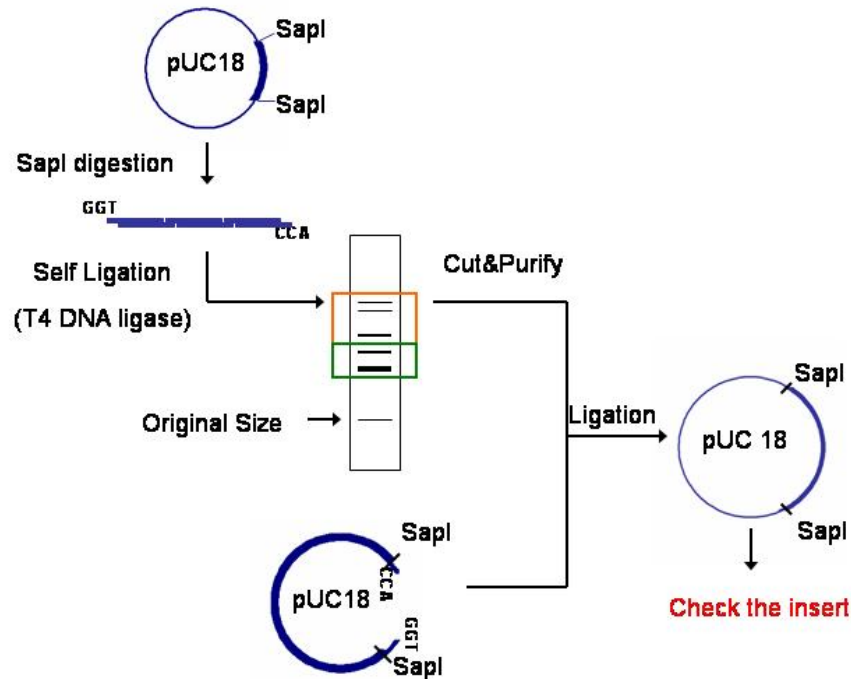


Figure 5.5 The self-ligation cloning method to derive longer repeats of DNA sequences of interest. DNA segments were digested from pUC18 vector and T4 DNA ligase was used to promote self-ligation of the DNA segments. The elongated DNA segments with desired length were selected, purified, and ligated into pUC18 vectors. The elongated DNA segments were further confirmed via restriction digestion and sequencing (111).

DNA segments in pUC18 with the correct sequence were digested with *Sap* I from pUC18 and ligated into a modified pET-19 expression plasmid (Figure 5.6) (111). The sizes of the MMP-K block protein polymers in mpET19-Trp vector were analyzed by agarose gel electrophoresis (Figure 5.7).

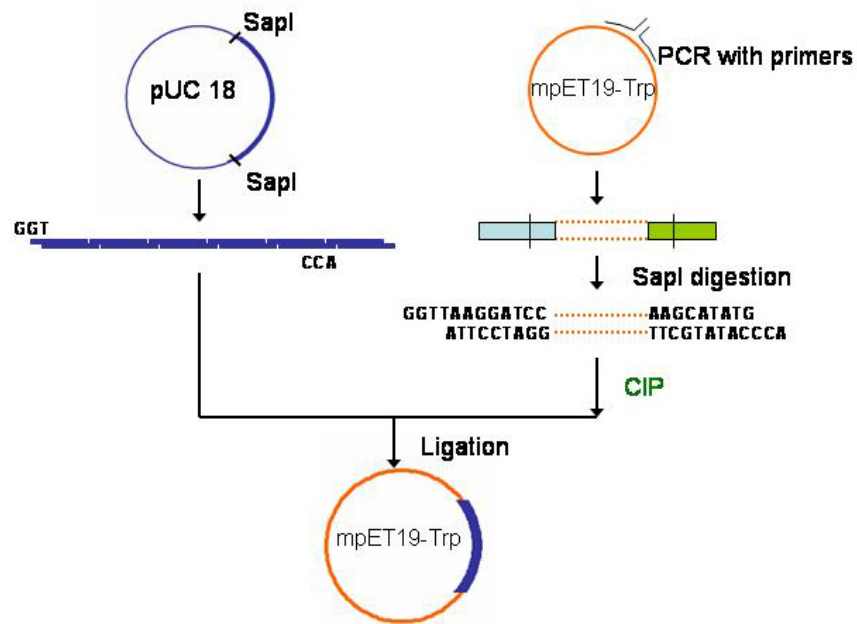


Figure 5.6 The transfer of DNA inserts in pUC18 to the modified expression plasmid, mpET19-Trp vector. (111)

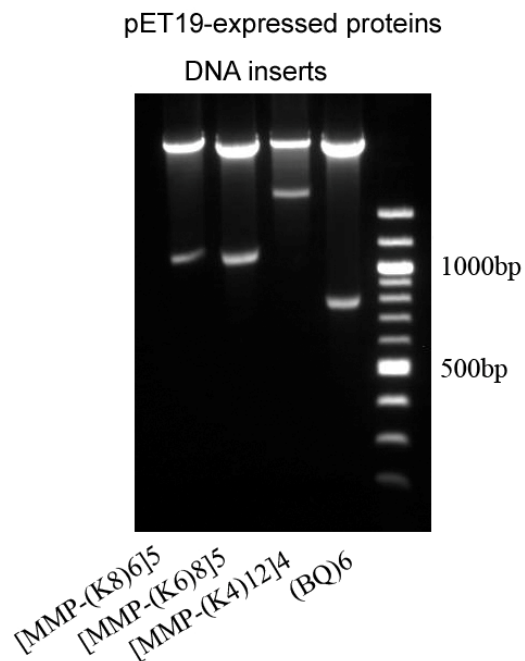


Figure 5.7 The agarose gel confirming the sizes of the constructed genes. DNA inserts in the mpET19-Trp plasmid as labeled were digested with NdeI and BamHI and was run in 1.5% agarose gel. 100 bp DNA marker was loaded on the most right lane.

5.3.3 The expression, purification and characterization of MMP-K proteins

The mpET19-Trp plasmids with desired gene insertions were transformed into the BLR (DE3) strain of *E. coli* for protein expression. Small-scale colony expressions were performed to select several colonies, check their expression level via dot blotting, and then choose the colony with the greatest expression for large scale production (111). [MMP-(K6)₈]₅ and [MMP-(K8)₆]₅ were successfully expressed and purified by nickel affinity chromatography. The expression and purification of these two protein polymers were confirmed in SDS-PAGE (Figure 5.8). In SDS-PAGE, these proteins appeared to be slightly larger than their actual sizes compared with the globular proteins used as protein markers, which is not an uncommon phenomenon that may be due to charge and solvent effects. Protein polymer molar mass was confirmed by MALDI-TOF MS analysis (Table 5.3). For unknown reasons, [MMP-(K6)₈]₅ and [MMP-(K8)₆]₅ showed similar molecular weights in SDS-PAGE, but the MW of [MMP-(K6)₈]₅ obtained from MALDI was slightly higher than the MW calculated based on the amino acid sequence. Since K8 was identified as the best lysine substrate polymer, [MMP-(K8)₆]₅ was chosen for the following studies. The yields of these two block polymers were both approximately 50-100 milligrams of purified protein per liter of bacteria culture.

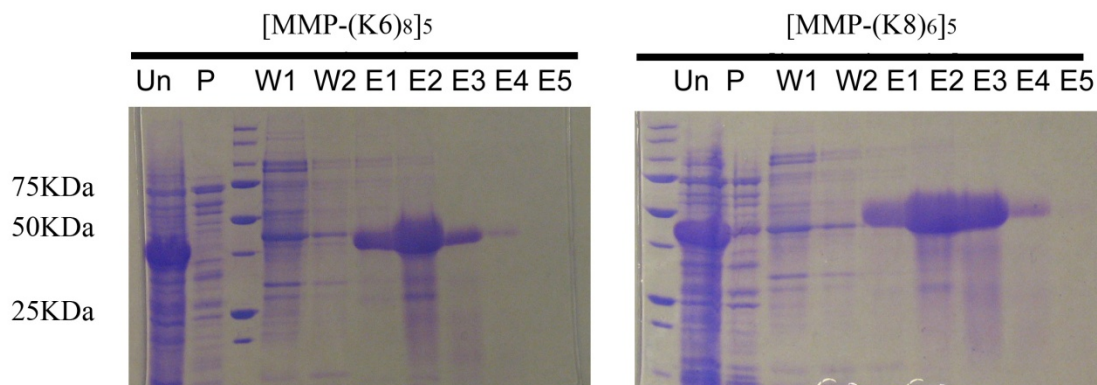


Figure 5.8 SDS-PAGE gels showing the purification process of the two MMP-K block proteins. The sizes of bold bands in the loaded protein marker were labeled on the left of the figure. “Un”: cell lysates before purification; “P”, protein marker; W1-W2, the first and the second column wash after resin binding; E1-E5, the first to the fifth elution of the desired proteins.

Protein	Expected MW (KDa)	MW from MALDI (KDa)
[MMP-(K6) ₈] ₅	31.607	35.001
[MMP-(K8) ₆] ₅	31.637	32.453

Table 5.3 Molecular weights (MWs) of MMP-K block polymers. The expected MW is calculated based on the protein’s sequence, and MW from MALDI was measured experimentally.

5.3.4 Evaluating the MMP-K protein as the MMP substrates

Typically, lysine residues (at their C-termini) indicate trypsin and plasmin cleavage sites in proteins (127). The K substrates protein polymers have been proven to be degradable by trypsin and plasmin (134, 135). To verify the sensitivity of [MMP-(K8)₆]₅ protein polymer to MMP degradation, protein degradation assays were performed with

plasmin and recombinant human MMP-2. As shown in Figure 5.9, both $(K8)_{30}$ and $[MMP-(K8)_6]_5$ were degraded after 4 h of incubation with plasmin, with virtually no full-length proteins detectable. However, for the action of recombinant human MMP-2, only $[MMP-(K8)_6]_5$ which comprises 5 MMP degradation sites per molecule was degraded, while $(K8)_{30}$ remained intact after 24 h of treatment with MMP-2. This degradation assay demonstrated the bio-functions of incorporated MMP degradation sites and the successful tuning of protein polymers' sensitivity to proteolytic degradation.

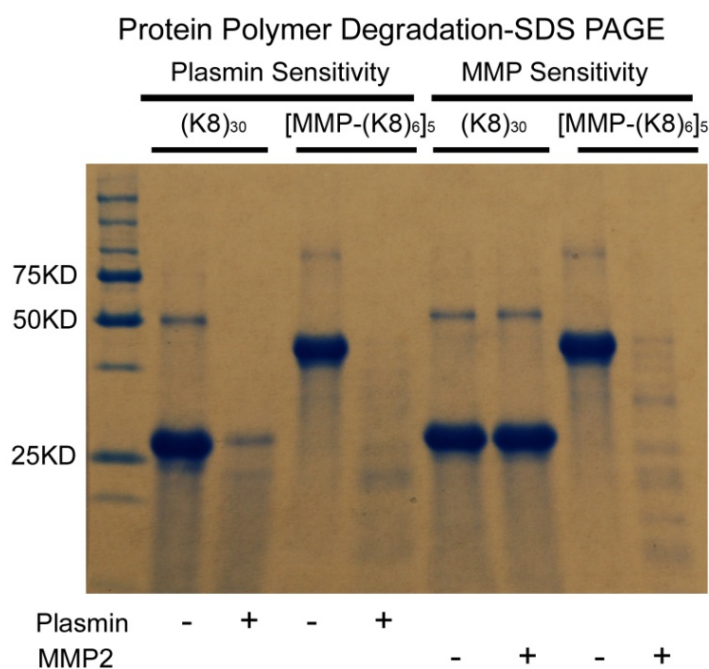


Figure 5.9 Protease degradation of $(K8)_{30}$ and $[MMP-(K8)_6]_5$. $(K8)_{30}$ and $[MMP-(K8)_6]_5$ before and after the protease treatment were run in SDS-PAGE gels. Protein polymers were treated with two proteases, plasmin and MMP-2 for 24h at 37°C. The gel was stained with Coomassie blue.

5.3.5 Enhanced cell invasion into the hydrogels incorporated with MMP degradation sites

For hydrogel studies, the purified proteins were further purified via RP-HPLC to remove the endotoxins that associated with these protein polymers. The protein polymer, [MMP-(K8)₆]₅, was successfully crosslinked with BQ protein polymers into hydrogels by TG, in a similar way as the lysine substrate K8₃₀. To test if the incorporated MMP degradation sites facilitate cell-mediated gel degradation, a hydrogel invasion assay was developed using transwells coated with protein hydrogels. 40 µl of the pre-gel solution was added on top of the membrane in the upper chamber of the transwell to form a thin layer of hydrogel covering the membrane within 2 minutes. MRC-5 fibroblasts were then seeded on top of the hydrogels and were allowed to proliferate and migrate for 5 days. Cells that reached the bottom of the transwell membranes after the incubation would then be stained and counted. Hydrogels made of K8₃₀+BQ and (MMP-K8₆)₅ +BQ were compared in this assay. MRC-5 cells were found to spread well on top of both gels after 12 h, even without the RGD motifs present. Five days later, much more MRC-5 cells reached the bottom of membranes covered with (MMP-K8₆)₅ +BQ hydrogel (Figure 5.10). This data suggested that functionalization of hydrogels with MMP degradation sites can promote the cell-mediated matrix reformation which is a desirable property for tissue engineering applications.

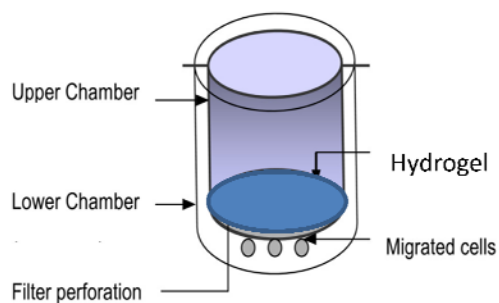
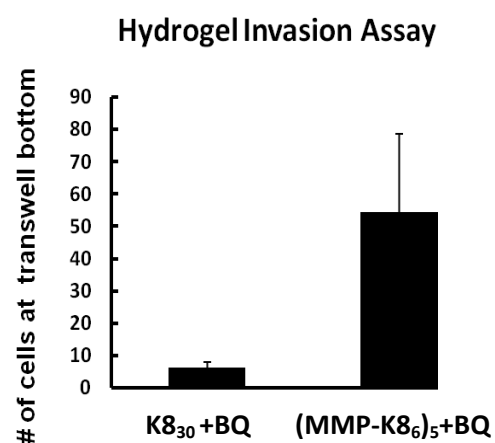
A**B****C**

Figure 5.10 Hydrogel invasion assay. (A) and (B), a cartoon and a real picture showing the hydrogel coating transwell; (C) Cells that reached the bottom of the transwell membrane in the hydrogel invasion assay were stained and counted. Two hydrogel compositions, K8₃₀+BQ and (MMP-K8₆)₅ +BQ, were compared, with triple repeats for each assay.

5.4 Conclusions

A MMP degradable module was cloned into the lysine substrate backbone, giving rise to three types of MMP-K block polymers, two of which have been successfully expressed and purified, [MMP-(K6)₈]₅ and [MMP-(K8)₆]₅. These two proteins were confirmed to be highly water-soluble and expressed at a similar yield to the lysine

substrate K8₃₀. The functionality of the MMP degradation sites in [MMP-(K8)₆]₅ was confirmed by its degradation upon recombinant human MMP-2 treatment. Moreover, [MMP-(K8)₆]₅ can be successfully crosslinked by TG into hydrogels with the glutamine substrate (BQ)₆, and this new hydrogel was found to better allow cell infiltration compared to the hydrogels without MMP degradation sites incorporated.

To better mimic the natural ECM, this hydrogel system can be further functionalized with other bioactive motifs, like cell adhesion signals, heparin binding peptides, cleavable epidermal growth factor sequences, *etc.* The work discussed here will be included in a manuscript for submission, entitled “Engineering of a novel enzymatically crosslinked protein hydrogel for improved biofunctionalities”, with the authors: S. Ding, W. Huang, D. M. Pinkas, N. E. Davis, A. E. Barron.

Chapter 6

**A preliminary study to develop a hybrid,
biocompatible hydrogel as biomimetic scaffolds**

6.1 Introduction

Tissue engineering is the process of creating functional tissues using cells combined with three dimensional (3D) scaffolds to facilitate cell differentiation, proliferation and organization. Moreover, recent studies have demonstrated the limitations of *in vitro* two dimensional (2D) culturing systems in capturing the physiological features (217). Hence, there is an increasing demand for three dimensional models that capture more of the relevant complexity of tissues. The extracellular matrix (ECM) functions as a scaffold to guide tissue reorganization, provides physical support to cells, and guides their behavior by providing both mechanical stimulation and soluble signals from adjacent and even distant cells. A particular challenge in tissue engineering is to develop biocompatible scaffolds that allow a safe loading of desired cells with cell-responsive coupling of chemical and mechanical signals.

The natural ECM comprises a scaffold of various collagens, fibrins, elastins and other structural proteins that is interlaced with many types of glycosaminoglycans including proteoglycans (e.g. heparan sulfate, chondroitin sulfates, keratan sulfates) and hyaluronic acid (HA). For natural ECM, its composition, architecture and degree of crosslinking dictate its mechanical properties both in bulk and locally to control how mechanical forces are transmitted to cells, and also contribute to the microenvironment surrounding the cells through their own bioactive moieties and their ability to bind growth factors, cytokines, and other diffusible compounds. Natural ECM has been found to be extremely variable in composition and mechanical properties (218). Hence, the interest in synthetic hydrogels that can be tailored to mimic specific ECM properties to provide biomimetic scaffolds in a well-controlled and reproducible manner. Due to the complexity associated

with natural ECM, there is a need to develop hybrid hydrogels with biocompatible gelation methods to better recapitulate the nature of ECM for tissue engineering applications.

As introduced in Chapter 5, natural materials, synthetic materials and recombinant protein-based biomaterials have been used to create hydrogels. Hyaluronic acid (HA) is comprised of alternate disaccharide units of [$\beta(1,4)$ -D-glucuronic acid- $\beta(1,3)$ -N-acetyl-D-glucosamine] linkages. It is the only non-sulfated glycosaminoglycan that is widely distributed in connective, epithelial, and neural tissues, such as skin and cartilage (219). Nowadays, HA can be easily synthesized in high molecular weights with a narrow molecular weight distribution and high purity via bacterial fermentation (220). HA is an essential natural ECM component that can directly interact with cell-surface receptors, modulate cell adhesion and motility, and facilitate ECM remodeling and has an important role in natural wound healing and morphogenesis (221). Moreover, HA is found to be biocompatible, non-immunogenic, and non-inflammatory, and has been extensively employed in the biomedical field for drug/gene delivery systems and tissue engineering (221, 222).

The Prestwich group has been developing semi-synthetic HA-based hydrogel systems for *in vivo* tissue engineering and *in vitro* cell cultures in research. First, highly thiolated HA (HA-DTPH) and gelatin (gelatin-DTPH) were generated and crosslinked into hydrogel films via disulfide bonds (219). Second, HA-DTPH can be crosslinked with PEG-diacrylate (PEG-DA) to form an injectable, cell-containing hydrogel for tissue engineering applications (223). This hydrogel was successfully used as a delivery scaffold of autologous mesenchymal stem cells (MSCs) for osteochondral defect

repairing in a rabbit model (224). This system also allows facile attachment of polypeptides which only need to contain cysteines (225) and non-covalent incorporation of native ECM proteins (226). For example, cysteine-tagged fibronectin functional domains were first coupled to one end of homobifunctional PEG derivatives (PEG-DA or PEG-divinylsulfone (PEG-DVS), and then together with bifunctional PEG crosslinkers (PEG-DA or PEG-DVS), reacted with thiolated HA (HA-DTPH) to obtain a hydrogel matrix for wound healing applications (with the crosslinking chemistry illustrated in Figure 6.1) (225). Similarly, a hybrid hydrogel with HA, gelatin, and chondroitin sulfate (CS) can also be formed with PEG-DA as the linker with a useful gelation time around 2-6 min. This sECM was able to grow 1-cm diameter healthy fibrous tissue when seeded with murine NIH 3T3 fibroblasts and grown subcutaneously in nude mice for 8 weeks (227). Third, co-crosslinking of thiolated HA with a small amount of a thiol-modified derivative of heparin can incorporate immobilized heparin into the hydrogel, mimicking the heparan sulfate proteoglycans (HS-PGs) of native ECMs (228). This HS-PG-mimetic ECM provides controlled-release and localized delivery of one or more growth factors, including bFGF, VEGF, angiopoietin-1, and *etc.* (229, 230) (228, 231).

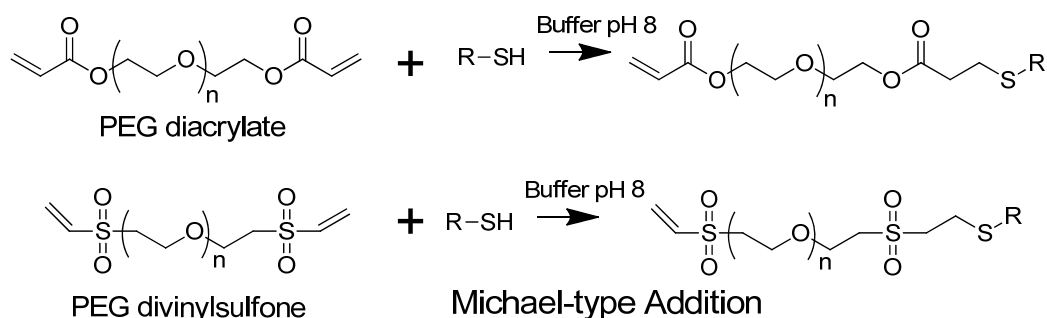


Figure 6.1 The coupling reactions between thiols and acrylate and vinylsulfone groups.

Synthetic polymers have also been widely used to form hydrogels for biomedical applications, including poly(ethylene oxide) (PEO), poly(vinyl alcohol) (PVA), poly(acrylic acid) (PAA), poly(propylene fumarate-co-ethylene glycol) (P(PF-co-EG)), and poly(ethylene glycol) (PEG) (109). PEG is one of the most popular synthetic polymers to develop hydrogels, due to its hydrophilic nature, biocompatibility and non-immunogenicity. However, PEG by itself has no biofunctionality, and thus it has been combined with other bioactive polymers or peptides to construct hybrid hydrogels. PEG-HA hybrid hydrogels have been explored by the Prestwich group, the Anseth group (232) and many others. However, the polyanionic surfaces of HA do not thermodynamically favor attachment to the anionic cell surfaces (219). In general, cells adhere better to polycationic surfaces. Hence, incorporation of more bioactive protein motifs together with HA and PEG may provide more favorable properties to better mimic natural ECM.

As discussed in Chapter 5, recombinant protein polymers provide a facile method to combine various desired protein domains into one single protein polymer in a well-controlled manner, and we have developed a block protein polymer with multiple MMP degradation sites combined with lysine-containing domains that can also promote cell adhesion. These recombinant protein polymers provide much better control than proteins purified from natural sources, such as collagen and gelatin. Herein, we intend to develop a hybrid biomimetic hydrogel which can combine these bioactive, recombinant proteins with HA and also take advantage of the hydrophilic and non-immunogenic properties of PEG. Hydrogels can be formed via various chemical, physical and enzymatic crosslinking methods (233-235). We propose to employ a biocompatible Michael-type

addition reaction with the 8-arm PEG-VS as the crosslinker to react with thiolated protein polymers and HA, giving rise to hybrid PEG-HA-protein hydrogels.

6.2 Materials and methods

6.2.1 Synthesis of the 8-arm PEG-VS crosslinkers

Karen Havenstrite in Blau's group at Stanford helped with the synthesis of the 8-arm PEG-VS (Figure 6.2). Briefly, 8-arm, 40 kDa PEG was dissolved in dichloromethane and NaH was added at 5 molar excess over the -OH groups at the end of the PEG arms. After hydrogen evolution ceased, 50-100 molar excess of divinyl sulfone was added, and the solution was magnetically stirred for 1 day at room temperature and under argon atmosphere. The solution was neutralized with acetic acid, and filtered through paper until clear. The solution volume was reduced to ~10 mL using a rotovap. The functionalized 8-arm PEG was precipitated using ice-cold diethyl ether, recovered via filtration and dried under vacuum. The precipitation was repeated for purification (215).

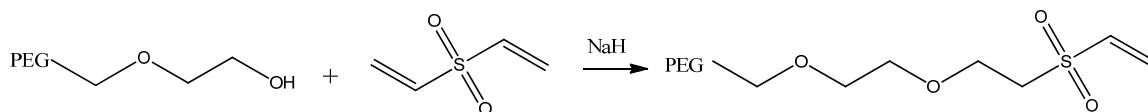


Figure 6.2 The chemistry of making PEG-VS.

6.2.2 Thiolation of the MMP-K proteins with SATP

N-succinimidyl S-acetylthiopropionate (SATP) was purchased from Pierce, and protein thiolation was carried out based on the provided protocol. Briefly, [MMP-(K6)₈]₅ was dissolved at 10 mg/mL in PBS (pH 7.4), and SATP was dissolved in DMSO at 25 molar excess of reactive lysine residues of the proteins. The two solutions were

combined, stirred to react at room temperature for 2 h, and then desalted. Hydroxylamine•HCl was dissolved in PBS (pH adjusted to 7.4 with base) and then reacted with SATP-modified proteins in the presence of EDTA at room temperature for 2 h for the protein deacetylation.

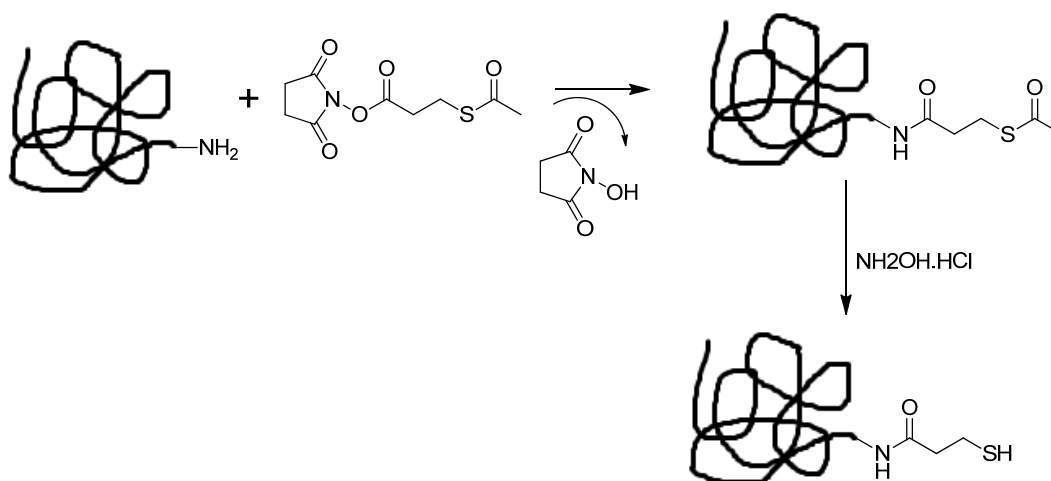


Figure 6.3 The thiolation of proteins with SATP. Proteins with amine groups were functionalized with SATP, and then reacted with Hydroxylamine•HCl for protein deacetylation to give free thiol groups.

6.2.3 Quantification of thiols with Ellman's reagents

Ellman's Reagent, 5,5'-Dithio-bis-(2-nitrobenzoic acid), also known as DTNB, was purchased from Pierce, and the quantification of thiols in HA (GlycosilTM from Glycosan Biosystems, Inc) and SATP-modified proteins was carried out based on the manufacturer's protocol. A solution of DTNB produces a measurable yellow-colored product when it reacts with sulfhydryls, and thus the sulfhydryl groups can be estimated in a sample by comparison to a standard curve composed of known concentrations of a sulfhydryl-containing compound. Cysteine hydrochloride monohydrate (Pierce) was

used as the standard. Briefly, the standards of known thiol concentrations and the samples with thiols to be measured were mixed with DTNB in a reaction buffer of 0.1 M sodium phosphate, pH 8.0, and then incubated at room temperature for 15 minutes. The absorbance of the colored solution was measured at 412 nm. The absorbance values obtained for the standards with known thiol concentrations were used to generate a standard curve. The experimental sample concentrations were derived by fitting their absorbance to the standard curve.

6.2.4 The hydrogel formation

The Michael-type addition reaction for hydrogel crosslinking occurs in 0.2 M TEA buffer, pH 8 (Sigma T0449). Based on thiol quantification, the molar ratio of VS groups in 10% (w/v) 8-arm PEG-VS (TEA buffer) to the thiols in 10% (w/v) SATP-modified proteins (TEA buffer) to the thiols in GlycosilTM (1ml, from Glycosan Biosystems, Inc) is ~ 2:4:1. For hydrogel formation, the total amounts of the VS groups and the thiol groups can be equivalent, while the amounts of HA and proteins adjusted based on interests.

6.2.5 Cell encapsulation and imaging evaluation

The multipotent mesenchymal cells derived from adipose tissues were kindly provided by the Longaker group at Stanford, and were cultured according to published protocols (236). In the initial cell encapsulation study, three types of hydrogels with various components were tested: 1) 8 μ l 10% PEG-VS + 12 μ l HA; 2) 8 μ l 10% PEG-VS + 12 μ l proteins; 3) 8 μ l 10% PEG-VS + 6 μ l HA + 6 μ l proteins. 20 μ l of the pre-gel solutions were quickly mixed with adipose stem cells at a cell density of 5 million/mL, and were then casted into a 3 mm-thick disk with a diameter of 8 mm. These cell-loaded gels were cultured up to 14 days with the culturing media changed once a day.

On day 7 and day 14, the viability of cells in the gels was visualized with the LIVE-DEAD staining kit (Invitrogen) based on its standard protocol. Cells in the gels were further fixed with formaldehyde (Sigma) and stained with Alexa Fluor® 568 phalloidin (Invitrogen) to visualize cellular actin skeletons. Images were taken with a fluorescent microscope (Nikon).

6.3 Results and discussion

6.3.1 The design of the hybrid, biomimetic hydrogel

Natural ECM can be considered as a complicated hydrogel network, with proteins and GAG as important components. To mimic this hybrid nature of ECM, the proposed hydrogel system consisted of three parts: 1) HA, which is an essential ECM component that can directly modulate cell behavior and influence ECM reformation. And its degradation is mediated by cell-produced hyaluronidase (232); 2) multivalent protein polymers which display multiple bioactive domains in a well-controlled manner; and 3) PEG-VS which functions as the crosslinker and enhance the biocompatibility of this gel system. PEG is typically degraded via hydrolysis. As illustrated in Figure 6.4, thiolated HA and thiolated protein polymers can be mixed in desired ratios to be crosslinked with 8-arm PEG-VS via the Michael-type addition reaction, which has been extensively studied by Lutolf *et al.* With the coupling of the Michael-type addition illustrated in Figure 6.1, this crosslinking can happen under physiological conditions with little toxicity to surrounding cells or tissues, allowing a facile encapsulation of cells with controlled release of various growth factors.

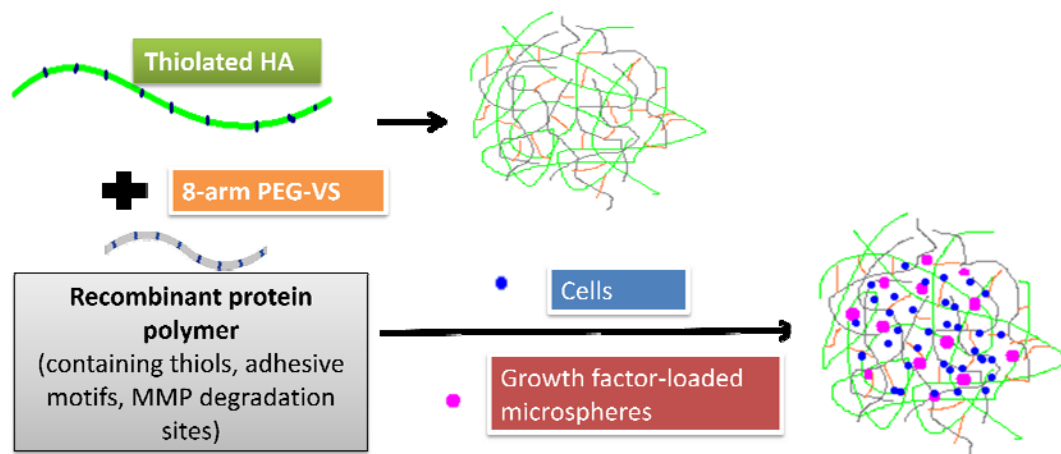


Figure 6.4 A cartoon illustrating the design of the hybrid, biomimetic hydrogel. Bioactive HA and recombinant protein polymers that display multiple thiol groups can be crosslinked by 8-arm PEG-VS into hydrogels. This crosslinking method is also compatible with incorporations of cells and/or growth factor-loaded microspheres.

HA is typically produced by bacterial fermentation with high-molecular weight, and can be degraded to low molecular weight in dilute HCl solution, for example, giving rising to $M_n \sim 120$ kDa with the polydispersity index ~ 1.97 (219). The Prestwich group has generated a method to produce thiolated HA, with the number of thiols per 100 disaccharide units being $\sim 42\%$ (223). Now, this thiolated HA is commercially available as GlycosilTM from Glycosan Biosystems, Inc (Figure 6.5).

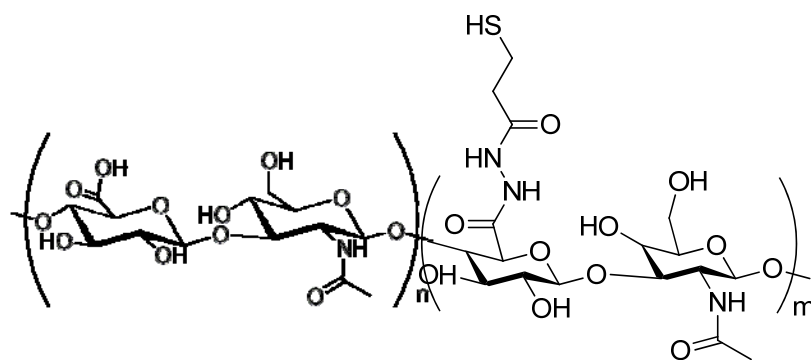


Figure 6.5 Thiolated HA, commercially available as Glycosil™ from Glycosan Biosystems, Inc.

Based on research from Lutolf *et al*, 8-arm PEG-VS of 40 kDa gives preferable mesh sizes in hydrogels when used as the crosslinker. The 8-arm PEG-VS used in this study were prepared by Karen Havenstrite based on published protocols (215, 216).

For the recombinant protein polymer part, these proteins can be genetically engineered to have cysteines in their sequences (126). In Chapter 5, we have developed a block protein polymer with multiple MMP degradation sites and lysine-containing blocks. These proteins are highly water-soluble with good yield. They have been used for multivalent displaying of MRI contrast agents via attachments to lysines (237). Herein, we propose to functionalize this MMP-K block polymer with thiols via the lysine residues.

6.3.2 Thiolation of MMP-K proteins and the characterization

N-succinimidyl S-acetylthiopropionate (SATP) was used to functionalize the MMP-K proteins with thiol groups by reacting with the amines in lysine residues, as illustrated in Figure 6.3. Ellman's reagent was used to quantify the thiol groups in Glycosil™ and SATP-modified proteins. The thiol concentration of the SATP-modified proteins was

determined to be 39.9 mM while the original lysine concentration was 76.7 mM, and thus the thiolation rate via the lysine residues can reach ~52%. GlycosilTM can be dissolved in 1 mL dH₂O to give rise to a HA solution in PBS (pH 7.4) for hydrogel construction and the solution was found to contain thiols at ~8.2 mM.

6.3.3 A biocompatible encapsulation of adipose stem cells with the hybrid hydrogel

To evaluate the biocompatibility of the proposed crosslinking method, a preliminary cell encapsulation study was performed with this hybrid hydrogel. Multipotent mesenchymal cells derived from adipose tissues were kindly provided by the Longaker group at Stanford. 8-arm PEG-VS which were dissolved in TEA buffer (pH 8.0) at 10% (w/v), can be mixed with thiolated HA and proteins in desired ratios to provide equal or greater amounts of thiol groups. This method allowed a fast gelation (within 2 min) to form transparent hydrogels. For cell encapsulation, the adipose stem cells were mixed with pre-gel solutions at a density of 5 million cells/mL, and the solutions were casted to form a 3 mm-thick gel disk with its diameter at 8 mm. The cell-loaded hydrogel disks were then cultured up to 14 days.

The LIVE-DEAD (Invitrogen) staining kit was used to stain the cells in gels on day 7 and day 14 to visualize the live and dead cells. As shown in Figure 6.6, cells with green fluorescence were alive, while cells stained with red were dead cells. We can see that, for all three gel compositions (PEG+HA, PEG+protein, PEG+HA+protein), most of the cells were viable for both 7 days and 14 days, which indicated that the gelation chemistry and the gel components were biocompatible. Moreover, on day 7, hydrogels containing proteins had better cell spreading than just PEG-HA hydrogels. On day 14, the hybrid PEG+HA+protein hydrogel had the highest number of viable cells with good cell

attachments. This preliminary data indicated that the hybrid, biomimetic hydrogel with both bioactive HA and proteins can provide a good environment for the cells to grow. To further visualize the cell spreading, cells in the gel were fixed and then stained with Alexa Fluor® 568 phalloidin. As shown in Figure 6.7, we found the presence of proteins in the hydrogel system promoted cell adhesions.

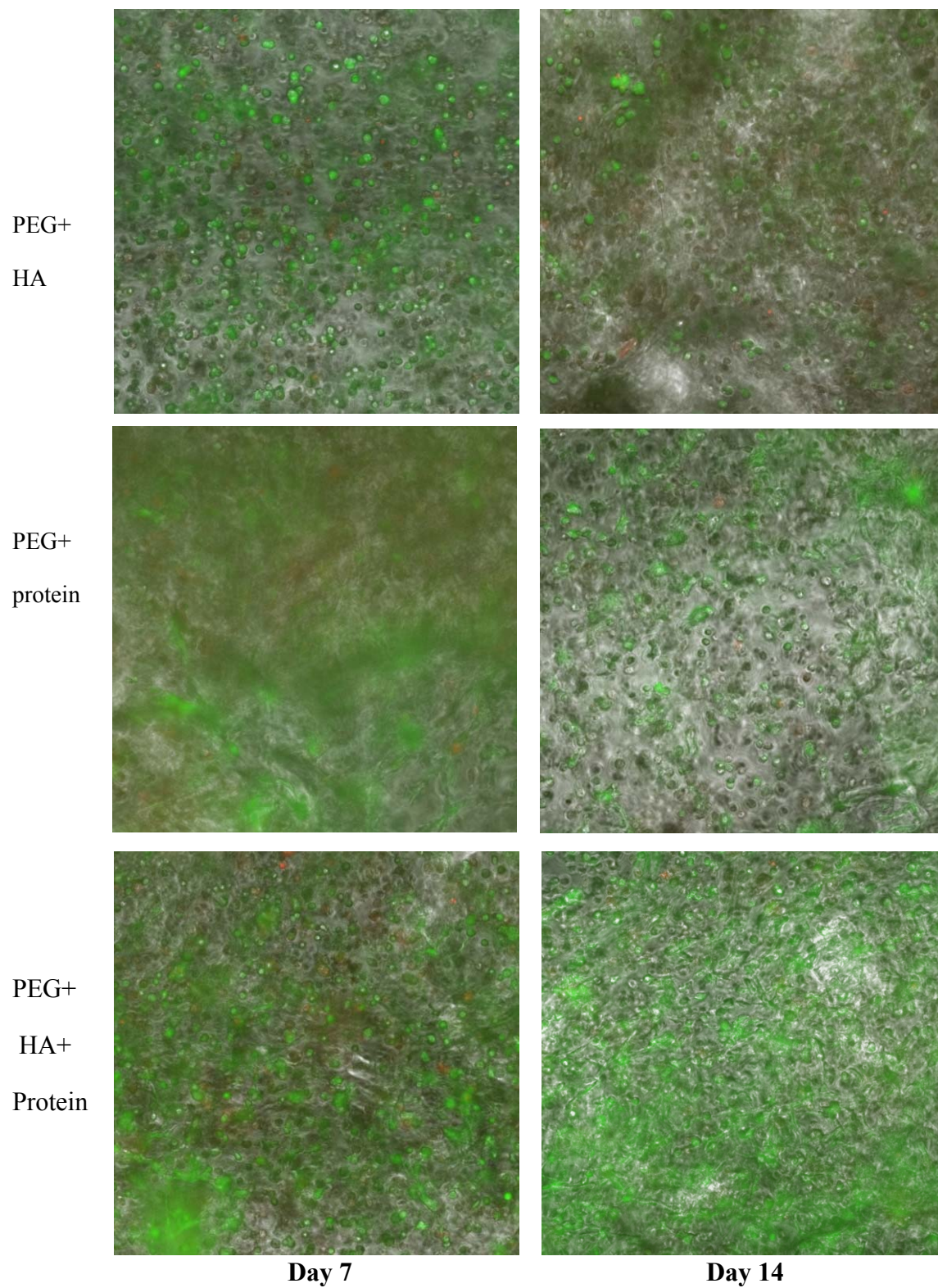


Figure 6.6 The LIVE-DEAD staining of cell-loaded hydrogels. Three types of gel constructs were studied, including PEG+HA, PEG+protein and PEG+HA+protein. The

same amounts of cells were loaded into each gel, and were evaluated on day 7 and day 14 after encapsulation. Each of the images was a merge of the white-field, green fluorescent and red fluorescent fields. Viable cells were stained with green, while the dead ones were stained with red. All the images were taken using fluorescent microscopy.

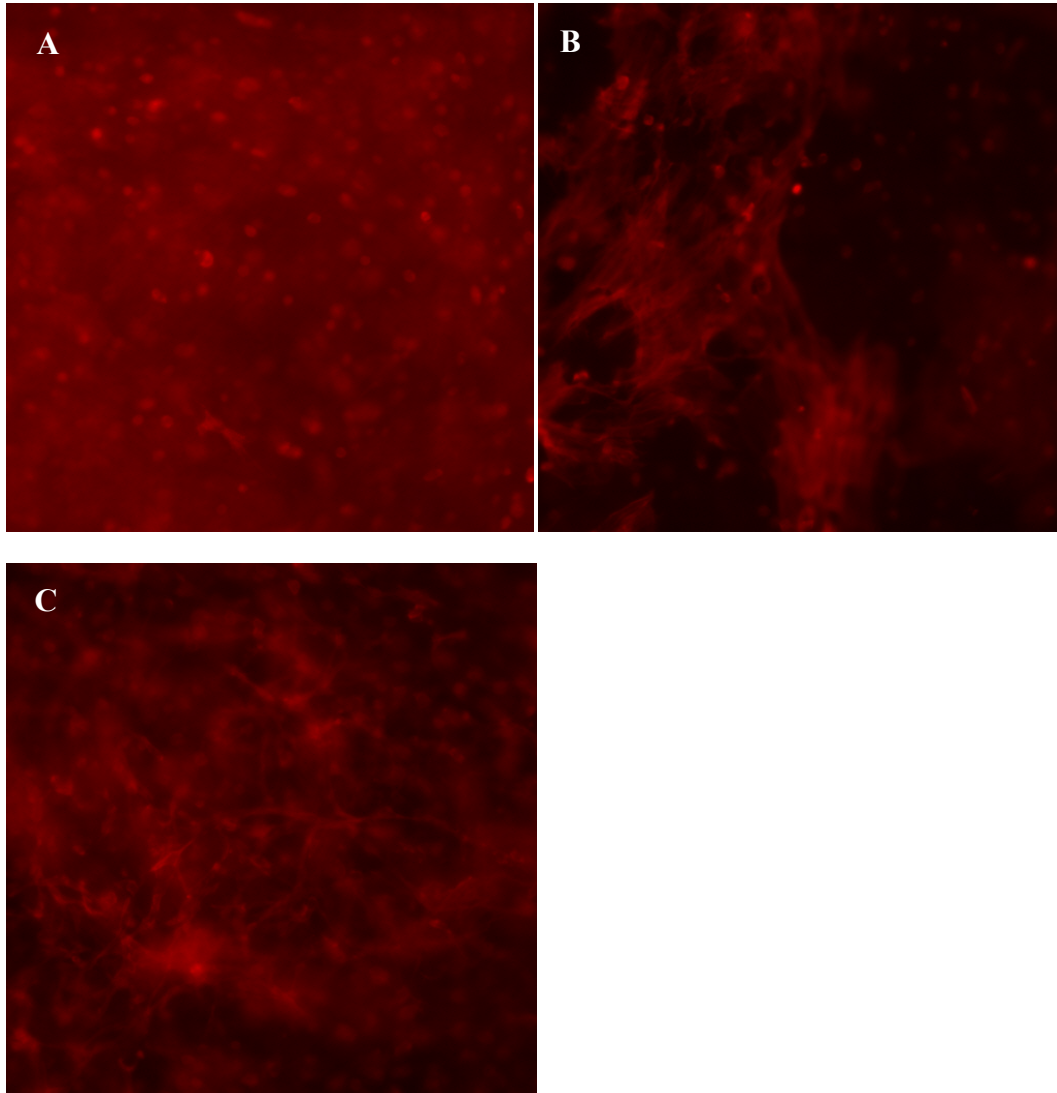


Figure 6.7 The actin staining of cells encapsulated in gels. On day 14, cells in the gel PEG+HA (A), PEG+protein (B), and PEG+HA+protein (C), were first fixed with formaldehyde, washed thoroughly and stained with Alexa Fluor® 568 phalloidin (Invitrogen), which can stain the cellular actin.

6.4 Conclusions

Based on the MMP-K recombinant protein polymers developed in Chapter 5, we proposed to develop a hybrid, biomimetic hydrogel that consisted of bioactive HA, recombinant protein polymers, and non-immunogenic PEG. In natural ECM, HA is an important component, which can directly interact with cells and is involved in various biological processes. Recombinant protein polymers provide a facile way to combine various recognized bioactive protein domains into a single protein polymer. PEG is one of the most widely used polymers for biomedical applications, known for its hydrophilicity and non-immunogenicity. They can be synthesized in a well-controlled manner and allow the incorporation of various reactive groups, broadening the chemistry that can be employed. In the hybrid hydrogel system, we proposed to combine the advantages of HA, recombinant proteins, and PEG, to better mimic the natural ECM. In this system, its degradation would also be mediated via three ways, the degradation of HA by cell-produced hyaluronidase, the hydrolysis of PEG, and the degradation of protein polymers via cell-produced plasmin and MMP.

The Michael-type addition reaction was proposed as the crosslinking method for the hybrid hydrogels. Bioactive HA and MMP-K proteins were chemically modified to display thiol groups, and their thiolation degrees were quantified via Ellman's reagent. 8-arm PEG-VS was used as the crosslinker. Solutions of these three components can be mixed to form hydrogels with a short gelation time. In a preliminary study, adipose stem cells were encapsulated in PEG+HA, PEG+protein and PEG+HA+protein hydrogels. The gelation and gel materials were found to be biocompatible with low toxicity to the

loaded cells, while the three-component PEG+HA+protein hydrogel was found as the best scaffold to promote cell adhesion and proliferation.

This hybrid hydrogel system may function as a good scaffold for tissue engineering applications or *in vitro* culturing systems. Future work will include better characterizations of the hydrogel properties (swelling ratios, mechanical properties, their degradation, different ratios between gel components) and exploring this system for a specific application.

Chapter 7

Conclusions and future directions

The development of biomimetic designs is a highly interdisciplinary research area which has the potential to address challenges in various biomedical applications. The first part of this dissertation focused on mimicking short bioactive peptides via synthetic peptidomimetic scaffolds, peptoids, and the second part focused on generating genetically engineered protein polymers for hydrogel development to mimic the ECM. Taken together, five different applications have been explored with various biomimetic designs. In the following sessions, a summary of contributions and recommendations for future directions are discussed for each application.

7.1 Development and evaluation of cationic, amphipathic peptoids for anticancer applications

7.1.1 Summary of contributions

Severe side effects and drug resistance developed in patients are two main drawbacks of current chemotherapeutics in cancer treatment. Interestingly, several host-defense peptides, most of which have a cationic, amphipathic structure, have been reported with selective anticancer activity and the ability to overcome multidrug resistance. In this dissertation, a library of cationic, amphipathic peptoids has been synthesized and evaluated for their anticancer activity via *in vitro* screening and a preliminary *in vivo* study in a primary, orthotopic xenograft mouse model of human breast cancer.

Our results indicated that, by just mimicking the cationic, amphipathic structural feature of host-defense peptides instead of their exact physicochemical moieties, these peptoids were able to display potent cytotoxicity against the tested cancer cell lines, and their activity was similar in MCF-7/TxT50 cells, in which the multidrug resistance was

developed. Moreover, these peptoids can be tuned in their structures to have some selectivity against the established cancer cell lines compared to the normal cell controls including nontransformed fetal fibroblasts, primary fibroblasts and red blood cells. These data indicated that peptoids can be an effective mimicking scaffold to capture the bioactivity of some peptides while gaining improved molecular stability against protease degradation.

Moreover, these peptoids may gain higher selectivity *in vivo* where cancerous cells keep proliferating but normal tissue cells don't. Their different proliferation rates may broaden the membrane differences between cancerous cells and normal tissue cells, and lead to higher selectivity of these peptoids. In addition, doxorubicin which is a clinically used chemotherapeutic agent didn't display any selectivity in our *in vitro* screening system.

Via an intratumor delivery in the primary orthotopic xenograft mouse model, the most potent peptoid with low selectivity, peptoid 1, was found to greatly inhibit the tumor growth and well tolerated locally. Taken together, these data have suggested a promising anticancer application of these cationic, amphipathic peptoids.

The other contribution of this work focuses on the killing mechanisms of these peptoids. By comparing the cytotoxicity and the acute plasma membrane damages caused by these peptoids, the fast killing of these peptoids mainly involved the plasma membrane damages. Moreover, these peptoids can penetrate into cells and be present in the cytoplasm for a long time (up to a week we have tested *in vitro*) without causing measurable cytotoxicity at lower concentrations. Our hypothesis is that these peptoids may

target intracellular organelles to cause chronic cytotoxicity at proper relatively low concentrations, but their acute killing mainly involves plasma membrane damages. Based on these mechanisms, the efficacy and biosafety of peptoids need to be studied carefully, and cancer types that are suitable for local drug treatment can be good models to evaluate these peptoids.

7.1.2 Recommendations for future directions

Two directions will be further explored for these anticancer peptoids.

First, we intend to evaluate the efficacy and biosafety of peptoids in an orthotopic bladder cancer mouse model. Bladder cancer remains a significant problem throughout the world. Carcinoma of the bladder is the second most common cause of mortality from genitourinary malignancies, second only to those caused by prostate cancer. Superficial bladder cancer is usually treated by surgical resections and intravesical chemo- and immunotherapy. However, muscle-invasive bladder cancer is frequently incurable and is associated with a high relapse rate after surgery and/or chemotherapy (238). Therefore, development of novel strategies for therapeutic targeting of muscle-invasive and metastatic bladder cancer is crucial.

We hypothesize that these anticancer peptoids may have clinical utility as potential treatments for bladder cancer. Previous data have indicated *in vivo* potency and good tolerance of these peptoids via local delivery, and we have demonstrated that treatments of the human bladder cancer cell lines UMUC3, 639V, and HT1376 with peptoid 1 resulted in fast, significant growth inhibitions (Data not shown). We propose to study the *in vivo* bladder toleration and drug penetration of peptoids via intravesical administration.

Cationic, amphipathic molecules normally have good tissue penetrations. Delivering fluorescently labeled peptoids to mouse bladders will enable studying the peptoid tissue penetrations. After examining peptoid toxicities associated with intravesical delivery, the *in vivo* anticancer efficacy of these peptoids will be examined in an orthotopic bladder cancer mouse model.

The second direction to explore is to further improve the selectivity of these peptoids, either by modifying these peptoids with targeting moieties or delivering them via targeted methods.

Two types of targeting moieties can be conjugated to promising peptoids to gain higher selectivity: (1) enzyme cleavable sequences, such as γ -glutamates that can be removed by prostate specific membrane antigen (PSMA), and HSKLQ that is sensitive to prostate specific antigen (PSA) (239, 240); (2) “tumor homing” moieties, such as the CNGRC peptide (165), the LHRH peptides (166), aptamers or antibodies (241), *etc.* Conjugations of these targeting moieties on the N terminal, C terminal or side chains of peptoids will be compared. The first group functions as pro-drugs, the full activity of which needs to be activated by tumor related enzymes; the second group tries to enhance their interactions and accumulations in tumors, and reduce their nonspecific membrane interactions, which can also be similarly achieved via targeted delivering methods for these peptoids.

7.2 Stable, efficient peptoid transporters--structures determine their uptake efficiency and mechanisms

7.2.1 Summary of contributions

Current research about cell penetrating peptides, or known as molecular peptoids, has been a diverse and controversial field. Peptides of various structural features have been found with cell penetrating capabilities and different uptake characteristics, and it is still unclear how their structures influence their uptake. By using peptoids as the peptidomimetic scaffold, we are able to control the structural features, including molecular hydrophobicity, charges, secondary structures, and *etc.*, in a simple and precise manner. Therefore, it is easier to carry out the structure-activity study using peptoids.

Our structure-activity study suggested that both guanidinium heads and bulky, aromatic side chains can influence the uptake of peptoid transporters depending on their molecular charge-to-length ratios. Moreover, cationic, amphipathic peptoid transporters have different internalization mechanisms from polyguandine comparators, as indicated by their different cellular distribution patterns and uptake behaviors at 4°C.

This work not only developed two efficient peptoid transporters, but also explored what structural elements may influence peptoid uptake efficiency, and also for the first time proved that the amphipathic transporters utilized different uptake mechanisms from polyguandine transporters.

7.2.2 Recommendations for future directions

Two efficient transporters were identified, the most efficient yet with high cytotoxicity, peptoid H-(Nlys-Nlys-Nspe)₄-NH₂, and the peptoid H-(Nbtg-Nbtg-Nspe)₄-

NH₂ with efficient uptake and low cytotoxicity. When killing is not the primary goal, transporters are desired to have low toxicity. Our future work involves specific applications of these peptoid transporters.

The applications of transporters to deliver various bioactive cargos are well reviewed previously (43, 44). Benefited from the low susceptibility of peptoids to protease degradation, these peptoid transporters can be used for similar applications, with the potential to gain an improved intracellular and *in vivo* efficacy compared to peptide transporters.

More specifically, we are interested in using the transporters to deliver hydrophobic small-molecule chemotherapeutic agents to overcome the multidrug resistance. Oligoarginines have been used to deliver Taxol with improved anticancer efficacy and the ability to overcome the drug resistance (171). Our peptoid transporters can be employed similarly to improve the solubility and tissue penetration of hydrophobic small-molecule chemotherapeutic agents. Especially for the most efficient and toxic peptoid H-(Nlys-Nlys-Nspe)₄-NH₂ which has been reported with anticancer activity at the low micromolar concentrations, it can be an ideal delivering vector for small-molecule chemotherapeutic agents due to its extremely high uptake efficiency and intrinsic ability to overcome the multidrug resistance.

7.3 Targeted prostate cancer imaging with a multivalent Bombesin-PEG probe

7.3.1 Summary of contributions

Truncated BBN(7-14) peptide provides a promising basis for developing radiometallated diagnostic or therapeutic radiopharmaceuticals to target GRPR/BB2/BB₂

positive prostate cancers. However, small peptide-based reagents suffer from a fast renal clearance from the blood circulation and thus short tumor retentions for *in vivo* applications. In this work, we synthesized 4-arm PEG conjugates functionalized with BBN(7-14) at the end of each arm, which can be further radiolabelled for targeted imaging of the GRPR/BB₂ positive prostate cancer. The PEG-BBN conjugates were shown to have a good and specific binding affinity to PC3 cells, and *in vivo* microPET imaging, quantifications, and biodistribution studies revealed that radiolabelled PEG-BBN conjugates had good and specific tumor uptakes.

PEG has been widely used to modify therapeutic molecules, to prolong the half-life of low-molecular weight drugs or peptides, enhance their stability in the plasma and reduce their immunogenicity. Our work, for the first time, employed high molecular weight PEG as a multivalent platform to deliver BBN for targeted cancer imaging. This method has been shown to improve their tumor uptake and have slower clearance from the body as compared to free small peptides.

7.3.2 Recommendations for future directions

Future work of this project involves more comparison of PEG-BBN conjugates with free BBN peptides in terms of their imaging efficiency and biodistribution. On the other hand, we are interested in developing a peptoid version of the BBN(7-14) peptide for targeted tumor delivery or imaging, aiming to retain its high affinity to GRPR on cancer cells while gaining improved molecular stability. This can be a challenging project, due to the lack of backbone hydrogen bonding and backbone chirality in peptoids, which may reduce the binding affinity of peptoid molecules to certain protein receptors.

Two methods can be used to develop such peptoids for this goal. First, single-residue mutations can be carried out on BBN peptides, replacing one residue with desired peptoid analogue at one time. This will also enable us to find out what residues are critical for BBN's affinity to GRPR. However, this method can be time-consuming, and the hybrid molecules may still suffer from fast degradation *in vivo*. Second, a large, random peptoid library can be synthesized and screened. Kodadek *et al* have used this method to develop peptoids that can antagonize VEGF receptor 2 activity (18). To use this method, a screening platform for GRPR binding needs to be set up first. And it can be risky to find peptoid hits with great affinity to GRPR. Despite the challenges in both methods, it can be very rewarding to develop a peptoid that can bind to GRPR with the good and specific affinity.

7.4 Recombinant protein polymer-based biomimetic hydrogels

7.4.1 Summary of contributions

Two hydrogel systems that are built with recombinant protein polymers have been discussed in this dissertation. The first is a TG crosslinking protein hydrogel system, which is comprised of just proteins. This system was initially designed by previous lab members in Barron group, and the work discussed here included the controlled cloning of MMP-K protein polymers to introduce the MMP-sensitive modules into this enzyme-crosslinking hydrogel system, to promote cell-mediated degradation of the hydrogel scaffold. The functionality of MMP-sensitive modules was verified, and hydrogels incorporated with MMP-sensitive modules promoted cell migrations in the gel.

The second hydrogel system is a hybrid, biomimetic gel comprising HA, PEG and bioactive protein polymers. Natural ECM is a hybrid gel network composed of protein fibrils, various polysaccharides and glycoproteins, signaling molecules, and *etc.* To better mimic natural ECMs, we intended to develop a hybrid hydrogel. As discussed in Chapter 6, HA, PEG and bioactive protein polymers all provide specific benefits as important components in this gel system. The work discussed is a preliminary study of this system, which showed a biocompatible crosslinking design to gelate the three components via the Michael-type addition, and a good support for cell proliferation and spreading to have all the three components present in the system.

7.4.2 Recommendations for future directions

Chapter 6 introduced a preliminary study of the hybrid, biomimetic hydrogel system, and future work involves characterizations of material properties and the utilization of this hybrid system for a specific application.

The hydrogel properties need to be examined carefully, including their gelation kinetics, the hydrogel swelling ratios, their mechanical properties, degradation rates, and the effects of different component ratios.

This hybrid, biomimetic hydrogel can be used as scaffolds for tissue regeneration or as scaffolds for *in vitro* three-dimensional cultures. HA is known to be important for wound healing, so this gel system can be good for related applications. Moreover, this system can be tailored to guide or instruct stem-cell fate both in culture and *in vivo* (242). More specifically, adipose-driven stem cells were safely encapsulated in this hybrid gel system and their proliferation was well supported. The ability to differentiate and

proliferate of these cells in these gels can be further characterized. Based on current knowledge, stem cells and biomaterials are two important components that can be employed for proper tissue regeneration. Biomimetic technologies may advance the stem cell-biomaterial-based clinical approaches to tissue regeneration.

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